

Reproductive innovation and angiosperm diversity: bridging micro- and macroevolution

Reprodukční inovace a diverzita krytosemenných rostlin: propojení mikro- a makroevoluce



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*In memory of my little brother, Omid,
forever missed and always loved.*

Table of contents

Declaration	1
Author contribution statement	2
Acknowledgment	3
Abstract	5
Abstrakt	6
Chapter 1: General introduction	7
1.1 Angiosperms evolution, the “Darwin’s abominable mystery”	7
1.2 Ecological drivers of reproductive trait divergence and speciation in angiosperms	8
1.2.1 Environmental selective pressures drives the evolution of reproductive traits in plants	8
1.2.2 Resource-dependent sexual investment in plants	10
1.2.3 Environmental selective pressures can drive RI and speciation in plants	10
1.3 Whole genome duplication and the evolution of traits and species in angiosperms	11
1.3.1 Polyploidization: a frequent and foundational phenomenon in plants	11
1.3.2 Polyploidization and the evolution of (reproductive) traits in plants	11
1.3.3 Polyploidization and its role in the evolution of RI, and speciation in plants	12
1.4 Reproductive strategies: a masterpiece of angiosperms’ traits and species evolution	13
1.4.1 Sexual selection: from elegant theory to evolutionary ambiguity	13
1.4.2 Sexual selection leads to the evolution of traits in plants	13
1.4.3 Sexual selection and the evolution of RI in plants	14
1.4.4 Sexual selection and speciation in plants	15
1.4.5 Selfing, a common evolutionary transition affecting diversification in plants	15
1.5 An interplay of ecological and sexual selection	16
Chapter 2: Objectives of the thesis	17
Chapter 3: Comments on material and methods	17
3.1 Model system	18
3.2 Phylogenetic reconstruction	19
3.3 Comparative analysis; bridging micro- to macroevolution	20
Chapter 4: Synthesis and discussion of the results	21
4.1 Environmental factors exert a limited impact on the evolution of reproductive traits	21
4.2 WGD drives trait shifts in neopolyploids but not diversification in established polyploids	22
4.3 Reproductive strategies shape reproductive trait variation and speciation patterns	24
4.4 Bridging micro- and macroevolution presents both opportunities and challenges	26
Conclusion	27
Reference	28
Case study 1	41
Case study 2	59
Case study 3	71
Case study 4	105
Case study 5	125

Declaration

I hereby declare that this thesis is my own original work, composed independently using the cited references. It has not been submitted, in whole or in part, for any other degree or academic qualification at this or any other institution.

Tímto prohlašuji, že tato disertační práce je mým původním dílem, které jsem vypracoval samostatně s využitím uvedených pramenů a literatury. Nebyla dosud předložena, ani zčásti, k získání jiného akademického titulu na této ani žádné jiné instituci.

V Praze, 31. 07. 2025

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Author contribution statement

I hereby declare that I have contributed significantly to all the papers included in this thesis. My specific contributions to each are outlined in the table below.

Contribution	CS1	CS2	CS3	CS4	CS5
Conceptualization and study design	✓	✓	✓	✓	✓
Sample/data collection and preparation	N/A	✓	✓	N/A	✓
Experimental work	N/A	✓	✓	N/A	✓
Data analysis, synthesis, and interpretation	✓	✓	✓	✓	✓
Writing – original draft preparation	✓	✓	✓	✓	✓
Writing – review/editing	✓	✓	✓	×	✓
Overall contribution (%)	70%	50%	80%	40%	80%

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I am your Doctor. "دکتر خودم"

Abstract

The remarkable diversity of angiosperms has long fascinated evolutionary biologists, yet the mechanisms at work remain incompletely understood. Since reproductive isolation (RI) underlies the formation of new species, the evolution of traits involved in sexual reproduction—such as those mediating mating and fertilization—is likely a key driver of reproductive divergence and, ultimately, angiosperm diversity. While some factors influence the evolution of reproductive traits at the population level, it remains unresolved whether and how they shape RI and ultimately speciation across clades. In particular, the macroevolutionary consequences of ecological drivers (e.g., abiotic and biotic environmental factors), whole-genome duplication (WGD), and reproductive strategies—including mating systems and sexual selection—have not been systematically integrated. This thesis bridges microevolutionary mechanisms (e.g., trait divergence and RI) with macroevolutionary outcomes (e.g., speciation) to uncover how ecological, genomic, and reproductive strategies drive angiosperm diversity.

To address these questions, I followed a multi-scale approach in this thesis. Firstly, I conducted field and greenhouse experiments to assess the impact of ecology and WGD on the diversity of reproductive traits and their impact on RI. In addition, I reconstructed two dated phylogenies of ~1,000 Brassicaceae species using nuclear and plastid data, fossil calibrations, and a Bayesian relaxed-clock framework. I then compiled trait data (e.g., pollen:ovule [P:O] ratio, style length, petal dimensions, mating systems, ploidy) from experiments and literature, combined with imputations via machine learning. This allowed me to assess whether rapid evolution of floral traits is associated with increased speciation rates, using trait rate estimates from BAMM and RRphylo, lineage-specific speciation rates from ClaDS, and Bayesian multilevel models (BMLMs) that incorporate mating system-specific slopes and phylogenetic structure via random effects for genus and tribe. I also tested how WGD and mating systems influence diversification using trait-dependent models (HiSSE, MuHiSSE). Together, these analyses integrate traits and diversification across scales.

I found that abiotic and biotic environmental factors—such as climatic conditions and pollinator diversity—had weak effects on pollen and ovule production, with most variation constrained by phylogeny rather than environment. Regarding WGD, newly formed polyploids in *Arabidopsis arenosa* showed enlarged floral organs, but these changes did not lead to any assortative pollinator visitation (i.e., no RI). At the Brassicaceae family level, polyploidy did not explain variation in reproductive traits and was not associated with increased speciation. Reproductive strategies had strong macroevolutionary effects. Selfing lineages exhibited reduced floral trait values (e.g., lower P:O ratio and shorter style), slower floral trait evolution, and slower diversification, primarily due to higher extinction. The values of sexually selected traits, by themselves, did not predict speciation, but the rates of their evolution did, especially among outcrossers. Specifically, evolutionary rates of the traits were consistently associated with faster speciation.

These findings reveal that genomic, ecological, and reproductive drivers differ markedly in their contributions to speciation and its underlying processes. Whole-genome duplication triggers immediate shifts in floral traits but has a limited long-term impact on diversification. Environmental factors had limited explanatory power, with variation in pollen and ovule production primarily structured by phylogenetic relatedness. In contrast to both genomic and ecological drivers, reproductive strategies—particularly the interaction between mating system and sexual selection—were the only driver to consistently influence both reproductive trait divergence, a key microevolutionary process, and speciation, offering new insight into angiosperm diversity and a potential resolution to “Darwin’s abominable mystery.”

Abstrakt

Druhová rozmanitost krytosemenných rostlin dlouhodobě fascinuje evoluční biology. Přesto zůstávají mechanismy, které tuto diverzitu generují, objasněné jen částečně. Vzhledem k tomu, že reprodukční izolace (RI) je podmínkou vzniku nových druhů, lze předpokládat, že evoluce znaků spjatých s pohlavním rozmnožováním—jako například znaky zprostředkující oplození—představuje klíčový hnací motor reprodukční divergence a diverzity krytosemenných rostlin. Zatímco některé faktory ovlivňují evoluci reprodukčních znaků na populační úrovni, není jasné, jak a zda tyto znaky formují RI a tím i speciaci napříč klady. Zejména makroevoluční důsledky ekologických faktorů, celogenomové duplikace (WGD) a reprodukčních strategií—včetně rozmnožovacích systémů a pohlavního výběru—dosud nebyly systematicky integrovány. Tato dizertace propojuje mikroevoluční mechanismy (např. divergenci znaků a RI) s makroevolučními procesy (např. speciací) za účelem odhalení, jak ekologické, genomové a reprodukční strategie ovlivňují diverzitu krytosemenných rostlin.

K zodpovězení těchto otázek jsem v této práci zvolil víceúrovňový přístup. Nejprve jsem provedl terénní a skleníkové experimenty, abych posoudil vliv ekologie a WGD na rozmanitost reprodukčních znaků a jejich dopad na RI. Dále jsem rekonstruoval dvě datované fylogeneze přibližně 1 000 druhů čeledi brukvovitých (Brassicaceae) pomocí jaderných a plastidových dat, fosilních kalibrací a Bayesovského modelu s rozvolněnými molekulárními hodinami. Shromáždil jsem data o znacích (např. poměr pyl:vajíčka [P:O], délka čnělky, rozměry korunních lístků, rozmnožovací systémy, ploidie) z vlastních experimentů i literatury a doplnil je imputacemi s pomocí strojového učení. To mi umožnilo testovat, zda je rychlá evoluce květních znaků spojena se zvýšenou mírou speciace, s využitím odhadů rychlosti evoluce znaků metodami BAMM a RRphylo, liniově specifických měr speciace z modelu ClaDS a Bayesovských hierarchických modelů (BMLMs) zahrnujících fylogenetickou strukturu s náhodnými efekty rodu a tribu. Rovněž jsem testoval, jak WGD a rozmnožovací systémy ovlivňují diverzifikaci pomocí modelů závislých na znacích (HiSSE, MuHiSSE). Tyto analýzy společně integrují znaky a diverzifikaci napříč škálami.

Zjistil jsem, že biotické a abiotické vlivy prostředí – jako klima a diverzita opylovačů—měly slabý efekt na produkci pylu a vajíček, přičemž většina variability byla podmíněna fylogenezí, nikoliv prostředím. Co se týče WGD, nově vzniklé polyploidie u druhu *Arabidopsis arenosa* vykazovali zvětšené květní orgány, ale tyto změny nevedly k preferenční návštěvnosti opylovači (tj. nedošlo k RI). Na úrovni čeledi Brassicaceae ploidie nevysvětlovala variabilitu v reprodukčních znacích a nebyla spojena se zvýšenou speciací. Reprodukční strategie měly silný makroevoluční vliv. Samoopylující linie vykazovaly snížené hodnoty studovaných květních znaků (nižší poměr P:O, kratší čnělka), pomalejší evoluci květních znaků a nižší diverzifikaci, především v důsledku vyšší míry vymírání. Samotné hodnoty znaků spojených s pohlavním výběrem nesloužily jako prediktor speciace, avšak rychlejší evoluce těchto znaků—zejména u cizosprašných druhů—byla konzistentně spojena s vyšší mírou speciace.

Tato zjištění ukazují, že genomické, ekologické a reprodukční faktory se výrazně liší ve svém vlivu na speciaci. Celogenomové duplikace vyvolávají okamžité změny v květních znacích, ale jejich dlouhodobý dopad na diverzifikaci je omezený. Environmentální faktory měly jen omezenou vysvětlovací sílu. Variabilita v produkci pylu a vajíček byla dána primárně fylogenetickou příbuzností. Na rozdíl od genomických a environmentálních faktorů byly reprodukční strategie—zejména interakce mezi rozmnožovacím systémem a pohlavním výběrem—jediným faktorem, který konzistentně ovlivňoval jak divergenci reprodukčních znaků, což je klíčový mikroevoluční proces, tak speciaci. To přináší nový vhled do rozmanitosti krytosemenných rostlin a možný klíč k rozřešení „Darwinovy příšerné záhady“.

Chapter 1: General introduction

1.1 Angiosperms evolution, the “Darwin’s abominable mystery”

The astonishing richness of angiosperms—“Darwin’s abominable mystery”—has long intrigued evolutionary biologists. What sparked the rapid and extensive radiation of flowering plants, and why do they so vastly outnumber other plant lineages? Today, angiosperms comprise nearly 90% of land plant species (~300,000 species in 400–500 families; Zuntini et al., 2024), from the minuscule *Wolffia* to the towering *Sequoiadendron* (Romano & Aronne, 2021; Sillett et al., 2021), and dominate nearly all terrestrial biomes (Folk et al., 2020). Their evolutionary success rests on deep-time innovations—morphological, physiological, reproductive, and genomic—that enabled unparalleled ecological expansion and adaptation (Endress, 2006; Bres et al., 2021; Bhadra et al., 2023). These innovations trace back to the rise of land plants, with key transitions in life cycles and organs largely complete by the Devonian (~359 Ma), setting the stage for plants to restructure energy and nutrient cycles and catalyze new ecological interactions (Kenrick & Crane, 1997; Pires & Dolan, 2012; Smart et al., 2023). As Benton et al. (2022) aptly observed, “The rise of angiosperms triggered a macroecological revolution.” Yet their diversity remains strikingly uneven—from relict lineages like *Amborellaceae* to hyper-diverse clades such as *Asteraceae* and *Orchidaceae* (Große-Veldmann et al., 2011; Wang et al., 2024a, Wang et al., 2024b). What explains this disparity, and do trait innovations reliably translate into higher speciation? These questions lie at the heart of this thesis.

At the core of angiosperm success lies a singular innovation: the flower. This complex reproductive structure, composed of sepals, petals, stamens, and carpels, emerged in the Early Cretaceous (~140 Ma), coinciding with a dramatic burst in angiosperm diversification (Sauquet et al., 2017; Magallón et al., 2019). Flowers enabled greater precision in reproduction and fostered intimate ecological partnerships with insects, birds, and bats, extending the adaptive reach of flowering plants (Armbruster, 2017; Pauw, 2019; Stephens et al., 2023). The staggering variation in floral traits—shapes, colors, rewards, architectures—epitomizes the kind of innovation thought to underlie their ecological and evolutionary success. Multiple factors have been proposed to explain this extraordinary radiation: adaptive radiation into new niches (Givnish, 2010), WGD that generate genetic novelty (Van der Peer et al., 2017), mating system shifts and sexual selection (Goldberg et al., 2010; Moore & Pannell, 2011), and transitions in pollination (Van der Niet & Johnson, 2012). Others highlight herbivore co-evolution (Van der Linden et al., 2021) and genome size variation (Bhadra et al., 2023). These may all contribute to trait disparity—but whether such diversity reliably promotes speciation remains unresolved.

To understand speciation in angiosperms, first, species must be defined. The Biological Species Concept (BSC), proposed by Mayer (1942), defines species as “*groups of actually or potentially interbreeding natural populations which are reproductively isolated from other such groups*” (Mayer, 1942, 1996). Under this framework, speciation occurs when RI gives rise to genetically and phenotypically distinct lineages (Dobzhansky, 1935; Mayer, 1942; Turelli et al., 2001; Mallet, 2010; Seehausen et al., 2014). RI can be prezygotic (e.g., geographic separation and pollinator shifts) or postzygotic, including hybrid inviability or sterility (Dobzhansky, 1970; Coyne, 1992; Coyne & Orr, 1998; Schemske & Bradshaw, 1999; Orr & Turelli, 2001). Such reproductive barriers may arise when divergent lineages evolve distinct traits—driven by ecological selection, sexual selection, genome duplication, or genetic drift—that reduce gene flow, either by impeding mating success (e.g., changes in pollination signals) or by lowering hybrid fitness due to local adaptation (Ritchie, 2007; Schluter, 2009; Nosil et al., 2009; Johnson, 2025). Yet whether RI scales predictably to macroevolutionary patterns of species richness remains uncertain. Speciation rate variation is often linked to RI tempo, and if diversification rate variation mirrors the tempo of RI, then clade-specific RI rates should predict species accumulation (Panhuis et al., 2001; Coyne & Orr, 2004; Jetz et al., 2012). However, studies in birds and *Drosophila* suggest that while clades vary in how rapidly they evolve prezygotic and

postzygotic barriers, these differences do not correlate with speciation rates (Rabosky & Matute, 2013), implying that RI does not necessarily drive speciation (Mallet, 2006).

Speciation-related traits are often assumed to hasten RI (Coyne & Orr, 2004). However, species accumulation reflects both speciation and extinction, constrained by ecological limits (Darwin, 1859; Rabosky, 2013). As ecological niches fill, speciation rates may decline—a process known as Darwinian diversity-dependence—which can reduce the macroevolutionary impact of trait-driven reproductive barriers (Rabosky & Matute, 2013). Thus, while RI helps maintain species boundaries, it does not necessarily drive long-term diversification (Rabosky, 2013). Successful speciation additionally requires ecological differentiation, geographic or environmental isolation, and lineage persistence (Harvey et al., 2019). These prompts a central question: under what ecological, genomic, or reproductive conditions does trait evolution drive long-term species diversity? In the following sections, I explore how ecological drivers, WGD, and reproductive strategies shape the evolution of reproductive traits and assess their potential to drive speciation across angiosperms (Figure 1). I focus on reproductive traits given their central role in mediating mating interactions and fertilization, making them prime candidates for generating RI and contributing to species divergence.

1.2 Ecological drivers of reproductive trait divergence and speciation in angiosperms

1.2.1 Environmental selective pressures drives the evolution of reproductive traits in plants

As sessile organisms, flowering plants are profoundly influenced by both abiotic and biotic environments (da Silva et al., 2021; Patra et al., 2024). Abiotic factors such as temperature, moisture, and soil shape trait variability and lineage success across habitats (Folk et al., 2020; Joswig et al., 2022). Biotic interactions—ranging from mutualism to antagonism—impose additional selective pressures and structure ecological networks, influencing evolutionary trajectories (Thompson, 1994; Herrera & Pellmyr, 2002; Van der Niet & Johnson, 2012; Armbruster, 2017). These pressures are especially evident in reproductive traits, which are shaped by both mutualistic and competitive interactions. Together, physical and biological environments scaffold repeated innovations in plant form and function—from drought tolerance to pollination specialization (Ricklefs & Relyea, 2014; Folk et al., 2020).

For example, specific root and leaf traits may be selected under drought or cold stress, as shown by experimental changes in root architecture and leaf physiology (Nehe et al., 2021; Raza et al., 2024; Li et al., 2025), while floral traits like flowering time, form, and nectar output can shift in response to abiotic stressors such as drought and heat, even in the absence of pollinators (García et al., 2023; Alquichire-Rojas et al., 2024). Abiotic constraints also shape the evolution of investment in floral size and reproductive effort (Charnov, 1982; García et al., 2023), and so do biotic interactions, especially with pollinators. Co-evolution drives specialized floral morphology and function that enhance pollen transfer (Van der Kooi & Ollerton, 2020; Trunschke et al., 2021). While pollinators seek rewards (Wright & Schiestl, 2009; Parachnowitsch et al., 2018), flowers evolve signals—visual, chemical, or deceptive—that influence pollinator behavior (Spaethe et al., 2001; Rakosy et al., 2012; Schiestl & Johnson, 2013; Eisen et al., 2023). Interestingly, selection on floral traits by abiotic constraints (such as water availability) appears to be as strong as selection exerted by pollinators, and both are likely to interact, antagonistically or synergistically (Caruso et al., 2019; García et al., 2023).

Together, these abiotic and biotic pressures illustrate how ecological context continuously shapes the evolution of reproductive traits—often setting the stage for trait divergence, RI, and potentially, speciation.

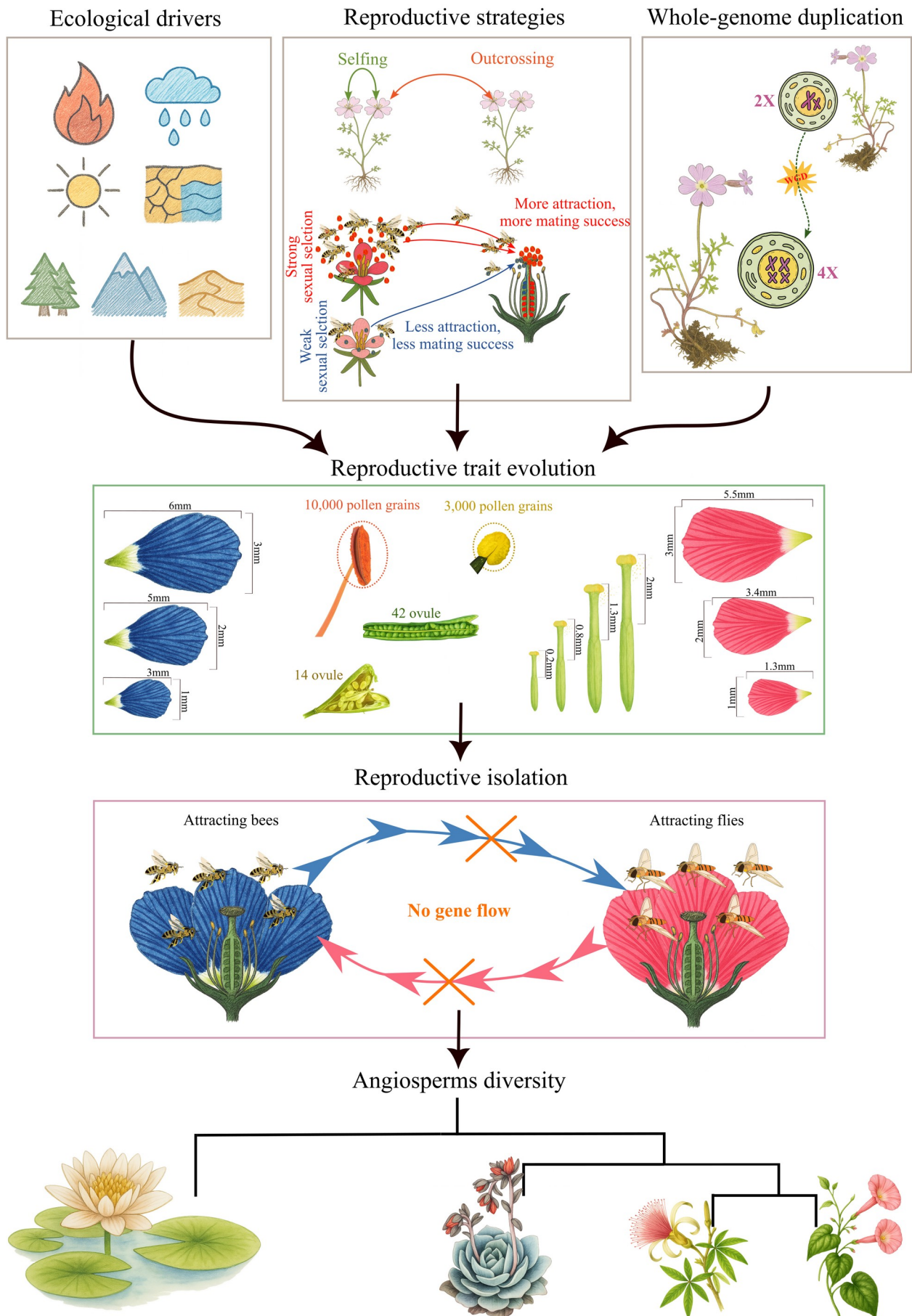


Figure 1. Unraveling “Darwin’s abominable mystery”: from evolutionary drivers to trait divergence, RI, and the flourishing of angiosperm diversity

1.2.2 Resource-dependent sexual investment in plants

As discussed in the previous chapter, ecological factors can shape reproductive trait evolution. In this section, I will focus in particular on how abiotic and biotic resources can impact the evolution of sexual investment. In flowering plants, sexual reproduction involves dividing resources between male and female functions (Charnov, 1982; Delph, 1999; Barrett & Hough, 2013). Ovules demand more maternal input than pollen, though total investment may be similar per flower due to high pollen grain numbers (Smith & Evenson, 1978; Gillet & Gregorius, 2020). This asymmetry favors conservative female strategies and opportunistic male ones in resource-poor versus rich environments (Lloyd & Bawa, 1984; Obeso, 2002; Harder & Johnson, 2025). The classic trade-off between offspring number and quality—more ovules vs. per-seed investment—shapes reproductive strategies (Baker, 1972; Greenway & Harder, 2007). Both modeling and experiments suggest that ovule availability, more than pollen, often limits reproductive success (Gillet & Gregorius, 2020; Harder & Johnson, 2025).

Abiotic stressors such as drought, temperature, and nutrient limitation strongly influence sexual investment. Although pollen is cheaper to produce than ovules, its output can decline under stress—for instance, *Trifolium willdenovii* reduces pollen production during drought (Wilson Rankin et al., 2020). Due to low pollen-transfer efficiency, excessive pollen often yields limited gains (Gong & Huang, 2014; Minnaar et al., 2019), and selection may favor reduced male allocation in resource-poor conditions (Obeso, 2002; Edward & Chapman, 2011). However, recent models suggest that the evolution of pollen output is not influenced by resource availability (Bochynek & Burd, 2024). Ovule production, which depends on maternal resource availability, is hypothesized to be sensitive to abiotic stress—potentially declining under poor conditions and increasing when resources are abundant. In support of this expectation, ovule number in natural populations of *Embothrium coccineum* (Proteaceae) was positively associated with rainfall across a strong precipitation gradient (Chalcoff & Aizen, 2016). In parallel, biotic resources like pollinator availability may also shape investment strategies. Pollen production is often negatively associated with pollinator availability (Harder & Johnson, 2023), suggesting that low visitation rates may favor increased pollen output as a compensation mechanism. Nevertheless, this hypothesis is contradicted by theoretical expectations and a recent modeling study showing that the evolution of pollen production is unimpacted by pollen transfer efficiency (Charnov, 1982; Bochynek & Burd, 2024), leaving the question open. Conversely, ovule production can increase under pollination uncertainty as a bet-hedging strategy (Friedman & Barrett, 2009, 2011). For example, alpine species, which experience irregular pollinator activity, produce more ovules (Arroyo et al., 2019).

These patterns reveal key differences: (i) pollen is a cheaper yet often inefficient investment; (ii) ovule production is costlier and more resource-dependent; and (iii) biotic and abiotic factors impose distinct selective pressures on each. Nevertheless, except for a few studies, it remains unclear to what extent and how distinctively biotic and abiotic resource limitation shapes pollen and ovule investment evolution across lineages.

1.2.3 Environmental selective pressures can drive RI and speciation in plants

Ecological differences—both abiotic and biotic—can drive populations along distinct evolutionary paths by shaping adaptive traits (Rundell & Price, 2009; Gorospe et al., 2025). Ecological RI can contribute to divergence between such populations, and can evolve via two main mechanisms:

- I. Divergent selection and local adaptation, which reduce hybrid fitness due to maladaptation to either environment—an extrinsic postzygotic barrier (Jarrett et al., 2025). In *Mimulus guttatus*, habitat-specific divergence between coastal and inland races persists despite gene flow, illustrating ecological RI (Lowry et al., 2008). Recent studies show that multi-locus interactions and epistasis accelerate the build-up of Dobzhansky–Muller incompatibilities during ecological divergence (Dobzhansky, 1937; Muller, 1942; Panhuis et al., 2001; Rundle & Nosil, 2005; Satokangas et al., 2020).

- II. Divergent biotic interactions, especially pollinator shifts, can reduce gene flow via pollinator-mediated isolation. This mechanism is well documented in *Mimulus* (Schemske & Bradshaw, 1999; Bradshaw & Schemske, 2003; Byers et al., 2014; Wenzell et al., 2025). Shifts in pollinator communities can generate ‘pollination ecotypes’ (Johnson, 2025), and meta-analyses confirm that divergent selection can rapidly lead to pre- and post-mating barriers (Jarrett et al., 2025). A recent test of the Grant–Stebbins model supports pollinator-driven floral divergence as a pathway to speciation (Moir et al., 2025).

These barriers can evolve in both sympatry and allopatry via ecologically driven selection (Sobel et al., 2010; Nosil, 2012; Jarrett et al., 2025). With accumulating genetic differences, such processes can lead to ecological speciation—where RI arises through adaptation to distinct environments (Rundle & Nosil, 2005; Schluter & Conte, 2009; Wang et al., 2020; Bock et al., 2023). Nevertheless, although evidence for ecology-driven RI in plants is mounting (Baack et al., 2015; Satokangas et al., 2020; Liang et al., 2023; Jarrett et al., 2025; Barbot et al., 2025), its macroevolutionary relevance remains uncertain. Do these barriers reliably restrict gene flow in nature? Do they significantly contribute to speciation at large scales? Addressing these questions requires integrating ecological, genetic, and phylogenetic data to bridge microevolutionary processes with plant biodiversity patterns.

1.3 Whole genome duplication and the evolution of traits and species in angiosperms

1.3.1 Polyploidization: a frequent and foundational phenomenon in plants

Polyploidy—the presence of more than two chromosome sets—is widespread in plants and has played a recurrent role in their evolutionary history. Polyploidization leave lasting genomic signatures, some of which align with major radiations or adaptive shifts (Jiao et al., 2011; Van de Peer et al., 2017; Edger et al., 2025). These events arise via unreduced gametes, producing either autopolyploids within species or allopolyploids through hybridization (Comai, 2005), though the boundary is often blurred due to genome restructuring and complex ancestry.

Despite its prevalence, polyploidization is not inherently beneficial. Experimental comparisons between neopolyploids and diploids frequently report reduced fertility, disrupted development, and lower overall fitness in newly formed polyploids (Ramsey & Schemske, 2002; Husband, 2000). While traits such as larger cell and organ size—the so-called gigas effect—often appear early, these differences tend to fade with time and are not consistently retained in established polyploids (Clo & Kolář, 2021). This challenges the notion of neopolyploids as inherently superior or more stress-tolerant than diploids, inconsistent with the ubiquity of polyploids. It also contradicts the fact that polyploids are more frequent in higher latitudes, where environmental conditions are less favorable for plant growth (Brochmann et al., 2004; Rice et al., 2019), raising an unresolved paradox.

Across deep evolutionary timescales, polyploidy contributes genomic novelty and reshapes lineage trajectories, yet its influence on diversification remains contested. Some phylogenetic studies suggest that neopolyploid lineages often face elevated extinction and reduced speciation rates (Mayrose et al., 2011), implying that few persist and radiate. However, this view has been challenged by more recent lineage-focused analyses. In the Brassicaceae, for example, polyploids show diversification rates similar to diploids and significantly influence clade-level richness, despite exhibiting higher turnover (Román-Palacios et al., 2020). These findings suggest that while polyploidization introduces early hurdles, its long-term impact on diversity may be substantial—especially under favorable ecological or lineage-specific contexts.

1.3.2 Polyploidization and the evolution of (reproductive) traits in plants

Polyploidization induces widespread genomic and regulatory restructuring, including gene silencing, chromosomal rearrangements, transposon activation, and epigenetic shifts (McCarthy et al., 2016; Xiao & Wang, 2025). These disruptions often cause shifts in morphology, physiology, and

development, but their evolutionary implications remain variable and context-dependent (Soltis et al., 2014).

One commonly reported outcome is the gigas effect—increased cell size leading to larger vegetative and floral organs (Müntzing, 1936). However, recent meta-analyses demonstrate that this effect is largely transient, present only in the earliest generations after genome doubling, and tends to disappear in established polyploids (Clo & Kolář, 2021). This challenges long-held assumptions that polyploidy consistently produces larger or more robust phenotypes. Likewise, changes in seed size or early growth advantages, though sometimes observed in neopolyploids, do not reliably confer long-term fitness benefits (Bretagnolle & Thompson, 1995).

Some traits do appear to diverge in certain taxa. In *Dianthus broteri*, for instance, ploidy level affects allocation to roots and xylem development (López-Jurado et al., 2025), and floral traits such as corolla size or scent may shift following polyploidization (Gross & Schiestl, 2015; McCarthy et al., 2016). Yet experimental work on neopolyploids suggests that such changes often do not translate into functional or ecological differentiation—they may be neutral, deleterious, or simply buffered by stabilizing selection (Ramsey & Schemske, 2002).

In crops like *Brassica napus*, homoeologous exchanges after polyploidization can affect traits tied to domestication (e.g., flowering time), but these outcomes are system-specific and shaped by strong human selection (Wang et al., 2024). Meanwhile, autopolyploids often show more subtle and less predictable trait shifts than allopolyploids (Clo & Kolář, 2021). While theoretical models suggest that polyploidy may facilitate trait evolution under novel environments, empirical evidence remains scattered and inconclusive (Hatakeyama & Ohbayashi, 2024). Overall, polyploidization induces widespread change—but whether these changes promote adaptive divergence is still an open question.

1.3.3 Polyploidization and its role in the evolution of RI, and speciation in plants

Polyploidy can facilitate the emergence of reproductive barriers and promote speciation, but its role remains debated. While ancient polyploidization are detectable across many angiosperm lineages (Jiao et al., 2011), most newly formed polyploids suffer from demographic disadvantages such as minority cytotype exclusion and reduced fertility (Fowler & Levin, 1984; Husband, 2000). Nonetheless, mixed-ploidy populations are common, aided by unreduced gamete formation and occasional triploid fertility (Ramsey & Schemske, 1998; Mason & Pires, 2015).

RI may arise via both pre- and postzygotic mechanisms, including the triploid block (Köhler et al., 2010) and shifts in floral traits. Polyploidization can influence floral morphology, color, or scent—traits that often mediate pollinator identity and behavior. These changes may reduce intercytotype gene flow by promoting assortative pollination. For instance, in *Chamerion angustifolium*, diploids and autotetraploids in mixed-ploidy populations received ~70–75% of pollen from donors of the same ploidy, reflecting strong pollinator-mediated prezygotic isolation (Husband & Sabara, 2004; Kennedy et al., 2006). However, much of the existing evidence comes from established polyploids, where floral divergence may reflect longer-term selection or ecological filtering rather than genome duplication per se. Synthetic neopolyploids—crucial for isolating the direct effects of polyploidization—rarely show consistent changes in pollinator behavior, even when floral traits are altered (Ramsey & Schemske, 2002). Thus, whether polyploidization alone generates functional prezygotic isolation remains unresolved.

At macroevolutionary scales, polyploidy has long been viewed as a potential engine of plant diversification (Soltis et al., 2009; Madlung, 2013). However, phylogenetic evidence is mixed. One influential study found that polyploids tend to have lower diversification rates and higher extinction risks than diploids—supporting the “dead end” hypothesis (Mayrose et al., 2011). Others, using

denser sampling or broader trait reconstructions, suggest that polyploidy can contribute to lineage expansion under certain conditions (Román-Palacios et al., 2022). Yet both approaches face methodological challenges, including incomplete ploidy data, inability to distinguish auto- vs. allopolyploids, and reliance on chromosomal proxies risks misclassifying mesopolyploids. Overall, the link between polyploidization and diversification appears clade- and context-dependent, and generalizations remain premature.

Despite its prevalence, polyploidization does not uniformly drive RI or species formation. Many floral and ecological traits evolve after, rather than because of, WGD. Whether polyploidy consistently initiates these processes—or simply provides genomic conditions that occasionally enable them—remains an open question.

1.4 Reproductive strategies: a masterpiece of angiosperms' traits and species evolution

1.4.1 Sexual selection: from elegant theory to evolutionary ambiguity

Sexual selection, first proposed by Charles Darwin in 1871, is a core evolutionary mechanism that acts through non-random differences in reproductive success driven by competition for mates (Kokko & Jennions, 2008). Unlike natural selection, it emphasizes traits that may be costly for survival but increase mating success—thereby enhancing reproductive fitness (Darwin, 1871).

Darwin distinguished two modes of sexual selection: intrasexual selection, involving competition within a sex (typically males) for access to mates, and intersexual selection, where one sex (usually females) prefers specific traits in mates (Darwin, 1871; Panhuis et al., 2001). These processes often lead to the evolution of striking traits—ornaments, displays, or mating behaviors—that improve competitive success or attractiveness. In plants, similar dynamics have been theorized in pollen–pistil interactions, where sexual selection may shape traits involved in male–male competition and female choice at the gametic level (Skogsmyr & Lankinen, 2015). Moreover, ecological and demographic factors such as spatial structure can influence the trajectory of sexual selection, leading to sex-specific physiological adaptations. Traits favored by sexual selection tend to increase in frequency, making this a potent force in shaping morphology, behavior, and even species divergence (Iglesias-Carrasco et al., 2025).

While the impact of sexual selection on trait evolution is well-established in animals, its role in RI and speciation remains under debate. Some comparative studies suggest a link between stronger sexual selection and higher species richness, supporting the idea that it may drive diversification (Janicke et al., 2018). Yet, other empirical and theoretical research reports mixed outcomes and highlights the difficulty of identifying clear causal mechanisms (Panhuis et al., 2001; Ritchie, 2007; Kraaijeveld et al., 2011; Mendelson & Safran, 2021). Parsing the effects of sexual selection from ecological or geographic drivers continues to pose a major challenge. In this context, flowering plants offer a promising but underexplored system: their diverse floral traits, reproductive modes, and well-documented pollination biology make them ideal for testing whether sexual selection contributes to reproductive barriers and macroevolutionary diversification.

1.4.2 Sexual selection leads to the evolution of traits in plants

Though once met with skepticism, the concept of sexual selection in plants—especially hermaphrodites—is now supported by growing evidence, including parallels with hermaphroditic animals (Moore & Pannell, 2011). Sexual selection in angiosperms operates at both pre- and post-mating stages. In plants, pollination functions analogously to mating in animals, and traits enhancing pollen delivery to conspecific stigmas act like male–male competition, boosting male reproductive success (Willson, 1979; Delph & Ashman, 2006). Indeed, floral traits such as petal size, biomass, and color are believed to reflect sexual selection on male function more than the female function. Darwin noted that male or hermaphroditic flowers typically have more elaborate perianths, aligning with

Bateman's principle, which predicts stronger selection on male traits due to the fact that male reproductive success is more limited by the number of mating events, while the female reproductive success is rather resource-limited (Willson, 1994; Collet et al., 2014). The '*fleurs-du-male*' hypothesis (Queller, 1997) similarly posits that floral traits evolve largely under male function selection. Supporting studies show that larger flowers often invest more in male structures and pollinator attraction (Paterno et al., 2020), and male reproductive success is more closely tied to trait variation than female success in species like *Sagittaria latifolia* (Kwok & Dorken, 2022) and *S. dioica* (Barbot et al., 2023). Even pollen morphology can be shaped by male–male competition for pollinator access (Cocucci et al., 2014). This excludes the case of pollen limitation, where mating may also be limiting for the female reproductive success. It is important to note that these processes also happen in hermaphroditic species, as male and female functions evolve independently to a certain extent.

After pollination, competition happens between pollen grains. When pollen exceeds ovule number, traits such as pollen viability, size, germination speed, and tube growth rate (PTGR) determine success (Arnold, 1994; Willson, 1994; Moore & Pannell, 2011). Faster germination and tube elongation can offer a competitive advantage, as shown in species like *Hibiscus moscheutos* and *Viola tricolor* (Snow & Spira, 1991a,b, 1996; Skogsmyr & Lankinen, 1999). Female choice may also shape outcomes via traits like style length, stigmatic receptivity, and biochemical signaling (Mazer et al., 2016, 2018). Longer styles may allow greater discrimination among pollen grains (Mulcahy & Mulcahy, 1975; Ramesha et al., 2011), yet findings are mixed. A meta-analysis found more non-random mating in species with short styles (Ruane, 2009), and modeling suggests germination speed—not tube growth—matters more in short-styled species (Travers & Shea, 2001).

In summary, sexual selection in plants shapes floral and reproductive traits through male–male competition and female choice. Yet key questions remain: How broadly does it explain trait variation across angiosperms? And how much of floral diversity reflects sexual selection versus other evolutionary forces? Finally, how much of this floral diversity is responsible for RI?

1.4.3 Sexual selection and the evolution of RI in plants

Sexual selection favors non-random mating, leading to unequal reproductive success and exaggerated traits that enhance fertilization. But does such trait variation drive speciation in angiosperms? One key pathway is via RI, essential under the BSC (see Chapter 1.1). Divergence in sexually selected traits may lead to RI either before mating—through pollinator shifts—or after mating, via barriers like failed pollen germination, disrupted tube growth, or altered pollen flow.

Pre-mating barriers arise when floral traits such as display size, nectar output, color, or morphology—often shaped by male–male competition—diverge across populations, altering pollinator behavior and reducing inter-lineage pollen flow. Under conditions of limited pollinator availability, intensified competition may drive pollinator shifts or specialization (Ratnieks & Balfour, 2021). These floral shifts can lead to assortative pollination and reduced gene flow, yet direct evidence linking such divergence via sexual selection to speciation remains scarce. Post-mating RI may occur when pollen is rejected by the pistil of another lineage, blocking fertilization at various stages, from germination and tube growth to guidance to the ovule. This may result from the coevolution of male traits and female preferences, in line with Fisherian runaway sexual selection (Fisher, 1930). Independent directional selection on traits like PTGR or style length may drive populations toward different optima, ultimately producing gametic incompatibilities. For example, reduced hybrid success in *Iris fulva* × *I. hexagona* due to slower PTGR supports such a mechanism (Arnold et al., 1993), though findings are variable. Asymmetric sexual selection, where its intensity is higher in one lineage compared to another, may also lead to unilateral incompatibility, where mismatched evolution of pollen competitiveness and female discrimination reinforces isolation. Whether these pre- and post-mating barriers commonly evolve via sexual selection or other confounding evolutionary factors, however, remains unresolved.

These mechanisms suggest that sexual selection may contribute to both pre- and post-mating RI in plants. Divergence in traits driven by male competition or mismatches in gametophytic traits from coevolution can reduce gene flow between lineages. However, direct empirical evidence linking sexually selected traits to RI remains limited. It is still unclear how often sexual selection alone produces isolating barriers or how widely it contributes to angiosperm speciation.

1.4.4 Sexual selection and speciation in plants

As explained in Chapters 1.4.2 and 1.4.3, sexual selection can drive trait divergence and RI. Yet, whether these sexually selected barriers translate into increased speciation rates across angiosperms remains largely unresolved and untested at macroevolutionary scales.

One strategy for testing the macroevolutionary consequences of sexual selection is to assess whether traits shaped by it—such as style length, P:O ratio, and floral size (e.g., petal length and width)—are associated with diversification across phylogenies. These direct, functionally grounded traits serve as quantitative proxies for sexual selection, offering a powerful alternative to traditional categorical variables like pollination mode or breeding system (Dodd, 1999; Goldberg et al., 2010). Some of these traits (P:O ratio, floral size) are linked to prepollination male–male competition, pollinator attraction, and male reproductive success (Paterno et al., 2020), consistent with the ‘*fleurs-du-male*’ hypothesis and findings that corolla width predicts mating success (Barbot et al., 2023). These traits can drive pollinator shifts and accelerate divergence (Cocucci et al., 2014; Paterno et al., 2020; Castañeda-Zárte et al., 2021). Other traits (style length) may reflect the male–female coevolution related to post-mating selection, which may lead to post-mating incompatibilities that reinforce RI (Arnold et al., 1993; Carney et al., 1996).

Incorporating these sexually selected traits into phylogenetic comparative frameworks provides a direct way to evaluate whether sexual selection contributes to speciation in angiosperms. This approach enables a broader test of sexual selection’s macroevolutionary role and its potential to explain large-scale patterns of floral trait evolution and lineage diversification.

1.4.5 Selfing, a common evolutionary transition affecting diversification in plants

While over 90% of plant species are hermaphroditic (Bawa, 1980), around 65% rely on outcrossing (Igic & Kohn, 2006). The transition from outcrossing to selfing is common in flowering plants and occurs when the benefits of reproductive assurance (Eckert et al., 2006) and selfing transmission (Fisher, 1941) outweigh inbreeding costs (Charlesworth, 2006). This shift often involves breakdown of the self-incompatibility (SI) system and emergence of self-compatibility (SC) (Takayama & Isogai, 2005), accompanied by a suite of traits termed the “selfing syndrome”—smaller flowers, reduced scent, and lower pollen output—thought to enhance autonomous reproduction (Sicard & Lenhard, 2011; Tsuchimatsu et al., 2020).

One key consequence of selfing is the relaxation of sexual selection (Gutiérrez-Valencia et al., 2022; İltaş et al., 2025). Efficient pollen transfer in selfers reduces male–male competition, weakening selection on floral display traits. Postpollination, competition among pollen grains also declines, as they originate from the same individual, limiting female choice. Genomic studies in *Arabis alpina* and *Arabidopsis lyrata* show reduced selection on male gametophyte genes and decreased pollen performance in selfing populations, suggesting that postpollination sexual selection is weakened. This relaxation may explain observed variation in sexual traits, complicating interpretations of floral trait evolution and its relationship to speciation (see Chapters 1.4.2–1.4.4). At the same time, selfing-associated changes—particularly the breakdown of SI and floral modifications—can contribute to RI by disrupting pollen transfer and reducing gene flow between selfing and outcrossing populations. For instance, in sympatric *Centaureum* species, mating system differences led to strong asymmetric RI due to unidirectional pollen transfer driven by floral display divergence (Brys et al., 2014). Similar

barriers have been reported in other systems (Tsuchimatsu & Fujii, 2020; Broz & Bedinger, 2021), suggesting that prezygotic incompatibilities arising from mating system transitions may promote lineage divergence.

The impact of mating system transitions on macroevolutionary shifts in diversification remains to be investigated. Nevertheless, in Solanaceae, SC species show higher speciation and extinction rates than SI species, resulting in lower net diversification overall (Goldberg et al., 2010). Recent work, again based on SC vs SI and not selfing/outcrossing, shows that diversification is shaped not just by self-compatibility systems, but by their interaction with ploidy and lineage-specific factors (Zenil-Ferguson et al., 2019). While SC is commonly linked to increased self-fertilization, many SC species continue to outcross to varying degrees (Igic et al., 2008; Steinecke et al., 2022), and thus, SC/SI status cannot reliably indicate mating strategy. This disconnect complicates macroevolutionary inference based solely on self-compatibility systems, especially as selfing-related traits may evolve independently of compatibility breakdown. Moreover, transitions to selfing are often accompanied by life history changes—such as enhanced colonization (Pannell, 2016; Makowski et al., 2024) and population differentiation (Huang et al., 2019)—which may promote speciation through ecological mechanisms. Consequently, there is increasing recognition that broad SI/SC classifications are insufficient and that trait-based proxies capturing functional mating behavior are needed to understand the evolutionary consequences of mating system transitions.

1.5 An interplay of ecological and sexual selection

Ecological drivers can shape RI by modulating sexual selection, especially when environmental variation affects traits involved in mate recognition (Rundle & Nosil, 2005). Yet a central question remains: does ecology amplify or constrain the speciation potential of sexual selection? Studies in animals increasingly suggest that sexual selection rarely acts independently. Instead, it often interacts with local adaptation—sometimes reinforcing, other times limiting divergence (Servedio & Boughman, 2017; Mendelson & Safran, 2021). A systematic review by Scordato et al. (2014) found that internal interactions—where natural and sexual selection act jointly on sexual traits—are more common than external interactions. This reinforces the idea that ecological context frequently shapes divergence in mating signals and preferences. As Safran et al. (2013) emphasize, habitat variation can affect the evolution of mating traits by altering either sexual signals or mate preferences.

As discussed in Sections 1.2 and 1.4, floral traits such as size, color, nectar output, and display architecture evolve under the combined influence of ecological and sexual selection. Abiotic stressors—such as drought and heat—can exert strong directional or stabilizing selection on floral traits by limiting the resources available for reproductive investment, independently reducing nectar volume, petal size, or floral longevity (García et al., 2023; Alquichire-Rojas et al., 2024). At the same time, these traits are subject to pollinator-mediated sexual selection, particularly through male–male competition for pollinator attention, which favors increased display and reward traits that enhance pollen export and mating success (Paterno et al., 2020; Barbot et al., 2023; see Section 1.4). When environmental conditions vary across populations—such as spatial differences in water availability—these joint selection pressures may drive divergence in floral phenotypes. In water-limited habitats, selection may favor reduced, resource-efficient flowers, while in mesic or pollinator-dense environments, sexual selection may maintain or enhance elaboration. If these divergent trait optima influence pollinator preferences or behavior, they can reduce gene flow and generate prezygotic isolation. Over evolutionary time, the interaction between abiotic ecological selection and sexual selection on floral traits may thus facilitate floral divergence, RI, and ultimately, speciation.

Yet, this interaction may also produce evolutionary conflict. Nectar production, for example, is shaped by both abiotic constraints and pollinator-mediated sexual selection, but these forces may act in opposing directions. In water-limited environments, selection often favors reduced nectar output to conserve resources—especially under drought stress, which imposes strong constraints on

reproductive investment (García et al., 2023; Alquichire-Rojas et al., 2024; see Section 1.2). At the same time, pollinators preferentially visit flowers offering greater rewards, exerting directional sexual selection for higher nectar volume or sugar content to increase male fitness via pollen export (Wright & Schiestl, 2009; Ratnieks & Balfour, 2021; see Section 1.4). When these selective regimes act in opposition—one favoring frugality, the other elaboration—floral traits may evolve along different trajectories across populations. In some cases, abiotic stress may override pollinator-driven selection, leading to smaller, less rewarding flowers; in others, strong pollinator dependence may sustain or amplify display traits despite environmental cost. Whether such conflicts constrain or facilitate speciation depends on the relative strength and direction of these selective regimes. Understanding how these tensions scale up to influence diversification across plant lineages remains a critical challenge for future research.

Chapter 2: Objectives of the thesis

The overarching aim of my PhD thesis is to evaluate how ecological drivers, WGD, and reproductive strategies, including mating systems and sexual selection, contribute to the evolution of reproductive traits, the emergence of RI, and ultimately, patterns of speciation in angiosperms. I specifically focus on bridging microevolutionary mechanisms (e.g., trait divergence, sexual selection, and RI) with macroevolutionary outcomes (e.g., speciation and diversification). To address this, I use the Brassicaceae as a model system—both at the species level (e.g., *A. arenosa*) and across the family—due to its ecological breadth, variation in ploidy and mating systems, and extensive trait data. My approach integrates experimental studies, field and trait measurements, and phylogenetic comparative methods.

To structure this investigation, I ask the following specific questions, grouped around the three major drivers of angiosperm reproductive trait evolution and diversification:

2.1 What is the role of environmental factors (abiotic and biotic) in shaping angiosperm diversity, and do they drive variation in reproductive traits, particularly pollen and ovule production?

2.2 What is the influence of WGD on the evolution of angiosperms?

2.2.1 Do WGDs drive the variation of morphological, mainly reproductive traits at the species-level?

2.2.2 Can these variations arise as reproductive barriers by increasing pollinator attraction to the neopolyploids?

2.2.3 Do these barriers lead to higher speciation in the lineages that have undergone duplications (e.g., *A. arenosa* polyploids)?

2.3 What is the influence of reproductive strategies (mating systems and sexual selection) on the angiosperm diversity?

2.3.1 Do mating systems and sexual selection explain the variation of reproductive traits?

2.3.2 Can these variations potentially arise as reproductive barriers?

2.3.3 Do these barriers lead to higher speciation in angiosperms? (e.g., Brassicaceae)

Chapter 3: Comments on material and methods

This chapter outlines the methodological framework used to address the central objectives of the thesis (Chapter 2). I employed both experimental and comparative approaches to evaluate how ecological drivers, WGD, and reproductive strategies contribute to the evolution of reproductive traits, the emergence of RI, and ultimately, patterns of speciation in angiosperms. I conducted field and greenhouse experiments to examine how environmental factors and polyploidization affect reproductive traits and contribute to isolation processes. To enable robust macroevolutionary inference, I reconstructed a comprehensive species-level Brassicaceae phylogeny incorporating approximately 25% of known species, which serves as the backbone for analyses testing whether

rapid floral trait evolution is associated with increased speciation rates and how WGD and mating systems influence diversification using trait-dependent models.

3.1 Model system

The Brassicaceae family served as my model system across all case studies, spanning species to clade levels. Brassicaceae is among the 15 most diverse angiosperm families, comprising over 4000 species in approximately 349 genera, grouped into 58 tribes (Hendriks et al., 2023; German et al., 2023). The family originated between the mid-Eocene (38.8 mya) and late Oligocene (24.5 mya) (Hendriks et al., 2023).

Several features make Brassicaceae a relevant model system for the questions of this thesis. First, the family has a nearly worldwide distribution—predominantly in the Northern Hemisphere and to a lesser extent in the Southern Hemisphere—with no presence in Antarctica. It spans a wide range of habitats, including terrestrial, montane, aquatic, freshwater, wetland, rocky, arid, coastal, desert, and temperate environments (Esmailbegi et al., 2018). Species richness is especially high in the Irano-Turanian Region (~900 species across 150 genera) and the Mediterranean Basin (~630 species in 113 genera) (Al-Shehbaz, 2002; Lysak & Koch, 2010). This extensive ecological and geographical breadth makes the family particularly suitable for investigating how environmental pressures shape the evolution of reproductive traits such as pollen and ovule production (CS3).

Brassicaceae has long served as a model for investigating the evolutionary significance of polyploidy in flowering plants (Walden et al., 2020; Mabry et al., 2020). Genomic studies of *A. thaliana* have revealed multiple ancient WGD events, likely shared by all core lineages except the early-diverging Aethionemeae tribe. One major duplication, dated to ~40–24 Mya, coincides with the radiation of core Brassicaceae and may explain the stark contrast in species richness (~3,700 species vs. a smaller Aethionemeae clade; Franzke et al., 2011). Additional lineage-specific WGDs, such as in *Brassica* and *Pachycladon*, have been linked to local adaptation and radiations (Lagercrantz & Lydiate, 1996; Joly et al., 2009), and recent analyses reveal widespread signatures of historical polyploidy, suggesting cryptic ploidy is more common than previously thought (Hendriks et al., 2023). To evaluate the macroevolutionary consequences of polyploidy with greater precision, I leveraged a newly curated dataset informed by Brassibase v2 expertise, allowing us to distinguish mesoneopolyploids and neopolyploids from mesopolyploids and diploids and to correct for common classification biases (Román-Palacios et al., 2020) (CS5). Complementing these large-scale analyses, I conducted experimental work on *A. arenosa* (CS2), a SC species with co-occurring diploid and autotetraploid cytotypes in southeastern Europe (Morgan et al., 2020). The tetraploid lineage originated within the last ~30,000 generations (Arnold et al., 2015), and crosses between cytotypes exhibit a strong triploid block, indicating prezygotic RI (Morgan et al., 2021). Together, these macroevolutionary and experimental approaches provide complementary insight into how polyploidization may influence speciation dynamics in Brassicaceae.

Thirdly, repeated shifts in mating systems—particularly from outcrossing to selfing—have played a central role in generating novel morphological and ecological traits in Brassicaceae (Igic et al., 2008; Sicard & Lenhard, 2011), potentially contributing to its early radiation into diverse ecological niches (Goldberg et al., 2010). These transitions have occurred independently across multiple lineages (Igic et al., 2008) and are frequently associated with range expansion, habitat disturbance, and polyploidy (Pannell, 2016; Makowski et al., 2024). The wide spectrum of mating strategies in the family—from strictly outcrossing to fully selfing taxa—highlights its reproductive flexibility and its potential role in driving both speciation and adaptation. Importantly, among outcrossing Brassicaceae, there exists a surprising and underappreciated diversity in floral morphology despite the family’s overall conserved floral ground plan. Detailed analyses have revealed extensive variation in traits such as petal size and shape, floral symmetry, organ number and elaboration, and inflorescence structure (Nikolov, 2019). This hidden floral disparity likely reflects divergent selection pressures from different pollinator

guilds and ecological contexts, even among species with superficially similar flowers. As such, it provides a rich landscape for investigating how sexual selection—through mechanisms such as pollinator-mediated male–male competition or female choice—acts on reproductive traits. This morphological and reproductive diversity, especially within the outcrossing lineages, offers critical variation in the targets and strength of sexual selection. Consequently, Brassicaceae represents an ideal model system for testing how mating system transitions and sexually selected floral traits interact to shape speciation dynamics and trait evolution across a major plant family (CS5).

Together, the Brassicaceae family’s global distribution, ecological diversity, mating system shifts, ploidy variation, and species richness asymmetry provide an exceptional framework for investigating how key evolutionary drivers shape diversification and trait evolution in angiosperms. Its economic and scientific importance has also led to intensive phylogenetic work (Walden et al., 2020; Lysak & Koch, 2010; Nikolov et al., 2019; Hendriks et al., 2023), which provided the backbone for the comprehensive Brassicaceae phylogeny reconstructed and used in this thesis (See Chapter 3.2).

3.2 Phylogenetic reconstruction

Despite several efforts to resolve the Brassicaceae phylogeny (Lysak & Koch, 2010; Nikolov et al., 2019; Walden et al., 2020; Hendriks et al., 2023), existing trees typically include fewer than 350 species, representing only a small subset of the family’s diversity. However, the comparative framework required for my analyses (CS3 and CS5) necessitated a more inclusive and taxonomically balanced phylogeny. Specifically, I aimed to cover at least 25% of Brassicaceae species, with increased representation from the larger tribes and inclusion of all species for which trait data were available.

To achieve this, I reconstructed a new species-level phylogeny using concatenated nuclear (ITS1, 5.8S, ITS2) and plastid (trnL/F and complete plastomes) sequence data. This approach has been shown to provide strong resolution at the tribal level ($\approx 75\%$) in previous studies, though usually with limited taxon sampling (~ 300 species) (e.g., Walden et al., 2020). I replicated this framework at a larger scale, combining model selection and maximum likelihood inference with a two-step molecular dating approach. First, I applied penalized-likelihood dating with treePL to generate an initial time-calibrated phylogeny, which served as the starting tree for subsequent analyses and reduced the need for computationally intensive time-bounded Bayesian inference. Second, I conducted full Bayesian molecular dating using BEAST2 under an optimized relaxed clock model, incorporating both fossil and secondary calibration points. This strategy allowed us to balance computational tractability with temporal precision: treePL enabled rapid estimation of divergence times across a large tree, while BEAST provided a statistically robust framework that accounts for rate heterogeneity and uncertainty. This combined approach ensured a chronologically well-supported phylogeny, essential for downstream analyses of trait evolution and diversification.

I also addressed a known challenge in Brassicaceae systematics: topological incongruence between plastome- and nuclear-based phylogenies, largely driven by frequent hybridization events (e.g., Hendriks et al., 2023). To mitigate this, I constructed an alternative “constrained tree” that incorporated genus-level relationships from the most recent phylogenomic dataset (Hendriks et al., 2023), based on target capture of 1081 gene-dataset. This constraint topology was enforced during tree inference and maintained during dating using a monophyletic genus constraint. This strategy allowed us to reconcile topological conflicts while preserving flexibility in species-level placement. Together, these two complementary phylogenies—a sequence-based and a topology-informed “constrained tree”—provided robust backbones for our comparative analyses, enabling both trait evolution and diversification studies within a unified and well-supported phylogenetic framework.

3.3 Comparative analysis; bridging micro- to macroevolution

“We are compelled at the present level of knowledge reluctantly to put a sign of equality between the mechanisms of macroevolution and microevolution, and, proceeding on this assumption...”
Theodosius Dobzhansky 1937

Macroevolution refers to evolutionary change at or above the species level, encompassing patterns and processes that shape the diversity of genera, families, and higher taxa. This contrasts with microevolution, which involves evolutionary dynamics within species or populations, such as genetic drift, selection, and local adaptation (Adams, 2021). The conceptual separation between these scales dates back to Filipchenko (1927), who proposed that macroevolution and microevolution were governed by fundamentally different mechanisms—a distinction also reflected in Darwin’s early insights on gradual change within populations versus the apparent discontinuities seen in higher taxonomic divergence (Darwin, 1859). A turning point in this debate came with Dobzhansky’s (1937) assertion that macroevolutionary change could be interpreted as the long-term outcome of microevolutionary processes. While this perspective remains central to the Modern Synthesis, the conceptual and empirical divide between the two levels of evolutionary analysis has persisted (Reznick & Ricklefs, 2009).

One productive strategy for bridging this divide is to examine traits or mechanisms that operate across both levels. Traits shaped by reproductive strategies within populations—such as P:O ratio, style length, or floral display—may also influence lineage divergence and diversification across clades (See chapter 1.4). Similarly, genomic mechanisms like polyploidization or transitions in mating systems that arise at the population level may leave lasting imprints on macroevolutionary patterns (See chapter 1.3 and 1.4.5). By focusing on integrative processes—specifically, traits and transitions that operate at both the microevolutionary (e.g., within-population trait variation, reproductive barriers) and macroevolutionary (e.g., lineage divergence, diversification) scales—I can better connect the mechanisms shaping short-term evolutionary change with long-term patterns of speciation and diversification.

Molecular phylogenies provide a powerful framework for studying macroevolution. They allow researchers to model the tempo and mode of diversification across lineages using birth–death models and their extensions (Nee et al., 1992; Morlon, 2014). These approaches have yielded major insights into speciation and extinction processes, but their reliability—especially in estimating extinction—is limited when using phylogenies of only extant taxa (Ricklefs, 2007; Rabosky, 2018). Because extinction leaves few detectable signatures in such trees, I focused my dissertation analyses specifically on speciation dynamics, which are better supported by extant-only phylogenetic data and are central to understanding lineage diversification.

To estimate lineage-specific speciation rates, I used the ClaDS2 model (Maliot & Morlon, 2022), which models lineage-specific diversification shifts with high resolution. This method is particularly suited to detecting fine-grained variation in diversification because it models speciation as a cladogenetic process, where each daughter lineage inherits a slightly modified rate from its parent, drawn from a lognormal distribution (Maliot & Morlon, 2022). Such an autocorrelated framework reflects empirical patterns of diversification heterogeneity and phylogenetic imbalance that are often underestimated by traditional birth–death models assuming constant rates or infrequent shifts (e.g., Rabosky, 2014). ClaDS2 is therefore well-suited for testing hypotheses about how reproductive traits associated with sexual selection influence diversification. Additionally, it accommodates incomplete taxon sampling through genus-level sampling fractions specified per species tip and remains computationally efficient even for large phylogenies, thanks to a data augmentation strategy (Maliot & Morlon, 2022; Afonso Silva et al., 2025).

I implemented ClaDS2 using the PANDA.jl package in Julia, with genus-level sampling fraction computed based on the latest Brassicaceae database (BrassiBase v2). This framework provided lineage-specific speciation rate estimates across Brassicaceae, which I used to investigate whether speciation is shaped by floral traits associated with sexual selection (e.g., P:O ratio). I tested whether speciation rates are associated with the magnitude of floral traits, the tempo at which they evolve, and the direction of their evolutionary change. This is based on the rationale that greater trait magnitude may reflect stronger selection, faster tempo may accelerate the accumulation of incompatibilities, and directional trends may indicate shifts in sexual selection intensity. These associations were evaluated using hierarchical models that incorporate clade-level structure and estimation uncertainty (Bürkner, 2017). Together, these analyses offered a comprehensive evaluation of the extent to which sexual selection proxies align with patterns of diversification across lineages (CS5).

To assess whether shifts in polyploidy and mating system influence speciation (CS5), I used Hidden State Speciation and Extinction (HiSSE; Beaulieu & O'Meara, 2016), modeling each trait separately. HiSSE extends earlier diversification models by incorporating hidden (unmeasured) states that may influence speciation independently or in combination with the observed trait, thereby reducing spurious associations. This framework allowed for a more robust evaluation of how reproductive strategies and genomic transitions contribute to macroevolutionary patterns, while accounting for background heterogeneity that might otherwise confound trait-dependent inferences.

Chapter 4: Synthesis and discussion of the results

4.1 Environmental factors exert a limited impact on the evolution of reproductive traits

The hypothesis that environmental factors shape reproductive trait evolution in flowering plants is grounded in long-standing theories of resource allocation and sexual investment (Charnov, 1982; Delph, 1999; Barrett & Hough, 2013). As outlined in Chapter 1.2.2, pollen and ovule production represent divergent strategies shaped by the costs and constraints of male and female function, respectively (Smith & Evenson, 1978; Gillet & Gregorius, 2020). Abiotic stressors have been proposed to constrain ovule production more tightly than pollen output (Chalcoff and Aizen, 2016), while biotic factors, particularly pollinator availability, may favor increased pollen production in response to mating limitation (Harder & Johnson, 2023). However, recent models and empirical studies challenge the generality of these patterns (Bochynek & Burd, 2024), raising questions about the consistency of ecological effects across species and environments.

To test these ideas in a macroevolutionary framework, I examined whether abiotic stress and pollinator diversity influence pollen and ovule production across 66 angiosperm species (CS3), grown under standardized greenhouse conditions. I quantified these traits and assembled habitat-level ecological data, including altitude, climate seasonality, climatic moisture index, soil nutrients, rooting conditions, and bee species richness (proxy for pollinator diversity). I modeled trait–environment relationships using GAMMs accounting for overdispersion and phylogenetic structure (see CS3).

Although theory predicts that ecological drivers shape reproductive traits, the CS3 showed only weak associations between environment and pollen or ovule number. Some factors—precipitation seasonality, climatic moisture index, altitude, and rooting conditions—were significant but explained little variance (3.3% and 3.8% for ovules and pollen, respectively). This suggests limited environmental effects, possibly context-dependent. In contrast, a broader analysis of 158 Brassicaceae species revealed strong macroevolutionary constraint. Ovule number showed a high phylogenetic signal (e.g., $\lambda = 0.95$), and pollen number showed a moderate to strong signal (e.g., $\lambda = 0.74$), consistent across topologies. Likelihood ratio tests confirmed that phylogenetic models outperformed those assuming independence, highlighting evolutionary history as the dominant structuring force.

These findings call for a reassessment of how ecological processes shape reproductive divergence and angiosperm diversification. As discussed in Chapter 1.2, ecological filtering, biotic interactions, and habitat differentiation are often seen as drivers of trait evolution, RI, and speciation. Yet our results show that key reproductive traits—pollen and ovule production—are evolutionarily conserved and minimally influenced by environment. This challenges the view that ecological selection broadly drives divergence in sexual allocation or prezygotic isolation. If such traits are phylogenetically entrenched, their role in RI or ecological speciation may be limited to specific cases. These results underscore the primacy of evolutionary history and caution against assuming ecological novelty equates to macroevolutionary innovation.

Despite offering robust insights, two limitations should be noted. First, bee species richness is a coarse proxy for pollinator diversity and omits factors like abundance, visitation rates, or behavior—possibly explaining the lack of association with male investment. More direct measures may uncover finer-scale effects. While the phylogenetic analysis spanned 158 species, trait–environment modeling used only 66 cultivated species. Though taxonomically diverse, this subset may not capture the full ecological or evolutionary breadth of Brassicaceae. Still, by removing environmental noise and applying phylogenetically informed models, our study offers a strong test of ecological hypotheses under controlled conditions.

While our results show that pollen and ovule production are conserved and weakly shaped by ecology, these traits represent just one facet of reproductive strategy. As noted in Chapter 1.2, other traits—floral morphology, phenology, nectar production, and pollinator-specific signals—may respond more strongly to environmental selection and influence RI and speciation. To clarify the role of ecology at the macroevolutionary scale, future studies must expand trait coverage and test environment–trait associations across a broader functional spectrum. Crucially, this research should link ecological divergence to speciation by combining phylogenetic analyses with data on isolation and diversification. The key challenge is to identify which ecologically responsive traits consistently mediate reproductive barriers and drive biodiversity at large evolutionary scales.

4.2 WGD drives trait shifts in neopolyploids but not diversification in established polyploids

The hypothesis that WGD facilitates the evolution of reproductive traits and promotes lineage diversification is a long-standing idea in plant evolutionary biology (Fowler & Levin, 1984; Soltis et al., 2014). As discussed in Chapter 1.3, polyploidization is expected to induce immediate genomic and phenotypic changes, particularly in reproductive structures, that may enhance pollinator attraction or cause prezygotic isolation (Ramsey & Schemske, 2002; Madlung, 2013). Such traits could, in theory, help newly formed polyploids overcome their initial rarity (minority cytotype exclusion) and initiate speciation. However, the consequences of WGD for trait divergence, RI, and diversification remain uncertain. Here, I evaluate these outcomes across three complementary case studies (CS2, CS4, CS5).

To assess whether WGD directly alters reproductive traits in newly formed polyploids, I conducted a meta-analysis of synthetic neopolyploids compared to diploid progenitors across 73 angiosperm genera (CS4). Using multilevel models on 15 reproductive traits, I found that floral and gametophytic traits often show immediate shifts following genome doubling. Tetraploids had significantly larger petals, ovaries, and pollen grains, and showed delayed flowering time relative to diploids. However, traits linked to reproductive success—such as seed number and germination—were reduced or unchanged, suggesting that the gigas effect does not necessarily confer functional or ecological advantage. These results indicate that while polyploidy causes widespread phenotypic change in synthetic neopolyploids, it does not reliably enhance early-stage fitness. This aligns with recent experimental and meta-analytic work showing that most neopolyploid lineages suffer reduced fertility and face strong stabilizing selection over time, and that WGD-associated phenotypic shifts tend to diminish as tetraploid lineages become established (Clo & Kolář, 2021).

To evaluate whether such trait changes promote pollinator-mediated RI, I tested whether synthetic polyploids of *A. arenosa* receive preferential pollinator visitation and exhibit assortative mating (CS2). Although floral traits were consistently enlarged in neotetraploids, they did not attract more or longer visits from insect pollinators. Instead, visitors frequently switched between diploids and neotetraploids, leading to high levels of interploidy pollen exchange. Most seeds from the neotetraploid mothers were triploid, indicating a lack of effective RI. These findings, together with few available studies testing the same question (Nghiem et al., 2011; Husband et al., 2016; Porturas & Segraves, 2020), challenge the assumption that floral divergence following WGD immediately fosters pre-mating barriers. Instead, polyploid-induced floral changes may even promote gene flow between cytotypes under natural conditions. While prezygotic isolation between diploid and established tetraploid *A. arenosa* populations has been reported (Morgan et al., 2021), our results suggest such barriers likely evolve post hoc and are not an inherent consequence of genome duplication.

Even if the role of WGD on RI might not be directly measurable, it may still impact diversification rates. To test this hypothesis, I fit trait-dependent diversification models across 550–688 Brassicaceae species (CS5). Using a curated ploidy dataset and refined classification criteria, I distinguished polyploids from mesopolyploids and diploids. The best-fitting model indicated that diploids diversify faster than polyploids, with polyploids often showing near-zero or negative rates. This supports earlier claims that polyploids may represent evolutionary dead ends (Mayrose et al., 2011). Same results were reported in Solanaceae where polyploids showed the lowest diversification in comparison to diploids (Zenil-Ferguson et al., 2019). Although some studies (e.g., Román-Palacios et al., 2020) reported richer polyploid clades, those estimates may be biased by unresolved ploidy origins.

Taken together, these findings offer a nuanced view of the macroevolutionary impact of WGD. Polyploidy clearly induces rapid and measurable changes in reproductive traits when examined in neopolyploids (CS4), yet these shifts do not establish functional pre-mating barriers (CS2), nor do they translate into increased diversification when tested in established polyploid lineages across Brassicaceae (CS5). This decoupling between phenotypic novelty and speciation potential contradicts the assumption that WGD directly accelerates lineage divergence. Instead, our results support an emerging view that WGD may only facilitate speciation under favorable genetic, ecological, or mating system contexts (Van de Peer et al., 2017; Román-Palacios et al., 2020).

While our synthesis challenges the idea that WGD consistently drives diversification, several caveats apply. First, CS5 did not distinguish auto- from allopolyploids, which may differ in genome stability or ecological range—a limitation shared by many studies due to scarce genomic data. This distinction is nontrivial, as differences in meiotic stability, ecological divergence, and establishment success between auto- and allopolyploids may lead to contrasting macroevolutionary outcomes (Soltis & Soltis, 2009). Second, although I used refined cytological criteria to classify ploidy and reduce the misidentification seen in studies like Román-Palacios et al. (2020), some mesopolyploids may still be misclassified. Third, all trait-dependent models used in this thesis assumed polyploidy is irreversible, consistent with most prior work (e.g., Mayrose et al., 2011), despite limited ability to detect diploidization. Yet evidence from Brassicaceae (Mandáková et al., 2010) shows that chromosomal reshuffling can obscure past WGD events. If diploidization happens undetected, our models may underestimate diversification in those lineages, highlighting the need for caution when interpreting results from models that assume fixed genomic states.

In summary, our results emphasize that the consequences of WGD are not fixed but contingent. While neopolyploidy induces widespread disruption, long-term evolutionary outcomes likely depend on lineage-specific buffering, selective environments, and interactions with other processes such as mating system evolution. Future work must explore how polyploidy interacts with other axes of reproductive variation, especially in natural contact zones and using phylogenetic models that account

for heterogeneous trait evolution across time and lineages. Only by linking trait evolution, reproductive barriers, and diversification in an integrated framework can I clarify the real contribution of WGD to angiosperm diversity.

4.3 Reproductive strategies shape reproductive trait variation and speciation patterns

Sexual selection is a well-established driver of trait divergence and RI in animals, yet its role in plant diversification remains underexplored (Moore & Pannell, 2011; Tonnabel et al., 2021). Floral traits linked to sexual selection—such as those mediating pollen competition or female choice—have been documented (Skogsmyr & Lankinen, 2002; Moore & Pannell, 2011; Mazer et al., 2016), but their macroevolutionary consequences for speciation remain unclear. Furthermore, the mating system itself may shape trait evolution and diversification while modulating the strength and efficacy of sexual selection, adding complexity to the picture. To address these questions, I combined conceptual synthesis (CS1) with macroevolutionary analyses (CS5).

In CS1, I reviewed how sexual selection operates in plants and developed a conceptual framework linking it to speciation. I showed that sexual selection can act at both pre- and post-mating stages, via male–male competition and female choice, and it may drive the evolution of traits like style, petal, and pollen performance, supported by theoretical and empirical evidence. These traits can establish prepollination barriers by influencing pollinator behavior and pollen competition, and postpollination barriers through style-mediated female choice, pollen tube growth dynamics, and compatibility mismatches. I further argued that mating system evolution alters selection regimes and modulates the strength of sexual selection, with selfing potentially reducing gene flow and relaxing trait selection. From this, I proposed testable predictions for how microevolutionary dynamics could scale up to macroevolutionary patterns of trait divergence and diversification—predictions I tested in CS5 using floral trait and mating system data across Brassicaceae.

To assess the role of the mating system in shaping reproductive traits and diversification, I analyzed over 750 Brassicaceae species (CS5). Mating system emerged as a central determinant of both how reproductive traits are expressed and how they evolve over macroevolutionary timescales. In selfing lineages, evolutionary trajectories consistently favored shorter styles, lower P:O ratios, and reduced petal size—a constellation of features long linked with the selfing syndrome (Sicard & Lenhard, 2011) and diminished sexual selection pressures (Gutiérrez-Valencia et al., 2022; İltaş et al., 2025). By contrast, outcrossing species tended to exhibit directional increases in petal size and P:O ratio over time. This disparity suggests that traits affecting pollinator attraction and pollen competition may be subject to stronger directional sexual selection in outcrossers, consistent with conceptual arguments developed in CS1. Surprisingly, style length did not follow this trend: despite theoretical expectations under a runaway selection model, we did not observe a pattern of directional increase in outcrossers. This may reflect functional constraints on the evolution of style length or indicate that it is not a primary target of male–female coevolution. Complementing these trait-based findings, diversification analyses showed that selfing lineages experienced reduced net diversification, a pattern largely attributable to higher extinction rates. These results align with prior work by Goldberg et al. (2010), who demonstrated that the loss of SI in Solanaceae—often corresponding with a shift to selfing—was associated with reduced diversification due to increased extinction. Zenil-Ferguson et al. (2019) similarly found that SC species in Solanaceae exhibited the lowest diversification rates, reinforcing the broader pattern that loss of outcrossing mechanisms often leads to evolutionary dead ends. Unlike both studies, my analyses are grounded in observable reproductive strategies (selfing vs outcrossing), offering a more ecologically and functionally integrative perspective. Overall, the evidence indicates that the mating system not only modulates the pace of reproductive trait evolution but also channels its direction, with long-term consequences for lineage diversification.

After removing the mating system as a confounding factor, I tested whether stronger sexual selection, as inferred from floral trait proxies, predicts speciation, under the assumption that stronger sexual

selection may lead to a faster emergence of reproductive barriers. Using BMLMs across Brassicaceae (CS5), I found that trait magnitudes alone were poor predictors of lineage-specific speciation rates. Only weak and often negative correlations were observed between trait values—such as P:O ratio and style length—and speciation rates. These findings suggest that higher trait investment, used here as a proxy for stronger sexual selection, does not necessarily translate into higher diversification. Instead, elevated trait values may reflect stabilizing selection or ecological trade-offs rather than drivers of reproductive isolation (Panhuis et al., 2001; Ritchie, 2007; Janicke et al., 2018; Mendelson & Safran, 2021). However, in our test case using pollen and ovule number, I found only weak associations with environmental predictors (CS3; see also Chapter 4.1), suggesting that these traits may not be the primary channels through which ecology inserts its impact. Moreover, strong sexual selection might in some cases reduce diversification by promoting convergence or exaggerated traits without reproductive divergence (Rodríguez et al., 2013; Servedio & Bürger, 2014), which may help explain the negative correlation between P:O ratio and speciation.

To further investigate the relationship between reproductive trait divergence and speciation, I moved beyond the magnitude of traits to focus on their evolutionary rates. Specifically, I asked whether faster changes in sexually selected floral traits are associated with higher speciation, particularly among outcrossers, where the scope of sexual selection may be more pronounced (CS5). I found that speciation was consistently associated with higher evolutionary rates across all traits examined, with particularly strong effects for P:O ratio, petal, and style length. This relationship was observed across both mating systems, but was especially evident in outcrossing lineages. These results suggest that speciation is more closely tied to the tempo of sexual trait divergence than to their static values. Divergence in traits such as petal dimensions, P:O ratio, and style length may create mismatches in pollinator preferences, pollen–pistil interactions, or floral cues, thereby reducing gene flow and fostering prezygotic isolation. Over time, the accumulation of such mismatches may establish persistent reproductive barriers that facilitate speciation (Schemske & Bradshaw, 1999; Ramsey et al., 2003; see also CS1). Further support for this interpretation comes from directional evolution analyses (CS5), which showed that outcrossers tend to evolve toward increased trait values across several reproductive traits, while selfing species exhibited downward trends. In correlative analyses, upward evolutionary trajectories were moderately associated with increased speciation rates, again primarily in outcrossers. These patterns align with the hypothesis that directional sexual selection in outcrossing species may enhance pre-mating or early post-mating isolation (Carney et al., 1996; Travers & Shea, 2001; Baek et al., 2015), whereas such effects appear attenuated in selfers.

While CS5 leveraged a comprehensive dataset and robust phylogenetic models, several caveats should be acknowledged. Trait imputation, though rigorously validated, introduces some uncertainty. Inferences about trait evolution and diversification also remain sensitive to phylogenetic structure, sampling completeness, and model assumptions. Although I identified strong effects of mating system and trait evolution on speciation, the role of ecological context remains only partially resolved. As detailed in Chapter 4.1, I explicitly tested environmental effects on a subset of reproductive traits—pollen and ovule number—and found these were conserved and only weakly shaped by abiotic and biotic variables. However, these traits capture just one dimension of reproductive investment. As outlined in Chapter 1.5, other traits shaped by ecological drivers, such as floral form, phenology, and pollinator cues, may interact more dynamically with sexual selection to influence divergence and speciation. Future work should expand the ecological scope of trait analyses and integrate both ecological and sexual selection pressures into trait-dependent diversification frameworks. Accounting for such environmental heterogeneity will be essential to determine whether ecological gradients facilitate or constrain sexual trait divergence and its contribution to plant speciation.

In summary, our findings provide the first empirical evidence that sexual selection leaves a macroevolutionary signal in plants by influencing speciation patterns across a major angiosperm family. Specifically, the evolutionary rate and direction of sexually selected traits—rather than their

magnitude—appear to predict speciation in flowering plants. Mating system not only shapes the expression of reproductive traits but also influences their evolutionary trajectories, with faster and directionally consistent trait shifts more tightly linked to speciation, particularly in outcrossers. Despite methodological uncertainties, I observed broadly consistent relationships across traits and models, lending credibility to the pattern. To understand how floral trait divergence contributes to reproductive isolation and species diversification, future work must consider how sexual selection interacts with ecological drivers—such as pollination systems, habitat heterogeneity, and floral phenology—which together shape the remarkable diversity of flowering plants.

4.4 Bridging micro- and macroevolution presents both opportunities and challenges

Linking trait divergence and RI within populations to broader patterns of speciation and diversification remains a central challenge in evolutionary biology and a key objective of efforts to connect micro- and macroevolutionary processes (Kearney et al., 2023; Rolland et al., 2023; Schluter, 2024). To address this challenge, this thesis investigated when and how reproductive trait evolution scales up to macroevolutionary consequences in angiosperms, as framed in Chapter 1.1 and explored through comparative and trait-based analyses in Chapter 3.3. Central to this inquiry is whether RI—often assumed to drive speciation—leads to long-term species accumulation, a question emphasized in recent calls for hypothesis-driven approaches to scaling evolutionary processes (Rolland et al., 2023).

As early evolutionary theorists such as Filipchenko (1927) and Dobzhansky (1937) debated, the relationship between micro- and macroevolution has long oscillated between assumptions of continuity and claims of mechanistic disjunction. Following Dobzhansky's premise, this thesis explores whether macroevolution reflects the accumulation of microevolutionary change—a question central to understanding “Darwin's abominable mystery” of angiosperm diversification introduced in Chapter 1.1. Yet the results show that not all trait divergence scales up in ways that reliably influence macroevolutionary outcomes.

Two drivers were directly tested for their ability to link micro- and macroevolutionary processes: WGD and reproductive strategies such as mating systems and sexual selection. Although WGD induces immediate and sometimes dramatic shifts in reproductive traits—as demonstrated in CS4 through meta-analyses of synthetic neopolyploids (e.g., increases in floral organ size)—these changes did not translate into detectable reproductive barriers in experimental settings (CS2). Complementing these findings, macroevolutionary analyses in CS5 showed that polyploid lineages do not diversify faster than diploids and, in some cases, exhibit lower net diversification—indicating that trait shifts induced by WGD do not necessarily translate into speciation. In contrast, reproductive strategies offered a clearer pathway for connecting scales. As developed in CS1, both sexual selection and mating system transitions shape reproductive trait evolution. Traits under sexual selection—such as style length, P:O ratio, and floral dimensions—mediate mate choice and compatibility, shaping both reproductive outcomes and lineage divergence by mediating mate choice, pollen competition, and compatibility. Mating system itself modulates the strength and direction of sexual selection and directly affects diversification: in CS5, I showed that selfing lineages tend to have lower diversification rates, while outcrossing lineages show stronger associations between trait divergence and speciation. These results highlight that both the mode of reproduction and, especially in outcrossers, the tempo of trait evolution contribute to speciation outcomes. Rapid trait divergence—more than trait magnitude—is what most strongly facilitates lineage splitting when bridging micro- and macroevolution. This aligns with the view that sexual selection acts as a dynamic driver of RI when divergence outpaces gene flow, echoing predictions from both population-genetic and comparative theory (Panhuis et al., 2001; Servedio et al., 2011).

These findings prompt a closer examination of when and why microevolutionary changes translate into macroevolutionary patterns. Bridging these scales requires moving beyond descriptive correlations and demands mechanistic understanding of how trait divergence affects gene flow, isolation, and diversification under ecological constraint (Rolland et al., 2023; Schluter, 2024). As

CS1 emphasized, such synthesis requires biologically meaningful traits, understood functions, and context-specific evolutionary dynamics grounded in ecology and genomics. Importantly, while trait divergence can facilitate isolation, and isolation can lead to diversification, neither outcome is guaranteed. The presence of reproductive barriers is not sufficient: species accumulation also depends on ecological opportunity, extinction risk, and lineage persistence. As Schluter (2024) notes, even rapid trait divergence may fail to produce speciation unless embedded in lineages with suitable ecological and demographic conditions (see also Rabosky & Matute, 2013).

This insight underscores a key limitation of trait-based approaches: divergence alone is not enough unless selection can act upon it under the right conditions. As discussed in Chapter 1.5, understanding when sexual selection promotes speciation requires evaluating how ecological constraints align—or conflict—with divergence. This interplay was partially explored in CS3, which tested whether abiotic and biotic conditions influence sexual investment traits such as pollen and ovule production. While central to reproductive allocation, these traits showed limited responsiveness to ecological gradients, suggesting limited ecological lability. Future research should explicitly examine how ecological selection interacts with mating-related or pollinator-mediated traits more directly involved in mate recognition or compatibility. Such studies would clarify when ecological variation amplifies or constrains the speciation potential of reproductive trait divergence, as anticipated in Chapter 1.5.

Altogether, this synthesis supports recent calls for a conceptual shift—from pattern cataloging to process inference—when bridging evolutionary scales (Rolland et al., 2023). Beyond documenting patterns of trait divergence, future research must test how divergence leads to isolation or speciation across ecological and phylogenetic contexts. Recommended strategies include simulation models that evaluate how sexual and ecological selection—independently or jointly—influence gene flow and speciation dynamics; comparative models that incorporate hidden trait or rate variation; and empirical studies linking trait function to reproductive outcomes. Traits like style length or P:O ratio will better connect scales when framed within clear hypotheses about their selective and ecological roles.

Conclusion

This thesis examined a central question in plant evolutionary biology: how ecological drivers, WGD, and reproductive strategies—specifically mating systems and sexual selection—shape reproductive trait evolution and drive speciation in angiosperms. It asked whether variation in key reproductive traits observed at microevolutionary scales translates into macroevolutionary diversification patterns. The goal was to identify which evolutionary drivers most consistently link trait divergence to speciation across evolutionary timescales. To address this, I combined a newly assembled phylogenetic backbone with comparative diversification analyses incorporating explicit mating system data.

Three innovations underpin this work. First, a time-calibrated phylogeny of over 1,000 Brassicaceae species provides an unprecedented framework for macroevolutionary analysis. Previous studies relied on limited or unresolved trees; this backbone supports robust trait-dependent diversification analyses. Second, this thesis presents the first macroevolutionary test of sexual selection in plants, bridging conceptual predictions with empirical data. Third, it integrates directly observed mating behavior (selfing vs. outcrossing) into diversification models—unlike prior work that relied on ambiguous SI/SC proxies. These advances create a foundation for linking reproductive strategies to diversification.

Findings revealed divergent roles for the three focal drivers. Abiotic and biotic environmental factors—such as climatic conditions and pollinator diversity—had minimal impact on pollen and ovule production, which were phylogenetically conserved. WGD caused immediate floral trait shifts in the neopolyploid but did not enhance pollinator attraction, generate RI, or boost diversification. This decoupling of novelty and macroevolutionary impact challenges the assumption that WGD directly

promotes speciation. In contrast, reproductive strategies—especially mating system transitions and sexual selection—had consistent impacts on both trait evolution and speciation. Selfing lineages showed reduced floral complexity, lower P:O ratios, and slower trait evolution. Outcrossing lineages exhibited more rapid, directional trait changes, closely tied to higher speciation rates. These patterns, seen across traits and methods, show that sexual selection leaves a detectable signature on plant macroevolution. Explicitly modeling mating systems proved critical for interpreting how trait divergence influences diversification.

Building on these findings, this thesis demonstrates that sexual selection—long theorized but rarely tested in plants—can influence speciation by shaping the tempo and direction of reproductive trait evolution. It is not the mere state of floral traits, but their evolutionary trajectories, that most strongly predict lineage diversification. While mating system transitions set the stage, it is the pace and directionality of trait change that appear to mediate speciation outcomes. These results establish reproductive strategies as central drivers of macroevolutionary dynamics in angiosperms. By linking trait evolution with diversification, this work offers a new empirical approach for addressing “Darwin’s abominable mystery” and sets the stage for future research that unites reproductive biology, ecology, and genome evolution in a fully integrative framework.

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Case study 1

Darwin's ‘mystery of mysteries’: the role of sexual selection in plant speciation

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ABSTRACT

Sexual selection is considered one of the key processes that contribute to the emergence of new species. While the connection between sexual selection and speciation has been supported by comparative studies, the mechanisms that mediate this connection remain unresolved, especially in plants. Similarly, it is not clear how speciation processes within plant populations translate into large-scale speciation dynamics. Here, we review the mechanisms through which sexual selection, pollination, and mate choice unfold and interact, and how they may ultimately produce reproductive isolation in plants. We also overview reproductive strategies that might influence sexual selection in plants and illustrate how functional traits might connect speciation at the population level (population differentiation, evolution of reproductive barriers; i.e. microevolution) with evolution above the species level (macroevolution). We also identify outstanding questions in the field, and suitable data and tools for their resolution. Altogether, this effort motivates further research focused on plants, which might potentially broaden our general understanding of speciation by sexual selection, a major concept in evolutionary biology.

Key words: sexual selection, plant speciation, reproductive isolation, reproductive strategies, macroevolution.

CONTENTS

I. Introduction	2
II. Sexual selection mechanisms and reproductive strategies in plants	2
(1) Premating sexual selection (pre-pollination stage)	3
(2) Postmating sexual selection (post-pollination stage)	4
(3) Reproductive strategies influencing sexual selection	5
(a) Pollination mode	5
(b) Sexual polymorphism versus hermaphroditism	5
(c) Selfing versus outcrossing	6
III. Sexual selection and reproductive barriers	7
(1) Pollinator shift in response to pre-pollination sexual selection	7
(2) Pollen rejection and post-pollination sexual selection	8
(3) Ecological speciation and speciation by sexual selection	10
IV. Sexual selection and plant speciation (macroevolutionary scale)	11
(1) Macroevolutionary research	11
(2) Bridging the micro- and macroevolution of speciation by sexual selection in plants	12
V. Conclusions	14
VI. Acknowledgements	14
VII. References	14

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I. INTRODUCTION

Why some evolutionary lineages are dramatically more species rich than others remains an enigma in biology. Sexual selection is often considered as one of the factors that promote speciation and diversification (Panhuis *et al.*, 2001; Kraaijeveld, Kraaijeveld-Smit & Maan, 2011; Maan & Seehausen, 2011), yet current theory and empirical results often suggest a surprisingly tenuous connection. For example, sexually dimorphic lineages tend to be more speciose (Futuyama & Kirkpatrick, 2017), and some theory exists on how sexual selection might lead to speciation (Fisher, 1930; Turelli, Barton & Coyne, 2001). But it has also been reported that sexual selection might be insufficient to govern the emergence of new species and sometimes may even hinder the speciation process (Servedio & Bürger, 2014; Servedio & Boughman, 2017; Irwin, 2020; Mendelson & Safran, 2021). Importantly, sexual selection can be hard to distinguish from other processes, mediated by environment and by geographic isolation, such as ecological or mutation-order speciation (Panhuis *et al.*, 2001; Kraaijeveld, Kraaijeveld-Smit & Maan, 2011; Maan & Seehausen, 2011). Thus, how to reconcile these mixed conclusions regarding the role of sexual selection on speciation remains an outstanding challenge, especially in plants. To address this, we need to study the processes that unfold across the continuum of evolutionary scales, from population divergence to the emergence of reproductive isolation and macroevolutionary diversification.

Sexual selection, defined as non-random variance in mating success (Kokko & Jennions, 2008), explains the evolution of sexual traits through their positive impact on reproductive success through a competitive advantage during mating, despite their often negative impact on survival (Darwin, 1871). It includes two dominant processes: the first is intrasexual (usually male–male) competition for mating; and the second is the choice of mates by the other sex, usually females (Darwin, 1871; Willson, 1979; Panhuis *et al.*, 2001; Rosenthal & Ryan, 2022). While sexual selection has been extensively studied in animals, its mechanisms and consequences remain largely understudied in plants.

Historically, the existence of sexual selection in plants has been controversial (Moore & Pannell, 2011), given that plant species are mostly hermaphrodites. Interestingly, sexual selection mechanisms have been demonstrated in hermaphroditic animal species (Beekman *et al.*, 2016) and, correspondingly, sexual selection is now generally recognised to operate in plants (Moore & Pannell, 2011). Yet, the consequences of sexual selection on plant evolution and plant speciation remain largely understudied.

In plants, whether sexual selection can lead to sexual trait divergence and the establishment of reproductive barriers has yet to be documented through empirical evidence. Divergent mate preferences in animals can act as a reproductive barrier (Boughman, 2001), and a similar process may take place in plants (Baek *et al.*, 2015; Li *et al.*, 2018). Nevertheless, the specificities of sexual selection in plants (the sexual mechanisms themselves, the diversity of

life-history traits, and the mating systems involved) may also contribute to the establishment of reproductive barriers that might be specific to plants.

Finally, an outstanding question in evolutionary biology is whether, and how, microevolutionary processes, such as sexual selection-driven reproductive barriers, translate into large-scale processes such as speciation and diversification (macroevolution; Jablonski, 2000). Darwin (1871) proposed that groups with more elaborate secondary sexual characters should be species rich, implying that the divergence of traits under sexual selection can lead to lineage diversification. While this hypothesis has been extensively tested in animals (Boul *et al.*, 2006; Mendelson, Imhoff & Venditti, 2007; Cally *et al.*, 2021), the role of sexual selection in large-scale speciation in plants remains surprisingly unaddressed (see Table 1).

In this review, we address the following questions: (i) what plant traits are under sexual selection, and how do reproductive strategies affect sexual selection; (ii) can sexual selection produce reproductive barriers and thus facilitate population divergence; and (iii) can this divergence lead to large-scale speciation? Some of these questions have been investigated in plants, but in isolation, and have yet to be evaluated together with a view to bridging the gaps between them. Our study details the connections between these research questions and aims to integrate previous work to formulate a more comprehensive perspective on plant speciation by sexual selection, ranging from micro- to macroscales.

II. SEXUAL SELECTION MECHANISMS AND REPRODUCTIVE STRATEGIES IN PLANTS

[A]though plants ‘neither sing nor dance’, they do exhibit female choice and male competition as intensely as animals do. (Ganeshaiah & Shaanker, 2001, p. 423).

Darwin (1871) proposed that sexual selection involves differences in reproductive success caused by competition for mating. By focusing on competition for mating, this concept therefore excludes processes related to natural selection acting on sexual traits. Darwin further suggested that sexual selection may involve either intrasexual (male–male) competition for mating, and/or intersexual selection (female mate choice). Female choice has been proposed to be costly and yet advantageous, as this cost may be compensated by an improved general fitness of the progeny (^{good genes} hypothesis; Byers & Waits, 2006) or by an improved reproductive success of male progeny due to the heritability of male competition abilities, hence indirectly increasing female fitness (*‘sexy son’* hypothesis; Weatherhead & Robertson, 1979). A link between male–male competition and female choice was further proposed under the concept of Fisherian runaway selection, which assumes that male traits evolve under recurrent directional selection by females; in other words, representing a case of coevolution between male traits and female preference (Fisher, 1930).

Table 1. Summary of current knowledge regarding the role of sexual selection in plant speciation. PTGR, pollen tube growth rate; SC, self-compatible; SI, self-incompatible.

Process	State of knowledge	Main evidence
Sexual selection mechanisms	Moderate amount of evidence	Floral and pollen traits evolve in response to male–male competition for pollinator attraction (Cocucci <i>et al.</i> , 2014; Paterno <i>et al.</i> , 2020) PTGR and pollen germination are associated with siring success (Snow & Spira, 1991a,b, 1996; Skogsmyr & Lankinen, 1999) Stressing the maternal plant increases random mating (Marshall & Diggle, 2001; Shaner & Marshall, 2003)
Reproductive strategies influencing sexual selection	Some evidence	Higher pollen performance in outcrossing <i>versus</i> selfing <i>Clarkia</i> species (Mazer <i>et al.</i> , 2018) Relaxation of sexual selection in selfing species (Gutiérrez-Valencia <i>et al.</i> , 2022)
Reproductive barriers promoted by sexual selection	Some evidence	Bilateral mating incompatibilities (Arnold <i>et al.</i> , 1993; Hodges & Arnold, 1995) Unilateral SI × SC incompatibility (Covey <i>et al.</i> , 2010; Back <i>et al.</i> , 2015) Negative correlation between style length and interspecific seed set in <i>Silene latifolia</i> (Nista <i>et al.</i> , 2015)
Sexual selection and local adaptation	No data	Suggestion to address the knowledge gap: testing for pleiotropic effects of sexual selection at (i) the trait level (e.g. artificial selection on pollen performance and measure its impact on adaptive traits such as drought response), and (ii) the gene/genome level (to search for possible links between genomic targets of local adaptation and genomic targets of sexual selection)
Sexual selection and plant diversification	No data or indirect evidence	Mating systems (SI <i>versus</i> SC) influence diversification rates (Goldberg <i>et al.</i> , 2010) Suggestion to address the knowledge gap: sexual selection directly measured at the population level could be correlated with speciation rates estimated from present-day phylogenies

The concept of sexual selection has been expanded to postmating processes, notably: (i) cryptic female choice (Eberhard, 1996), a process in which the fertilisation of certain sperm is favoured to the detriment of other sperm; and (ii) postmating male–male competition (competition between sperm cells for fertilisation of the egg cell; Eberhard, 2009). These expansions of the concept of sexual selection are particularly pertinent in the case of plants, which, for obvious reasons, do not show courtship behaviours, male fights for access to mating or visible female choice (Willson, 1979).

Sexual selection in plants (mostly angiosperms) acts at the pre- and postmating stages similarly to in the animal kingdom. In plants, the deposition of pollen on the stigma (i.e. pollination) is considered analogous to mating in animals. Sexual selection can therefore act before mating on traits that promote pollen transfer to conspecific stigmas, which corresponds to male–male competition (Willson, 1979; Delph & Ashman, 2006). Sexual selection in plants also takes place after pollination, whereby the traits mediating pollen performance as well as the mechanisms of female selection may impact the reproductive success of both males and females (Queller, 1984; Willson, 1994; Moore & Pannell, 2011; Tonnabel *et al.*, 2021). Interestingly, plants show a much wider diversity of reproductive strategies than animals, likely associated with dramatic differences in the intensity of sexual selection (Friedman & Barrett, 2011; Collet *et al.*, 2014). This makes plants an important taxon for the study of general principles that govern sexual selection and speciation, potentially providing key insights for evolutionary theory. Below, we review the currently known mechanisms

of sexual selection in plants before and after mating and evaluate how reproductive strategies can affect sexual selection.

(1) Premating sexual selection (pre-pollination stage)

Using flowers to attract pollinators is a major part of pre-pollination sexual selection. Angiosperm flowers show an enormous diversity (e.g. petal size, biomass, petal and stamen number, colour, and scent) (Armbruster, 2014; Davis *et al.*, 2014; Vasconcelos *et al.*, 2019; Nikolov, 2019). Darwin (1877) noted that the perianth (petals and/or sepals) size was often larger in functionally male or hermaphrodite flowers than in female flowers in taxa with unisexual flowers.

This observation is in line with an influential theory, ‘Bateman’s principle’, which postulates that sexual selection acts on male mating success more than on the success of females (Willson, 1994; Collet *et al.*, 2014; Paterno *et al.*, 2020). It hypothesises that the reproductive success of females is limited by resources, and the frequency of mating therefore has minimal impact on their reproductive fitness, whereas male reproductive success is limited by mating opportunities and thus should increase with the number of mating events (Bateman, 1948; Collet *et al.*, 2014). When applied to plants, mating success is a crucial bottleneck for males: it was estimated for one plant community that less than 3% of pollen produced reaches a conspecific stigma (Gong & Huang, 2014). Thus, from the male perspective, the more pollinators that visit the plant, the higher the chances that pollen from a particular male will access a

female, and the higher their reproductive success. By contrast, from the female perspective, as only a fraction of all pollen is required to fertilise all seeds, a low pollinator visitation rate may be sufficient (Elle & Meagher, 2000; Collet *et al.*, 2014). This imbalance will mean that floral traits influencing both pollen import and export by pollinators will be selected to a great extent by the male function (the ‘fleurs-du-male’ hypothesis; Queller, 1997). As a consequence, male fecundity should be strongly associated with floral traits, such as conspicuous flowers (Stanton *et al.*, 1991; Paterno *et al.*, 2020). Consistently, recent research showed that larger hermaphrodite angiosperm flowers show increased investment in male organs and pollinator attraction (larger petals), whereas smaller flowers tend to be female (Paterno *et al.*, 2020). In other words, there is male–male competition over pollinator visitation.

Sexual selection might also shape traits that facilitate pollen removal and transfer. Studies have shown the possibility of intrasexual selection through pollen–pollen competition during the movement of pollen to and from the pollinator’s body (pollen transfer) (Cocucci *et al.*, 2014; Duffy & Johnson, 2014; Lynn *et al.*, 2020). For example, Cocucci *et al.* (2014) showed that pollinaria (aggregated pollen) in the Apocynaceae subfamily Asclepiadoideae physically compete for access to attachment sites on the body of pollinators.

Therefore, increasing mating opportunities appears to be under strong selection mostly for male parts of the plant (Willson, 1979; Willson & Burley, 1983). Divergent premating sexual selection may lead to the divergence of floral traits between related lineages, and if so, this could result in the establishment of reproductive barriers and, eventually, to plant speciation. This hypothesis will be further explored in Section III.

(2) Postmating sexual selection (post-pollination stage)

When pollen is deposited on the stigma, there arises the possibility of competition between males (pollen) to reach the ovules (Mulcahy, 1979; Arnold, 1994; Willson, 1994; Moore & Pannell, 2011) and of female choice, i.e. the selection of pollen by females (Mazer *et al.*, 2018). At this stage, therefore, traits related to both male performance and female choice may be under sexual selection.

Competition between males may occur when the number of pollen grains deposited on a stigma exceeds the number of ovules (Delph, Weinig & Sullivan, 1998; Winsor, Peretz & Stephenson, 2000), which is the case for the majority of species (Gong & Huang, 2014; Ilta, Le Vève, Slovak & Lafon Placette, in preparation). In male–male competition for access to ovules, pollen performance traits including pollen viability, pollen size, pollen germination rate, chemical interference, and pollen tube growth rate (PTGR) are likely to be under selection (Snow & Spira, 1991a; Spira *et al.*, 1992; Johnston, 1993; Delph *et al.*, 1998; Snow, Spira & Liu, 2000; Lankinen & Skogsmyr, 2002; Delph & Ashman, 2006; McCallum & Chang, 2016). Of these, pollen germination rate and PTGR are the most studied

performance traits that might mediate pollen–pollen competition in the style (Mazer *et al.*, 2010; McCallum & Chang, 2016). In *Hibiscus moscheutos* (Malvaceae) and *Viola tricolor* (Violaceae), experiments showed that pollen with higher germination rates and PTGRs can reach unfertilised ovules faster and achieve greater fertilisation success than those with lower rates (Snow & Spira, 1991a,b, 1996; Skogsmyr & Lankinen, 1999). Thus, sexual selection could act on the PTGR as a competitive trait.

After the deposition of pollen on the stigma, female choice might affect the success of pollen donors. Although the relative siring success of different pollen donors was not affected by the maternal plant in one study (Marshall, Shaner & Oliva, 2007), stressing the maternal plants led to more random mating (Marshall & Diggle, 2001; Shaner & Marshall, 2003), suggesting that healthy females can affect seed paternity. The traits involved in pollen selection by females may include style length, stigmatic area, stigma receptivity, and chemical signalling in the stigma and style (Mulcahy, 1979; Armbruster *et al.*, 1995; Lankinen & Strandh, 2016; Mazer *et al.*, 2016; Tonnabel *et al.*, 2021), of which style length and chemical signalling are the best investigated traits.

Pistil traits may enhance pollen competition as a female choice mechanism (Lankinen & Green, 2015). For example, there is evidence that a longer style provides females with a mechanism to discriminate among pollen with different performance levels; it might lead to a longer interaction between pollen tubes and pistil that allows females to control pollen tube growth in the style *via* molecular interactions (Mulcahy & Mulcahy, 1975; Lankinen & Skogsmyr, 2002; Ramesha *et al.*, 2011). Hence, sexual selection might favour females with longer styles (Ö. Ilta, A. Le Vève, M. Slovak, C. Lafon Placette, in preparation). However, while a meta-analysis confirmed that style length was significantly correlated with rate of non-random mating, the degree of non-random mating was highest in species with shorter styles (Ruane, 2009), arguing against the proposal that a longer style may increase female selectivity. In a mathematical modelling study, Travers & Shea (2001) used measurements of *in vivo* pollen performance (PTGR and pollen germination rate) and style length. They predicted that faster pollen germination would increase siring success when style length was shorter, and faster PTGR would increase siring success when styles were longer (Travers & Shea, 2001). Thus, the higher degree of non-random mating in species with shorter styles found by Ruane (2009) might be explained by a greater role of pollen germination rate rather than PTGR in male–male competition.

Female control of pollen germination and tube growth likely involves complex chemical signalling in the stigma and style, allowing the female sporophyte to recognise compatible pollen, support the growing pollen tube in the style, and help the pollen tube to target the ovary (reviewed in Tonnabel *et al.*, 2021). Germination of pollen at the stigmatic surface is controlled *via* a complex signalling interaction between specific pollen proteins/molecules and proteins at the stigmatic surface (reviewed in Wheeler, Franklin-Tong & Franklin, 2001;

Zheng *et al.*, 2018; Johnson, Harper & Palanivelu, 2019), leading to female provisioning of the pollen grain with resources, especially water for the hydration process and germination of the pollen on the stigma. After this stage, the continued chemical interactions between pistil and pollen tubes control pollen tube growth into the style transmitting tract (Zheng *et al.*, 2018). When pollen tubes reach the style, guiding signals and nutrients from the pistil extracellular matrix guide the pollen tube towards the ovary. In the final step, signals produced by the ovules guide pollen tubes towards the ovule and lead to fertilisation (reviewed by Dresselhaus & Franklin-Tong, 2013; Johnson *et al.*, 2019).

At each of these steps (pollen germination, pollen tube growth, and pollen tube guidance) female choice has the potential to act at a molecular level (Tonnabel *et al.*, 2021). For example, there is evidence for mate recognition mechanisms such as the self-incompatibility (SI) system, in which the female can accept/reject compatible pollen at the germination (sporophytic SI; e.g. Brassicaceae) and tube growth stage (gametophytic SI; e.g. Solanaceae), and this can affect the siring success of pollen donors carrying alleles recognised as self by the female plant (Richman & Kohn, 2000; Doucet, Lee & Goring, 2016). Nevertheless, while this may appear to be an example of female choice and fits the definition for sexual selection of non-random variance in mating success (Kokko & Jennions, 2008; Tonnabel *et al.*, 2021), this may not be the case. Darwin (1871) proposed that sexual selection acts on differences in reproductive success caused by competition for mating, and in the SI system, female choice is only based on 'self' *versus* 'non-self' criteria (even though 'self' may be other individuals with the same S-allele) in an inbreeding-avoidance strategy. This does not involve any competition for mating, even though rare S-alleles are beneficial to males as they increase the probability to be accepted by females against more frequent S-alleles (Durand *et al.*, 2020), resembling male–male competition. The consequences are however different, with self-incompatibility leading to balancing selection on S-alleles, rather than directional selection expected in a sexual selection scenario. Also, self recognition may impact gene flow and lead to species barriers in nature (Pickup *et al.*, 2019; Harkness & Brandvain, 2021), which we will consider in Section III. In general, evidence for sexual selection acting on female chemical control of pollen germination/tube growth/tube guidance, i.e. female choice of male competitive traits ('sexy-son' hypothesis) or male traits directly improving female fitness ('good genes' hypothesis), remains sparse.

Therefore, in principle, sexual selection could operate after pollination through both male–male competition and female choice. Males with higher pollen performance traits may be at an advantage against other males (Mazer *et al.*, 2010; Baskin & Baskin, 2015), while females, *via* pistil traits and chemical signalling, could exhibit choosiness among different pollen (Lankinen & Strandh, 2016; Mazer *et al.*, 2016; Madjidian *et al.*, 2020). We will further explore below whether such mechanisms can establish postmating reproductive barriers, and potentially lead to speciation.

(3) Reproductive strategies influencing sexual selection

Compared to animals, and probably due to their sessile nature, plants (particularly angiosperms), have evolved myriad unique reproductive strategies that are likely to influence how sexual selection works, as well as its intensity. In this section we evaluate the impact of various life-history traits on the potential for sexual selection.

(a) Pollination mode

Plants are immobile and thus rely on biotic (pollinators) and abiotic (wind and water) vectors for transferring their (male) pollen to the (female) ovary (Friedman & Barrett, 2009), and these different vectors might change the intensity of sexual selection. In particular, the use of wind *versus* animal pollination is likely to affect pre-pollination male–male competition, as the differences between these two strategies lie mostly in the premating phase.

In animal-pollinated plant species, conspicuous flowers and inflorescences (e.g. larger petals; Paterno *et al.*, 2020) or floral scents used to signal the availability of nutritional rewards function to attract pollinators and thus contribute to male reproductive success (Arnold, 1994; Willson, 1994; Delph & Ashman, 2006; Cocucci *et al.*, 2014; Paterno *et al.*, 2020). Thus, for animal-pollinated species, a large set of traits, ranging from floral display, reward, or pollen structures enabling attachment to pollinators, are likely to improve male mating success under male–male competition for pollinators. This is likely to facilitate the evolution of premating sexual traits *via* sexual selection. By contrast, in wind-pollinated plants pollination takes place randomly, and male reproductive success is linearly correlated with male reproductive investment (Charnov, 1979; Aljiboury & Friedman, 2022). As a result, higher production of pollen or traits affecting pollen dissemination (e.g. aerodynamic properties) are the only traits that can increase the probability of fertilising ovules in competition with other males (Cruden, 1977; Eppley & Pannell, 2007; Friedman & Barrett, 2009; Tonnabel, David & Pannell, 2019), as most of the pollen produced will be wasted (Darwin, 1876; Ackerman, 2000). Therefore, in wind-pollinated species, sexual selection has very few opportunities to act at the premating stage, and likely does so mostly on pollen production and dissemination.

After the pollen reaches the stigma (postmating), sexual selection is likely unaffected by pollination mode. In a meta-analysis of 51 studies on 23 different species from 17 different families (mostly Brassicaceae, Poaceae, and Cucurbitaceae), no significant difference was found in the non-random siring rate between wind- and animal-pollinated species (Ruane, 2009), suggesting that the pollination mode does not affect the intensity of postmating sexual selection.

(b) Sexual polymorphism versus hermaphroditism

Most angiosperm species are hermaphrodites, with both female (ovules) and male (pollen) parts on the same flower

(Barrett, 2002; Charlesworth, 2006). However, some species have evolved from hermaphroditism towards gynodioecy and androdioecy, i.e. the co-occurrence of hermaphrodite and female or male individuals, respectively (Darwin, 1877; Pannell, 2002), or even towards the complete separation of the sexes (male and female individuals only; dioecy) (Charlesworth, 1999). Dioecious species are found in only 6% of angiosperms (Renner, 2014), although it has evolved from hermaphroditism multiple times independently (Käfer, Marais & Pannell, 2017).

In hermaphroditic species, the evolution of sex-specific traits is likely a trade-off between male and female interests due to resource and morphological constraints, similarly to trade-offs proposed in hermaphroditic species of animals (Schärer & Janicke, 2009). Thus, the partial or complete separation of the sexes is likely to favour the evolution of fitness optima, such as higher pollen production and pollen transfer for males (Wilson & Harder, 2003) or higher seed production for females (Freeman, Kliff & Harper, 1976). Sexual selection is expected to lead to fitness optima, with more competitive males (more conspicuous flowers, higher pollen performance) (Kwok & Dorken, 2022) and more selective females (longer styles, stricter chemical female choice) compared to hermaphrodites. This means that the effects of sexual selection may be clearer in dioecious species compared to hermaphrodites. Modelling studies have shown that males are more likely to evolve showy flowers than females in dioecious animal-pollinated plants, especially with pollinator abundance (Vamosi & Otto, 2002). However, a morphological analysis on tropical forest trees found that the flowers of almost all animal-pollinated dioecious species were relatively small, unspecialised, and of inconspicuous colour in comparison to hermaphroditic plants (Bawa & Opler, 1975; Bawa, 1994). While this apparently contradicts the suggestion that intrasexual selection will act more on male flower traits in dioecious compared to hermaphroditic species, this finding might be explained by the fact that the small flowers of dioecious species were associated with visitation by a wide range of insect pollinators while the hermaphroditic species were visited by larger specialists (Bawa & Opler, 1975; Bawa, 1994). Attracting a wide range of pollinators might improve pollen transfer and thus male reproductive success more than relying on a few specialised pollinators.

In androdioecious and gynodioecious species, as unisexual individuals co-occur with hermaphrodites, the sex ratio may be biased towards one or other sex (Pannell, 2002), and this may influence the intensity of intrasexual competition for mating. Interestingly, a reciprocal effect was found in dioecious species with an XY sex-determination system: more intense pollen competition led to a sex ratio more biased towards female progeny. This was explained by pollen grains with an X chromosome outcompeting pollen grains bearing a Y chromosome during mating (Delph, 2019). Just as for fully dioecious species, gyno/androdioecy may allow the specialisation of individuals into solely male or female function, respectively, and under sexual selection we can expect

exacerbation of traits for male–male competition or female choice. For example, in the genus *Silene*, some female flowers of gynodioecious species have larger stigmatic surfaces and longer stigma papillae than in hermaphroditic flowers (Bock, 1976; Dulberger & Horovitz, 1984).

(c) *Selfing versus outcrossing*

While more than 90% of plant species are hermaphrodites (Bawa, 1980), almost 65% of all species are outcrossing, 11% selfing, and 24% show mixed mating (Igic & Kohn, 2006).

The well-known evolutionary transition from outcrossing to selfing in flowering plants (reviewed in Igic, Lande & Kohn, 2008) can take place when the advantage of reproductive assurance [e.g. reproducing under pollen limitation (Darwin, 1876; Eckert, Samis & Dart, 2006) and selfing transmission (Fisher, 1941)] outweighs the cost of inbreeding depression (i.e. reduced offspring fitness compared with outbreeding; Charlesworth, 2006). This transition frequently involves breakdown of the SI system, and hence the evolution of SC (Takayama & Isogai, 2005). The so-called ‘selfing syndrome’ of floral traits that evolve after the transition to selfing is distinguished by small and inconspicuous flowers, reduced scent, reduced pollen production and a lower pollen:ovule (P/O) ratio (reviewed in Sicard & Lenhard, 2011; Willi, 2013; Tsuchimatsu *et al.*, 2020).

The transition from outcrossing to selfing is likely accompanied by relaxation of sexual selection (Cutter, 2019; Gutiérrez-Valencia *et al.*, 2022). At the premating stage, the high pollen transfer efficiency ensured by selfing strongly reduces male–male competition for pollen transfer (Harvey & May, 1989; Willis, 1999). Therefore, a selective pressure on males for attractive traits such as flower size or scent will be relaxed, leading to the reduction of these traits as part of the ‘selfing syndrome’. At the postmating stage, as both pollen and the female sporophyte are from the same individual and thus carry the same pool of alleles, variation in male competitiveness/female choice is virtually null, likely making sexual selection inefficient. In addition, the mating success or failure of a specific pollen grain is less likely to impact fitness in a scenario where male gametophytes are virtually identical to each other (due to homozygosity and lack of efficient recombination) and to the female gametophytes. One prediction is that male competitive traits (pollen germination, PTGR, etc.) and female choice traits (style length, chemical selection of pollen grains) would be under weaker selection in selfers compared to outcrossers. Also note that natural selection may actually favour changes that accompany the transition to selfing as a resource allocation trade-off. For example, in the selfer *Arabidopsis thaliana* it was shown that the reduction in pollen number is caused by an allele which has been under positive selection (Tsuchimatsu *et al.*, 2020). Additionally, it is likely that any form of asexual reproduction, which occurs in about 80% of angiosperms (Klimeš *et al.*, 1997), will have a similar effect as selfing of reducing the strength of sexual selection.

A study that examined the divergence in pollen performance in two *Clarkia* sister species, predominantly outcrossing *C. unguiculata* and self-pollinating *C. exilis*, revealed that *C. unguiculata* had a higher PTGR than *C. exilis* (Mazer *et al.*, 2018). Similarly, in another model system, *Mimulus guttatus* (outcrosser) had a higher PTGR than *M. nasutus* (selfer) (Diaz & Macnair, 1999). It was also shown that *M. nasutus* pollen did not outcompete *M. guttatus* pollen when *M. guttatus* was used as the maternal plant, suggesting a role of female choice. Indeed, style length comparisons between selfer and outcrosser species revealed that outcrossers had longer styles (Diaz & Macnair, 1999; Runions & Geber, 2000), consistent with the idea that sexual selection is weaker in selfing species in comparison to outcrossers. Interspecific crosses between several SI and SC *Arabidopsis* species revealed that pollen of SI species could germinate on an SC stigma, while in the reciprocal crosses pollen of SC species did not germinate when the maternal plant was an SI species (Li *et al.*, 2018). This suggests that females of SI species are choosier than SC females in the signalling crosstalk between pollen and stigma. A similar case of SI/SC unilateral pollen rejection was reported in the genus *Solanum*, involving the inhibition of pollen tube growth and pollen tube targeting to the ovules rather than of pollen germination (Baek *et al.*, 2015; Lafon-Placette *et al.*, 2016). Importantly, many phenotypic and genetic changes occur after the transition to selfing due to factors other than sexual selection, and disentangling these factors remains a challenge for future research.

III. SEXUAL SELECTION AND REPRODUCTIVE BARRIERS

Speciation is the process by which genetically and phenotypically distinct lineages emerge *via* the evolution of reproductive isolation (Dobzhansky, 1935; Mayer, 1942; Turelli *et al.*, 2001; Mallet, 2010; Seehausen *et al.*, 2014). Isolating barriers are classified into prezygotic (e.g. geographical isolation, gametic isolation, or isolation by different pollinators) and postzygotic (hybrid inviability or sterility) (Dobzhansky, 1970; Coyne, 1992; Coyne & Orr, 1998; Orr & Turelli, 2001). A prezygotic barrier means that zygotes are never created, whereas a postzygotic barrier means that individuals of two species may produce zygotes, but these hybrid offspring are inviable or infertile (Dobzhansky, 1970; Coyne, 1992; Coyne & Orr, 1998; Orr & Turelli, 2001). Prezygotic barriers in plants can occur before mating, such as pollinator shifts [e.g. genus *Mimulus* (Schemske & Bradshaw, 1999); *Satyrium longicauda* (Castañeda-Z´arate, Johnson & van der Niet, 2021)], or after mating by pollen rejection through blocking pollen germination on the stigma (Li *et al.*, 2018), delaying or stopping pollen tube growth in the style (Baek *et al.*, 2015), or misguiding pollen away from the ovary (Lafon-Placette *et al.*, 2016). It has been suggested that sexual selection could play a role in plant speciation

(Panhuis *et al.*, 2001; Ritchie, 2007) by driving the evolution of these reproductive barriers (Lafon-Placette *et al.*, 2016). In this following section, we evaluate whether sexual selection is a plausible driver of these reproductive barriers in plants at the pre- and postmating stage.

(1) Pollinator shift in response to pre-pollination sexual selection

To mate, plants are reliant on vectors (pollinators) for transferring pollen to the stigma. Most angiosperms (87.5%) use animals as vectors (Friedman & Barrett, 2009; Ollerton, Winfree & Tarrant, 2011), and pollinator-mediated selection on floral traits is argued as a mechanism underlying plant adaptation and speciation. It is thought that flowers evolved in angiosperms to attract pollinators (Stebbins, 1970; Kiestner, Lande & Schemske, 1984; Crane, Friis & Pedersen, 1995; Thien *et al.*, 2009). A shift in pollinators *via* floral trait divergence could act as a strong hybridisation barrier; if two lineages no longer share the same pollinator, gene flow between them will be limited or absent. For example, a change in floral reward chemistry from nectar to oil is thought to have led to a pollinator shift in the long-spurred African orchid, *S. longicauda* from moth to oil-collecting bee pollination (Castañeda-Z´arate *et al.*, 2021). Similarly, in monkeyflowers, *Mimulus lewisii* is predominantly pollinated by bees, while another, *M. cardinalis*, is hummingbird pollinated (Ramsey, Bradshaw Jr & Schemske, 2003). These differences in pollinator preferences have been explained by floral trait divergence between the two species, which differ in petal colour and nectar volume (Schemske & Bradshaw, 1999; Ramsey *et al.*, 2003).

According to Bateman's principle, male reproductive success will be limited by mating opportunities and therefore the evolution of floral traits is largely driven by selection on males (see Section II.1; Bateman, 1948; Willson, 1994; Collet *et al.*, 2014; Paterno *et al.*, 2020). In other words, increasing pollinator attraction (higher visitation rate) will benefit males (pollen transfer) more than females (Stanton *et al.*, 1991; Queller, 1997; Paterno *et al.*, 2020). It is important to note, however, that in some taxa or under certain ecological circumstances, plants can experience pollen limitation (Knight *et al.*, 2005). In this case, pollinator visitation may also be a limiting factor for female reproductive success, leading to female–female competition for pollinators. In particular, pollinator shift has been proposed to occur in response to decreased abundance of a given pollinator (Thomson & Wilson, 2008) since a shift to a new pollinator should increase the chances of pollination. Thus, pollinator shifts may take place in response to high male–male (and female–female) competition for pollinator attraction (Fig. 1), with low resources (i.e. pollinator visitation) leading to higher intrasexual competition, and hence selecting for flower traits such as more conspicuous display and higher rewards (Ratnieks & Balfour, 2021). We argue that this will lead to floral trait divergence and ultimately to the establishment of reproductive barriers between individuals showing the ancestral and

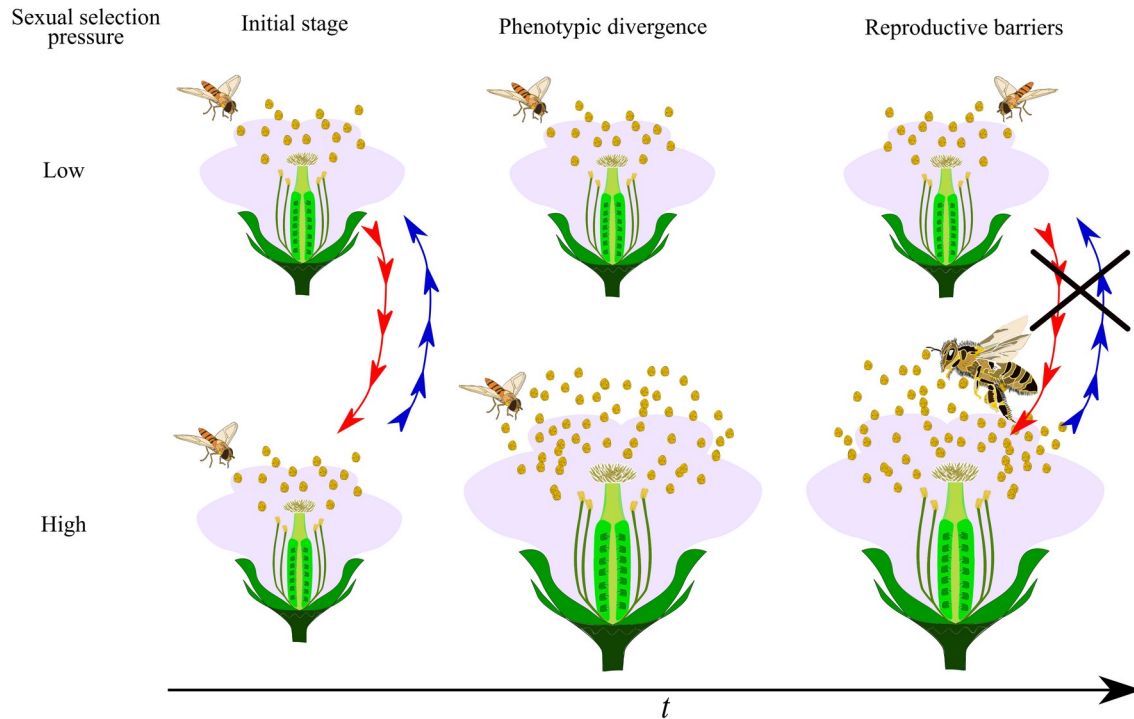


Fig. 1. Pollinator shifts in response to male–male competition (pre-pollination sexual selection). Higher male–male competition for pollinators, for example resulting from local pollinator scarcity, may select for a larger floral display and higher production of pollen grains (bottom row). This may lead to floral divergence (phenotypic divergence) and, ultimately, to the establishment of reproductive barriers between individuals with the ancestral and derived pollinator syndromes. t is evolutionary time; red and blue arrows indicate gene flow between the two lineages; black cross indicates a barrier to gene flow.

derived pollinator syndromes. This process may act in interaction with other speciation mechanisms, such as reinforcement, i.e. positive selection for a barrier to prevent costly hybridisation. Interestingly, changes in pollinator attraction between *Phlox* species have been shown to act as a reinforcement mechanism (Hopkins & Rausher, 2012; Hopkins *et al.*, 2014). In a contact zone between *Phlox cuspidata* and *P. drummondii*, a change in flower morphology in *P. drummondii* promotes assortative mating (Hopkins & Rausher, 2012). In this contact zone, male–male competition for pollinators could occur both intra- and interspecifically, and as a result, sexual selection may be more intense. In this context, an allele affecting flower colour that promotes assortative mating would potentially increase mating success (pollinator visitation). Thus, we speculate that pollinator shifts may occur as a result of an interaction between sexual selection and reinforcement mechanisms.

(2) Pollen rejection and post-pollination sexual selection

Pollen from a given lineage may be rejected by the pistil from another lineage, leading to a postmating reproductive barrier. Pollen rejection can take place at different stages: (i) at the germination stage when pollen lands on the stigma (stigmatic pollen rejection); (ii) as pollen tubes grow through the

style (stylar pollen rejection); and (iii) during targeting of pollen tubes to the ovary (ovarian pollen rejection) (Baek *et al.*, 2015; Lafon-Placette *et al.*, 2016; Li *et al.*, 2018). Below, we consider these stages of pollen rejection as reproductive barriers and discuss the possible role of sexual selection among other evolutionary scenarios for establishing these barriers.

While interspecific pollen rejection at the stigma, style or ovary stage may appear to be a form of female choice and sexual selection, this may not always be the case. The process of genetic drift, i.e. random changes in the number of allelic variants in a population (Lynch *et al.*, 2016), may be an alternative explanation for the establishment of pollen rejection. Sexual reproduction involves the location of suitable mates, and the chemical communication between pollen and the female organs (stigma, style, ovules) fulfils this role. If two lineages evolve separately, this chemical communication is likely to evolve separately simply by genetic drift. If sufficient evolutionary time elapses, pollen–female communication will likely diverge enough between lineages that pollen receptors from one lineage will not match female ligands from the other, eventually leading to pollen germination/tube growth/tube guidance incompatibilities in hybridisation events, and hence a reproductive barrier. For example, Swanson *et al.* (2016) compared pollen performance in two *A. thaliana* accessions: Columbia (Col), and Landsberg erecta

(Ler). In cross-pollination using pollen of Col and Ler with Col pistils, Ler pollen on a Col pistil showed reduced germination and tube growth, leading to significantly higher siring success for Col pollen (98%) than Ler pollen (11%) in a mixed pollination experiment (Swanson *et al.*, 2016). *A. thaliana* is a selfer, and hence is expected to be under relaxed sexual selection (Cutter, 2019). Thus, while this case initially seems to provide an example of female choice, sexual selection is unlikely to be the underlying evolutionary driver and the best scenario is neutral divergence of pollen–pistil communication between Col and Ler by genetic drift, leading to a reproductive barrier.

However, pollen rejection could also be a result of independent runaway sexual selection (Fisherian model; see Section II.1) in two lineages (Ritchie, 2007). For example, in plants, style length as a female choice trait and a faster PTGR as a male trait could coevolve to reach an optimum (Ö. İltis, A. Le Vève, M. Slovak, C. Lafon Placette, in preparation). Some studies have found positive correlations between pistil length and PTGR (Williams & Rouse, 1990; Herrero & Hormaza, 1996), whereas studies in *Hibiscus* and *Ipomopsis* failed to find a correlation (Snow & Spira, 1996; Alarcón & Campbell, 2000). If female preference (style length, chemical signalling) and male traits (pollen germination, tube growth, chemical reception) coevolve separately in two lineages they might reach two different optima, and as a result, the two lineages will exhibit bilateral mating incompatibilities (Fig. 2). For example, crossing experiments between two predominantly outcrossing iris species, *Iris fulva* and *I. hexagona*, showed a low number of hybrids, while further analysis using mixed pollination (1:1 pollen mixture of both species) to test for interspecific pollen competition resulted in no hybrid progeny, suggesting that the low number of hybrids was due to selective fertilisation by intraspecific pollen (Arnold, Hamrick & Bennett, 1993). Further research revealed that a reduced PTGR of heterospecific pollen may explain failed hybridisation between these two irises (Hodges & Arnold, 1995). The same results were found for two sympatric *Hibiscus* species, *H. moscheutos* and *H. laevis* (Klips, 1999). However, artificial crosses in *Helianthus* gave contrasting results in which a lower number of hybrids could not be explained by PTGR differences (no intraspecific and interspecific difference in PTGRs Rieseberg, Desrochers & Youn, 1995). Disentangling the roles of divergent runaway selection *versus* neutral divergence by genetic drift as a cause for pollen rejection remains difficult to do, as both have the same outcome, i.e. bilateral pollen rejection. Genomics may help resolve this issue, by identifying the genes responsible for bilateral pollen rejection and assessing selective processes acting on them. Indeed, reproductive genes may evolve faster than other genes simply because they are expressed in a sex-specific manner and therefore selection on them is relaxed: when carried by the other sex, these alleles will be masked to selection (Dapper & Wade, 2016). However, this may be only partly true in plants, since the haploid state in the gametophytic phase is subject to stronger purifying selection than the diploid state, as shown for

pollen-specific genes in *Capsella grandiflora* (Arunkumar *et al.*, 2013). In this study, pollen-specific genes also showed higher rates of positive selection than sporophytic genes. Thus, if genes responsible for bilateral pollen rejection would also show signs of positive selection, one could safely assume that this hybridisation barrier arose as a result of divergent runaway selection rather than drift. This hypothesis remains however to be tested.

Differential sexual selection intensity (i.e. divergence in sexual selection) might be another explanation for the establishment of barriers through pollen rejection in plants. For example, the species *Silene latifolia* shows a wide natural variation in style length, and *S. latifolia* style length was negatively correlated with seed set when pollinated with *S. diclinis* (Nista, Brothers & Delph, 2015). This could suggest that populations of *S. latifolia* have experienced different intensities of sexual selection that have led to divergence in style length that translates into different strengths of hybridisation barrier with a related species. Interestingly, the decreased seed set was found between sympatric populations of the two species, suggesting that sexual selection may play an important role in preventing hybridisation in sympatry.

In general, we predict that if two lineages experience different intensities of sexual selection, pollen competitive traits and female choosiness will diverge quantitatively, and as a consequence, these two lineages will face unilateral incompatibilities (Fig. 3). An extreme case of such divergence is relaxed sexual selection inherent to the transition to selfing, as male–male competition will be much reduced in a selfer as compared to an outcrosser (Cutter, 2019). An investigation in wild tomatoes identified a unilateral prezygotic barrier that occurs between the pollen of a self-compatible (SC) species (*Solanum lycopersicum*) and the female of a self-incompatible (SI) related species (*S. pennellii*), whereas the SI pollen was able to fertilise an SC female (Covey *et al.*, 2010; Baek *et al.*, 2015). These results suggest that SI females are choosier than SC females, and that SI pollen is more performant than SC pollen; this has been coined the ‘weak inbreeder/strong outbreeder’ hypothesis (Brandvain & Haig, 2005). As a result, the SI population with a higher intensity of sexual selection is more likely to reject the SC pollen and therefore hybridisation is prevented. Note that it is not clear whether divergent sexual selection alone is involved in this hybridisation barrier, or whether this type of reproductive isolation could simply be a by-product of the SI system. Indeed, some studies suggest that molecular components of the SI system are directly involved in the SI × SC unilateral incompatibility, notably in *Solanum* (Markova *et al.*, 2016; Pease *et al.*, 2016). Work on several *Arabidopsis* species showed that the locus responsible for the SI × SC unilateral incompatibility was not linked to the S-locus (Li *et al.*, 2018), suggesting that this barrier is not a by-product of the SI system. Finally, a study surveying cases of unilateral gene flow in the wild found that all cases of SI × SC species pairs with available information showed unilateral gene flow (Pickup *et al.*, 2019). This suggests that SI × SC unilateral

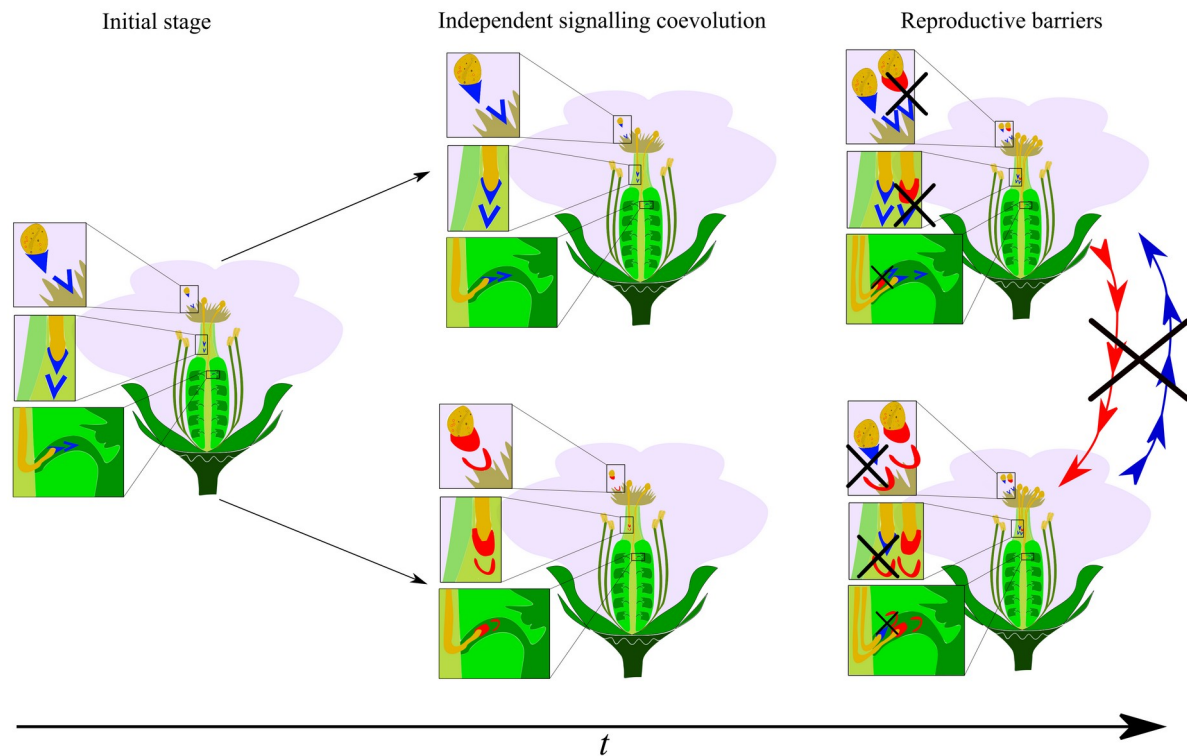


Fig. 2. Pollen rejection in response to independent coevolution of female preference and male traits (post-pollination sexual selection). The separate coevolution of female choice (ligands) and male trait (receptors) in two lineages might establish bilateral incompatibilities, as the ligand and receptor of each lineage will no longer match. Blue triangle/V-shape indicate the receptors and ligands for the ancestral allele; red semicircle/U-shape indicate the receptors and ligands for the derived allele; t is evolutionary time; red and blue arrows indicate gene flow between the two lineages; black cross indicates a pollen rejection/barrier to gene flow.

incompatibility, and thus potentially sexual selection, could play an important role in species boundaries in nature.

Taken together, plants can show prezygotic barriers before mating through the evolution of traits for different pollinators (pollinator shifts) (Schemske & Bradshaw, 1999; Ramsey *et al.*, 2003; Castañeda-Z'arate *et al.*, 2021) and after mating due to pollen rejection (Sarker, Elleman & Dickinson, 1988; Dickinson, 1995; Hulskamp, Schneitz & Pruitt, 1995; Zheng *et al.*, 2018; Tonnabel *et al.*, 2021). Before mating, intense male–male competition for pollinator visitation and pollen transfer may drive rapid divergence of floral traits and potentially lead to pollinator shifts, providing a strong reproductive barrier between lineages. After mating, many evolutionary scenarios can explain pollen rejection as a reproductive barrier. Sexual selection (independent runaway selection or divergent intensity of sexual selection) may be a plausible scenario, but one should first rule out the neutral divergence of molecular mechanisms as an evolutionary cause for interspecific pollen rejection. To achieve this, we propose that (i) the traits involved should be investigated for a role within species in male–male competition/female choice as defined by sexual selection theory; and (ii) pollen rejection should be explained within a sexual selection theoretical framework, i.e. with evidence for independent runaway selection or divergent sexual selection intensity.

To disentangle neutral divergence *versus* independent runaway selection as an explanation for pollen rejection, we predict that pollen rejection established by runaway-like mechanisms should arise faster in clades with more intense sexual selection.

(3) Ecological speciation and speciation by sexual selection

Animal studies have suggested that sexual selection alone may be insufficient to drive the emergence of new species, and that it is likely to act in concert with local adaptation and ecological speciation (Servedio & Bürger, 2014; Servedio & Boughman, 2017; Irwin, 2020; Mendelson & Safran, 2021). Sometimes, sexual selection may be antagonistic to the latter processes. Below, we evaluate possible interactions between sexual selection, natural selection and speciation in plants.

A clear case of an interaction between sexual selection and natural selection is when traits/genes under sexual selection are also involved in local adaptation. For example, pollen production and performance are affected by different environmental variables such as drought, nutrient availability, or herbivory (Delph, Johannsson & Stephenson, 1997; Liu *et al.*, 2023). Adaptations to such environmental variables

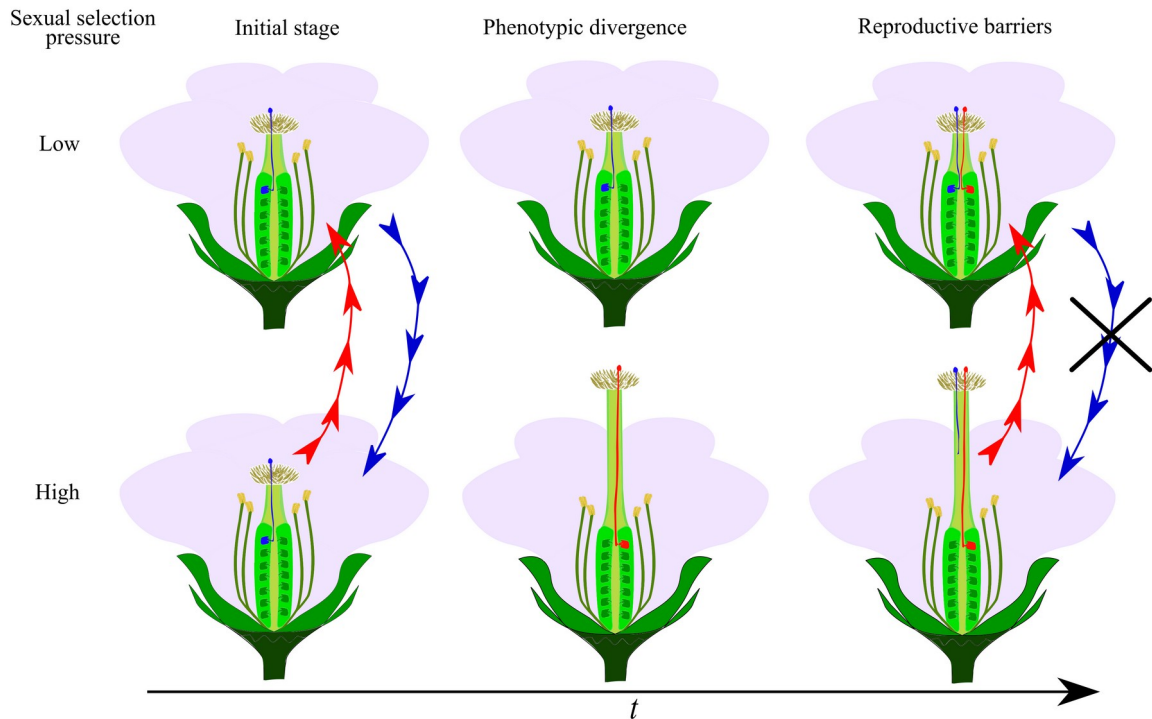


Fig. 3. Pollen rejection in response to divergent sexual selection intensity. Two lineages experience different intensities of sexual selection, and as a response pollen competitive traits (pollen tube growth rate, PTGR) and female choosiness (style length) diverge quantitatively. As a result, these two lineages show unilateral incompatibilities, with the pollen of one lineage being unsuccessful in the style of the other lineage, but the reciprocal is not true. Red circle, pollen for the ancestral trait; blue circle, pollen for the novel trait; t is evolutionary time; red and blue arrows indicate gene flow between the two lineages; black cross indicates a pollen rejection/barrier to gene flow.

may include pollen traits that are actually deleterious to pollen performance (e.g. accumulation of chemical compounds; Descamps, Quinet & Jacquemart, 2021), or adaptation may impact genes with pleiotropic effects, affecting all cells, including pollen. Natural selection would thus favour lower pollen performance while sexual selection would favour higher pollen performance, leading to antagonistic effects between the two. This could explain why, in natural populations, genetic variation is found in pollen performance: directional selection such as sexual selection is expected to decrease variation for this trait, but if natural selection acts antagonistically, this would maintain polymorphism (Delph *et al.*, 1997). Conceivably, sexual selection may slow down ecological divergence between populations by retarding local adaptation, and this could prevent ecological speciation.

Alternatively, sexual and natural selection may act synergistically on plant speciation. This could happen when male traits are a proxy for the fitness of the progeny, also known as ‘good genes’ selection (Byers & Waits, 2006). It is likely that pollen tube growth, a cell growth process with high metabolic demands (Selinski & Scheibe, 2014), is correlated with other aspects of the life cycle involving plant growth. In fact, genes expressed in pollen are also involved in other stages of plant life (Beaudry *et al.*, 2020). If there is female selection for pollen performance, then it is likely that it will also

select for progeny fitness, fitting the ‘good genes’ scenario. In this case, sexual and natural selection may act in the same direction, and this is likely to accelerate the ecological divergence between two populations, and ultimately, ecological speciation.

While these considerations are key to understanding the role of sexual selection in plant speciation, they are admittedly speculative. Further empirical research is needed to unravel how sexual selection might interact with local adaptation and how this interplay could affect speciation, including whether these processes typically act in synergy or antagonism and under what circumstances.

IV. SEXUAL SELECTION AND PLANT SPECIATION (MACROEVOLUTIONARY SCALE)

(1) Macroevolutionary research

Sexual selection is known to influence evolution above the species level (macroevolution) (Futuyma & Kirkpatrick, 2017). While the mechanisms behind the effects of sexual selection on large-scale speciation and extinction remain subject to research, a significant body of empirical results for plants support correlations between a variety of sexual traits

(sexual dimorphism, reproductive system, flower colouration) and diversification rates inferred from molecular phylogenies (Hodges & Arnold, 1995; Dodd, Silvertown & Chase, 1999; Kay *et al.*, 2006; Goldberg *et al.*, 2010).

Diversification refers to the net difference between speciation and extinction and can be estimated from the branching patterns of present-day phylogenies (Nee, May & Harvey, 1994; Ricklefs, 2007; Title & Rabosky, 2019). Consequently, differences in diversification, as defined here, might be due to differences in speciation, extinction, or some combination of these. Early studies correlated sexual traits with diversification rates and sometimes with differences in species richness across sister clades within a phylogeny (Heilbuth, 2000; Paradis, 2012; Jetz *et al.*, 2012; Rabosky & Matute, 2013; Maddison & FitzJohn, 2015; Rabosky & Goldberg, 2015; Futuyma & Kirkpatrick, 2017). While these correlations seem tentatively to suggest a link between sexual selection and diversification, later work revealed that such correlations might have other causes. These correlations could result from an asymmetry in diversification rates across trait values (due to sexual selection) but also from an asymmetry of transitions between trait values (due to trait evolution) (Maddison, 2006; Käfer *et al.*, 2017). To tease apart these effects, a range of methods have been developed that model the two asymmetries simultaneously (FitzJohn, 2010; Beaulieu & O'Meara, 2016). While these methods have their limitations (e.g. Type I errors when the assumption of time-constant diversification rates is violated) (Machac, 2014; Rabosky & Goldberg, 2015), empirical research based on this wide range of methods [sister-clade comparisons, binary state speciation and extinction (SSE), quantitative SSE, hidden SSE] has identified a compelling battery of traits that suggest signatures of sexual selection at the macroevolutionary level.

Flower symmetry, floral structure, nectar production, and the presence of an inflorescence represent several well-known examples of sexual traits that seem to be statistically associated with faster diversification. Other traits that might also be interpreted as relating to sexual selection include animal pollination, bilateral floral symmetry, and the presence of nectar spurs, all of which have also been hypothesised to promote speciation rates because of their likely effects on mating specificity in plants. Empirical research confirmed that self-incompatibility (Goldberg *et al.*, 2010), floral nectar spurs (Hodges & Arnold, 1995), and bilateral floral symmetry (Kay *et al.*, 2006) promote diversification, at least in some plant taxa. Moreover, Dodd *et al.* (1999) used sister-clade comparisons to show that animal pollination is associated with faster diversification than abiotic modes of pollination (e.g. wind pollination). These findings correspond with much empirical research implicating a role of plant–pollinator interactions in plant speciation (Stebbins, 1975; Grant, 1981; Coyne & Orr, 2004; Kay *et al.*, 2006).

Identifying traits that fit the theoretical and mechanistic definition of sexual selection is not trivial, as not all reproduction-related traits will be under sexual selection.

Moreover, different traits could have mutually contradictory effects. For example, even though selfing is often thought to promote diversification (Igic *et al.*, 2008), Goldberg *et al.* (2010) reported that selfing might simultaneously increase both speciation and extinction rates, at least in the family Solanaceae. Selfing might therefore produce a short-term advantage through increased speciation, which might be balanced by the long-term extinction risk, making outcrossing advantageous in the long term. Importantly, transitions to selfing from outcrossing are related to many other life-history and population genetic changes, such as assured reproduction, increased capacity to colonise new regions, and population genetic structure, all of which may contribute to speciation independently of sexual selection. This example illustrates the complexity of attempting to isolate the effects of any particular trait and to determine whether it operates through sexual selection on plant speciation. This is a shortcoming of the correlative research that typifies macroevolutionary studies, and this field could benefit from the more mechanistic results available from population genetics, involving selection strength, genetic architecture of traits, etc.

Identifying the mechanisms behind widely known statistical correlations (e.g. between sexual dimorphism and diversification) has been notoriously difficult, partly because previous studies rarely investigated the mechanisms across scales that bridge the gap between microevolutionary processes and macroevolutionary patterns.

(2) Bridging the micro- and macroevolution of speciation by sexual selection in plants

Evolution has been extensively studied below (microevolution) as well as above the species level (macroevolution) and while much research exists within each of these two levels, the chasm between them has been hard to close (Reznick & Ricklefs, 2009). One way to bridge the gap between the microevolutionary processes and macroevolutionary patterns might be to focus on sexual traits or reproductive strategies that are linked to sexual selection below the species level but also correlate with diversification patterns above the species level. These traits (e.g. the P/O ratio, style length) might serve as a common currency to translate sexual selection and its cascading effects across the levels of evolutionary investigation. Possible disconnects between these levels would indicate that sexual selection within species does not translate into macroevolutionary patterns, presumably because its effects are less important than other selective processes (e.g. temperature-driven speciation rates, extinction rates mediated by geographic range size; Fig. 4) (Brown *et al.*, 2004; Jablonski, 2008). By considering traits under sexual selection, we might achieve an integrative perspective that encompasses population-level processes (strength of sexual selection) which translate into divergence between populations, the emergence of reproductive barriers, speciation and ultimately, the evolution and diversification of new clades.

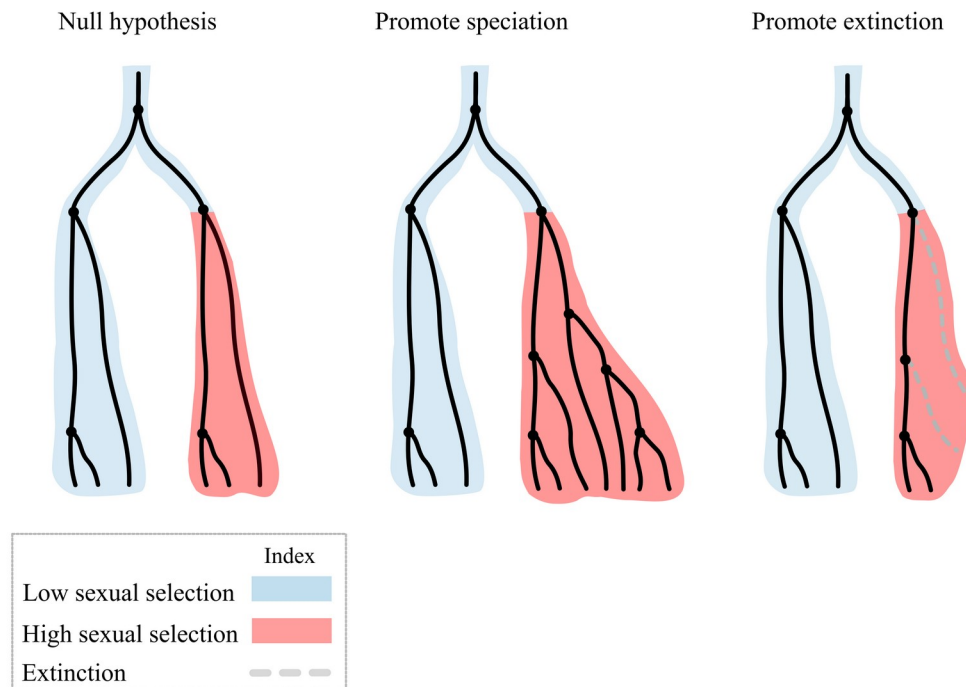


Fig. 4. Influence of sexual selection on evolution above the species level in plants. The null hypothesis is that branching patterns in clades with low or high sexual selection intensity are homogeneous. Alternatively, higher sexual selection in one clade may lead to higher branching, promoting speciation, or it might lead to lower branching, promoting extinction.

Another way to bridge micro- and macroevolutionary levels might be to consider the mechanisms of sexual selection, and to postulate likely links. For example, premating male–male competition may lead to floral divergence (Cocucci *et al.*, 2014; Lynn *et al.*, 2020; Paterno *et al.*, 2020), causing pollinator shifts (Ramsey *et al.*, 2003; Castañeda-Z'arate *et al.*, 2021) that, in turn, result in faster speciation rates. Runaway selection may lead to faster divergence of pollen–pistil communication mechanisms that result in the emergence of pollen–pistil incompatibilities (Arnold *et al.*, 1993; Carney, Hodges & Arnold, 1996; Klips, 1999) and speciation. Finally, sexually driven divergence in pollen performance and female choosiness (Díaz & Macnair, 1999; Runions & Geber, 2000; Mazer *et al.*, 2018; Madjidian *et al.*, 2020) may produce unilateral pollen–pistil incompatibilities (Covey *et al.*, 2010; Baek *et al.*, 2015) which result in speciation. For example, the change in pollination mode from biotic to abiotic is correlated with a decreasing speciation rate (Dodd *et al.*, 1999). We may mechanistically connect the levels of evolutionary investigation by stating that premating barriers arise faster as a consequence of sexual selection having more opportunities to act (floral attraction traits) in animal-pollinated species compared to wind-pollinated ones, and this may lead to faster speciation rates due to frequent pollinator shifts. Also, as a similar intensity of postmating sexual selection was found in wind- and animal-pollinated species (Ruane, 2009), and as speciation rates do vary between

these species (Dodd *et al.*, 1999), we may predict from this example that postmating barriers driven by sexual selection such as pollen rejection do not play a major role in plant speciation.

While each of these mechanisms have been supported by empirical work, no studies to our knowledge have attempted to demonstrate a chain of these processes within the same study system. Yet, suitable systems for such work exist (e.g. well-studied plant families with abundant phenotypic and phylogenetic data such as Brassicaceae or Solanaceae), that would permit the examination of sexual selection from micro- to macroscales.

Despite the promise of the trait-based approach, we caution that some traits might approximate sexual selection better than others. While reproductive strategies might provide the first approximation of the effects of sexual selection, future studies might benefit from more direct proxies, such as style length and stigmatic surface area as proxies for female choice. The P/O ratio might serve as a direct proxy for the intensity of premating male–male competition. The ratio between pollen deposited on the stigma and ovule number might be a useful proxy for the intensity of postmating male–male competition. Measuring these more direct proxies and relating them to life-history traits and diversification across a whole clade (e.g. Brassicaceae, Solanaceae) is feasible and may produce compelling results. Importantly, since these more direct proxies have specific effects, we will be able to make accurate predictions as to how they lead to the

formation of reproductive barriers, and hence bridge the micro- and macroevolutionary scales.

V. CONCLUSIONS

- (1) While a role of sexual selection in plants is now relatively well accepted, its role in plant speciation remains largely unexplored (see Table 1). Compared to animals, plants show a wider range of reproductive strategies, making them a unique system to study sexual selection mechanisms and their impacts on speciation.
- (2) Many authors have associated pollen rejection, a common reproductive barrier in plants, with female choice/sexual selection. However, the most parsimonious explanation for pollen rejection is neutral divergence in the mechanisms for mate compatibility. For a hybridisation barrier to fit into a theoretical frame of sexual selection, one requires evidence for a role of the mechanism in *bona fide* sexual selection within a lineage together with evidence for divergent sexual selection between reproductively isolated lineages.
- (3) Whether reproductive barriers driven by sexual selection translate into large-scale speciation remains unknown in both plants and animals, despite the use of relatively direct proxy traits for sexual selection mechanisms in numerous macroevolutionary studies in the latter. In plants, we first need to identify traits that act as direct proxies for sexual selection mechanisms. Reproductive strategies (pollination mode, sexual polymorphism, mating system) may prove too indirect and introduce too many confounding factors to be used to investigate the role of sexual selection in plant speciation.
- (4) Only by using direct proxy traits with a clear role in sexual selection mechanisms, will we be able to bridge micro- and macroevolution in order to understand if and by which mechanisms sexual selection plays a role in plant speciation. This will require evidence for a role in sexual selection below the species level and a role in diversification patterns above the species level, together with the ability to link these processes mechanistically at all evolutionary levels of investigation.
- (5) Such work may reveal that sexual selection plays an important role in plant speciation, or may instead show that classical drivers of speciation such as geographic or ecological isolation are dominant.

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Case study 2

OIKOS

Research article

Polyploidy-induced floral changes lead to unexpected pollinator behavior in *Arabidopsis arenosa*

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The paradox between the ubiquity of polyploid lineages in plants and the early obstacles to the establishment of polyploids is a long-studied yet unresolved question in evolutionary biology. It is assumed that to successfully persist after emergence, newly formed polyploids need to display certain fitness advantages and show a certain extent of reproductive isolation with their diploid progenitors. In this study, we tested whether immediate floral changes following polyploidization can improve pollinator visitation and enable pollinator preference leading to assortative mating, i.e. build a premating reproductive barrier between diploids and polyploids. For this purpose, we generated synthetic tetraploids of *Arabidopsis arenosa* and measured insect visitor behavior on diploids and synthetic tetraploids. We found that the increased floral size that accompanied polyploidization did not lead to a measurable increase in visitor preference, with insects visiting diploid and tetraploid plants at equal frequency. Despite this observation, tetraploids set more fruits than diploids, suggesting a positive impact of polyploidization on pollen transfer via other means. In addition, polyploidization did not lead to assortative mating but instead promoted interploidy pollen exchange, since visitors preferentially switched between cytotypes rather than preferring one. Consistent with this switching behavior, most of the progeny from tetraploid plants were triploid. Our data suggest that polyploidization has an immediate impact on mating in plants, but in a more complex way than has been assumed previously.

Keywords: *Arabidopsis arenosa*, floral display, interploidy barrier, pollinator behavior, polyploid establishment, reproductive success

Introduction

Polyploidy is widespread across angiosperms, with a large number of successfully established polyploid lineages (Wood et al. 2009, Barker et al. 2016). This remains



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a paradox, as upon emergence, polyploids face numerous obstacles to their establishment, such as reduced mating opportunities, also known as minority cytotype exclusion (Levin 1975). Polyploids may overcome their starting disadvantage by displaying novel traits increasing their fitness. One of these traits is the so-called gigas effect, where polyploids often have larger cells, leading in particular to an increase in the size of their floral organs (Müntzing 1936, Levin 1983). Flower size is known to influence flower visitation by pollinators, either increasing the visitation rate (Klinkhamer et al. 1989, Andersson 1991, Conner and Rush 1996), decreasing it (Andersson 1988, Klinkhamer and de Jong 1990) or having no obvious impact on it (Vaughton and Ramsey 1998, Ohashi and Yahara 2002). Thus, larger flowers due to the gigas effect may influence pollinator visitation of newly formed polyploids (neopolyploids), potentially compensating for the rarity of mating opportunities of neopolyploids. Experimental work has shown different floral morphology and pollinator behaviors between natural polyploids (i.e. polyploids established in populations in the wild) and their presumed diploid progenitors, but not necessarily increased visitation of polyploids by pollinators (Segraves and Thompson 1999, Palmqvist et al. 2021). However, such polyploids have undergone natural selection, and the direct effect of whole genome duplication cannot be derived from such experiments. Whether the increase in flower size by the gigas effect is enough to affect the visitation of newly formed polyploids by pollinators remains to be tested.

The floral changes brought on by whole genome duplication may impact the foraging preference of pollinators and potentially lead to assortative mating, enabling a reproductive barrier between diploids and newly formed polyploids. Reproductive isolation is an important factor contributing to the establishment of a polyploid lineage from diploids. Triploid block, or the inviability of hybrid seeds between diploids and polyploids, is often evoked as an instant reproductive barrier between cytotypes (Köhler et al. 2010). However, triploid block is often 'leaky' with interploidy seeds surviving to differing extents (Burton and Husband 2000, Meeus et al. 2020, Morgan et al. 2021). Thus, other mechanisms are likely to contribute to reproductive isolation between diploids and newly formed polyploids in early stages, such as assortative mating driven by pollinator shift. Differences in floral morphology and pollinator preference between diploids and established polyploids have been shown, leading to assortative mating (Segraves and Thompson 1999, Palmqvist et al. 2021). Again, these may be explained by late-acting selective processes rather than polyploidy itself. A few studies used synthetic polyploids to address whether assortative mating mediated by pollinator preference for a specific cytotype could arise immediately after polyploidization, in *Acacia mangium*, *Chamerion angustifolium* and *Trifolium pratense* (Nghiem et al. 2011, Husband et al. 2016, Porturas and Segraves 2020). All three studies did not find assortative mating following polyploidization, questioning whether

assortative mating can be established directly after the formation of new polyploids, and the low number of available studies on this topic calls for additional work.

From all possible systems to study these questions, *Arabidopsis arenosa* (Brassicaceae) appears as a relevant one. *Arabidopsis arenosa* is self-incompatible (Säll et al. 2004) and, as the rest of the Brassicaceae family, is likely to be a generalist when it comes to pollinator attraction (Gómez et al. 2016). It is a species complex including diploids and established autotetraploids. Their distribution ranges from south-eastern Europe to Scandinavia, with a primary contact zone between the cytotypes in central Europe (Carpathian Mountains) and a few secondary contact zones in southeast-ern Europe (Monnahan et al. 2019, Morgan et al. 2020). The origin of tetraploids is estimated to have occurred in the Carpathian Mountains 11 000–30 000 generations ago (Arnold et al. 2015) or, alternatively, 226 000 generations ago (Marburger et al. 2019), making tetraploid *A. arenosa* a relatively young lineage. This may explain why no significant niche shift has been observed in the contact zones between cytotypes (Morgan et al. 2020). Nevertheless, triploid block is strong in this species, at least in the direction of diploid mother $\times$ tetraploid father (Morgan et al. 2021). Also, common garden open pollination experiments have shown that, despite being in physical proximity, plants of the two cytotypes barely produced any triploid seeds (Morgan et al. 2021), suggesting that prezygotic barriers arose during establishment of the tetraploid lineage. This raises the question whether such barriers arose as a direct byproduct of polyploidization, or whether these barriers arose later on during the tetraploid lineage evolution.

In this study, we tested whether newly formed polyploids show an advantage in insect visitor attraction due to the changes mediated by whole genome duplication in *A. arenosa*. A higher reproductive output may offset the deleterious impacts of minority cytotype exclusion by increasing the size of the polyploid's brood, even if the progeny are triploid due to the mating with surrounding diploids (triploid bridge; Ramsey and Schemske 1998). We also tested whether polyploidy-induced floral changes induce assortative mating between individuals of the same ploidy.

Material and methods

Plant material and polyploid induction

In this study, we focused on a diploid population of *A. arenosa* from the western Carpathian lineage (Kolář et al. 2016b), which was shown to be the ancestor of the natural autotetraploids (Arnold et al. 2015). Seeds of the diploid population were collected in the Veľká Fatra Mountains, Slovakia (48°50'4.128"N, 19° 0'48.528"E) in August 2018 (Supporting information). Autotetraploid individuals of *A. arenosa* were derived from the diploid population treated with colchicine by following the method modified

from a previous study (Lafon -Placette et al. 2017). First, seeds collected from the field were surface-sterilized (with 5% NaClO, 0.01% (v/v) Triton X-100 and sterile water) and sown on an agar plate (containing: 1× MS-Salts, MES hydrate and 0.8% (w/v) plant agar; pH 5.8). Then, plates were transferred and kept under controlled conditions (12 h at 23°C/day and 12 h at 13°C/night) for two weeks to initiate germination. After two weeks, germinated seeds were transferred to small pots filled with damp soil and were grown for two weeks under the same conditions to provide time for the seedlings to be large enough for the colchicine treatment. Following that, the shoot apical meristems of diploid seedlings were treated with 0.025% colchicine solution by dropping 10 µl on to them with a pipette. Colchicine-treated seedlings were then transferred to the vernalization chamber and were grown for six weeks in short-day conditions (8 h of light per day) at 4°C to induce flowering. Following vernalization, colchicine-treated seedlings were transferred to long-day conditions (16 h of light per day) at 21°C day/15°C night. Ploidy levels of colchicine-treated individuals were estimated using flow cytometry. We found a rare occurrence of tetraploids among all the treated plants, in addition to chimeras (different stems with different ploidies) and other ploidy levels (Supporting information). Successfully induced autotetraploid individuals were selected to produce F1 seeds, to reduce the toxic effect of colchicine on the phenotype. For that, manual pollinations were performed within the populations for both diploids and induced autotetraploids. To avoid genetic differences between the diploids and induced autotetraploids, the same maternal and paternal donors were used to perform manual pollination for both diploids and induced autotetraploids (Supporting information). This was possible since these plants were chimeras, producing both diploid and tetraploid stems. For all the crosses, designated female partners were emasculated before anthesis, and the pistils were hand-pollinated two days after emasculation. Even though *A. arenosa* is self-incompatible (Säll et al. 2004), emasculation was performed to avoid any potential mentor effect (forced self-pollination made possible with the deposition of foreign pollen). For both crosses, 20 manual pollinations were performed. Five weeks after pollination, fully matured siliques from the first generation of diploids and induced autotetraploids were harvested and were stored at 4°C under dry and dark conditions for six weeks to break dormancy. These seeds were used to grow the plants for the experiment. Seeds were germinated and the plants were vernalized applying the same conditions as explained above. For acclimatization, seedlings were then transferred to big pots filled with damp soil and exposed to long-day conditions (16 h of light per day) at 21°C day/15°C night for two weeks. During this period, the ploidy level (relative DNA content) of all induced autotetraploid plants was estimated using DAPI flow cytometry (Doležel et al. 2007), and unambiguously tetraploid plants (Supporting information) were kept for the experiment. Finally, all of the plants were transferred to the field for the visitor experiment.

Ploidy level detection

We measured the ploidy level of induced plants and their progeny using DAPI flow cytometry (Doležel et al. 2007). For the induced polyploid plants, we collected petals from different stems of each plant, to account for potential chimeras. For their progeny, we collected rosette leaves. All the samples were measured individually, using *Carex acutiformis* (Vesely et al. 2012) as an internal standard. Samples were prepared using Otto I and Otto II buffers containing the fluorescent DAPI dye according to the protocol of Doležel et al. (2007). The samples were analysed on a CyFlow ML instrument (excitation source: UV LED chip 365 nm). The fluorescence of at least 3000 nuclei (for ploidy level detection) per sample was recorded, and we used a maximum coefficient of variation (CV) of 5% as the threshold for valid measurements. All data were evaluated using the Partec FloMax 2.4d program (Partec GmbH, Münster, Germany). A comparison with DAPI flow cytometric data of diploid and tetraploid *A. arenosa* from Kolář et al. (2016a) allowed for unambiguous ploidy determination.

Flower visitors experiment

Experimental design

A common garden for the purpose of the visitor experiment was established in Průhonice, Czech Republic (49°59'41.042"N, 14°33'55.27"E). This common garden has a similar climate to the original location of the *A. arenosa* populations (central Europe), and tetraploid *A. arenosa* grows naturally within a few kilometers of this common garden (Padilla-García et al. 2023). Therefore, we did not expect major changes in the insect visitor community between the common garden and the original locations of the *A. arenosa* populations, even though we were unable to exclude it. Two experimental plots for the purpose of floral visitation observation were established; these two plots were 70 m apart from each other. Each plot consisted of 16 diploid and 16 synthetic tetraploid plants of *A. arenosa* arranged in a regular chessboard pattern, with four blank spaces left in the center of the square design (Supporting information). Blank spaces were included due to the unavailability of enough healthy representatives of both cytotypes to fill the full square design in both plots. This design of the plot ensured that the representatives of both cytotypes were equally accessible to floral visitors. The distance between a pair of two laterally adjacent plants within a plot was set to 50 cm. The surface of the plots was formed by open soil, as we removed all vegetation prior to the experiment to avoid interaction of the experimental plants or floral visitors with other plant species within the plot.

Plants were moved to the experimental plots from the cultivation chamber prior to anthesis, typically once they had formed a well-developed basal rosette. The plants were buried in their original pots in the ground according to the scheme in the Supporting information; we did so as planting them in the common garden substrate may have damaged their root system and impacted their subsequent performance.

All experimental plants within both plots were treated with identical watering management. From the time of establishing the plots until the time of flowering, the plants showed individual differences in growth vigor, and we classified them as small, medium or large; we labelled as 'small' and 'large' any plants that were obvious and extreme outliers in their growth (reduced and increased, respectively) by simple observation, and labelled as medium the other plants. We did not move extreme plants to different positions in the plots.

Data collection

To assess visitor foraging, the visit duration (i.e. time spent per plant) and visitation sequence (i.e. the identity of plants visited during a visitation bout) for each individual visitor were recorded. We used a digital stopwatch to estimate visit duration, measured as the time a floral visitor spent on a plant before moving to another plant in the plot, and we recorded the visitation sequence until a floral visitor left the experimental plot. These measurements were done 'blindly' without knowledge of the ploidy of the plants to avoid any bias related to collector expectations. For each plot, three observation periods with at least 30 visitations each were set across a time range of 2.5 weeks (17 June–5 July 2022) between 11:00 and 16:00. The date and hour of the visitor observations, as well as weather conditions, were recorded.

Analysis of insect visitation

To determine whether floral visitors preferentially spend more time visiting plants of a given ploidy level, we used a linear mixed effects model with visit duration (natural log transformed) as the response variable, ploidy, plant size (small, medium or large) and plot as fixed effects, and bout ID and plant ID as random effects. The model was fit using the package 'lme4' (Bates et al. 2015), and the statistical significance of the fixed effects was calculated with 'lmerTest' (Kuznetsova et al. 2017) in R ver. 4.2.2 (www.r-project.org).

To test whether floral visitors sequentially visited plants independently of ploidy level, we calculated a 2×2 transition matrix for all visitor movements between two plants within a single pollination bout. This transition matrix was calculated for each of the two plots separately. We used a X^2 test to evaluate the hypotheses that transitions between ploidy levels are 1) dependent only on the frequency of plants of a given ploidy in the plot, and 2) a function of the frequency of plants of a given ploidy considering only the nearest neighbours (plants adjacent to a focal plant). Due to the experimental design, each plant was surrounded by a plant from the opposite ploidy in the vertical and horizontal axes, and by a plant of the same ploidy along the diagonals (Supporting information). Thus, we used a correction factor when calculating the expected transition matrix to account for the fact that plants along the diagonal (same ploidy) are further away than those along the vertical and horizontal axes (different ploidy). From the Pythagoras theorem, the distance along the axes is equal to $h = \text{square root}(2) \times c$, where c is the horizontal and vertical distance among plants, and h is the diagonal distance. Finally, we also calculated [Gegear](#)

and Thomson (2004) constancy index CI, where values range from -1 (complete inconstancy) to 0 (random foraging) to $+1$ (complete constancy).

Plant traits measurements

All the plants used for the phenotypic analyses (floral traits, fruit set and progeny characterization) were also included in the pollinator experiment.

Floral traits

Ten randomly selected plants of both ploidy levels were used for morphometric measurements of floral traits. We collected three well-developed flowers per individual plant (total 60 flowers). Immediately after collecting, individual flower parts, i.e. petals, sepals, stamens and pistil, were separated and attached onto a paper sheet of contrasting color with transparent adhesive tape. Sheets with flower parts were scanned with an attached scale bar. From the scanned material, and two weeks after collection, seven floral traits comprising all floral organs were measured on each analyzed flower using ImageJ 1.53e (Schneider et al. 2012): petal length, petal width, sepal length, sepal width, pistil length, filament length and anther length. For measurements of each morphological trait, we chose the longest flower part within a particular flower (i.e. the longest petal, sepal, filament and anther). As flowers are not biological replicates, we averaged the values of three flowers to obtain a single value per plant, and this was used for statistical analyses.

A Shapiro–Wilk test and Mardia's skewness and kurtosis test were used to determine multivariate normality of the dataset comprising the seven morphological traits. A MANOVA was used to test the significance of morphological difference between plants of the two ploidy levels, followed by the post hoc univariate ANOVA tests to determine the differentiation among each trait individually.

Statistics were performed in R ver. 4.2.2 (www.r-project.org) utilizing the packages 'mvnrmtest' (Jarek 2012) and 'mvnrmTest' (Zhou and Shao 2014), while the package 'ggplot2' (Wickham 2016) was used for creation of box plots.

Fruit set

At the end of the flowering period, we measured the total number of peduncles and the number of developed siliques (longer than 1 cm) on 20 randomly selected plants per ploidy out of all the open-pollinated plants of the experimental plot. The measurements were done in a double-blind manner to avoid any bias. The relative fruit set was calculated by dividing the total number of developed siliques by the total number of peduncles for each plant.

A Wilcoxon rank sum test was used to compare the relative fruit set between diploids and tetraploids using the 'R Stats' package.

Seed progeny germination rates

Open pollinated, mature siliques were collected from five mother plants in the experimental plot per cytotype and

kept in a paper bag. Seeds collected from these siliques were arranged on white paper and were photographed under a stereo microscope (Leica M205 C with Fusion Optics). Normal-looking seeds were identified as brownish-yellow and plump, with variable sizes. Collapsed seeds were dark brown/black, shriveled and irregularly shaped. Counting seeds for each category was done using ImageJ (Schneider et al. 2012). The rate of normal-looking seeds was obtained by dividing the number of normal-looking seeds by the total number of seeds.

After phenotype assessment, seeds were surface sterilized (with 0.5% NaClO and 0.01% (v/v) of Triton X-100) and plated on MS medium (containing: 1 × MS-Salts and 0.8% (w/v) plant agar, at pH 5.8). Then, plates were transferred and kept for two weeks in a growth chamber (12 h at 23°C/day and 12 h at 13°C/night). Germination rates were assessed by counting seeds with a ruptured seed coat and protruding radicles.

A Wilcoxon rank sum test was used as non-parametric test to compare the rate of normal-looking seeds/germination between diploids and tetraploids using the 'R Stats' package.

Seedling progeny ploidy measurements

Following seed germination, we investigated the ploidy level of germinated progenies (three-week-old seedlings). To do this, we collected rosette leaves of 10–15 seedlings per mother plant (five mother plants) for each cytotype and measured the ploidy level following the same procedure as explained in the 'Ploidy level detection' section.

Results

Larger floral organs in synthetic polyploids

The Shapiro–Wilk and Mardia's skewness and kurtosis test confirmed multivariate normality of the floral dataset. A MANOVA revealed significant morphological differentiation between plants of the two ploidy levels ($F(7,12) = 18.85$, $p < 0.0005$; $n = 10$ plants per cytotype), and the inter-cytotype difference in individual traits was consistently confirmed also in post hoc tests, with p-values below 0.005 in all seven variables (Fig. 1). On average, flower parts of tetraploids were 38% longer or wider than diploids. The most significant difference was observed in petal width (on average, tetraploids had 1.8 times wider petals compared to diploids), while the smallest relative difference was observed in filament length, with tetraploids having on average 1.2 times longer filaments than diploids. The number of flowers opened at the same time could influence overall floral display and attractiveness to visitors but did not significantly differ between diploids and tetraploids (average of 10.9 flowers for diploids and 13.4 for tetraploids).

Higher mating success in synthetic polyploids

The number of fruits produced per plant was significantly higher in induced autotetraploids than in diploids (Wilcoxon rank sum test [W] = 97, p -value = 0.009338, $n = 19$ –20 per ploidy). A similar trend was observed for the flower number between cytotypes, but the differences were not significant ($W = 123.5$, p -value = 0.06365, $n = 19$ –20 per ploidy).

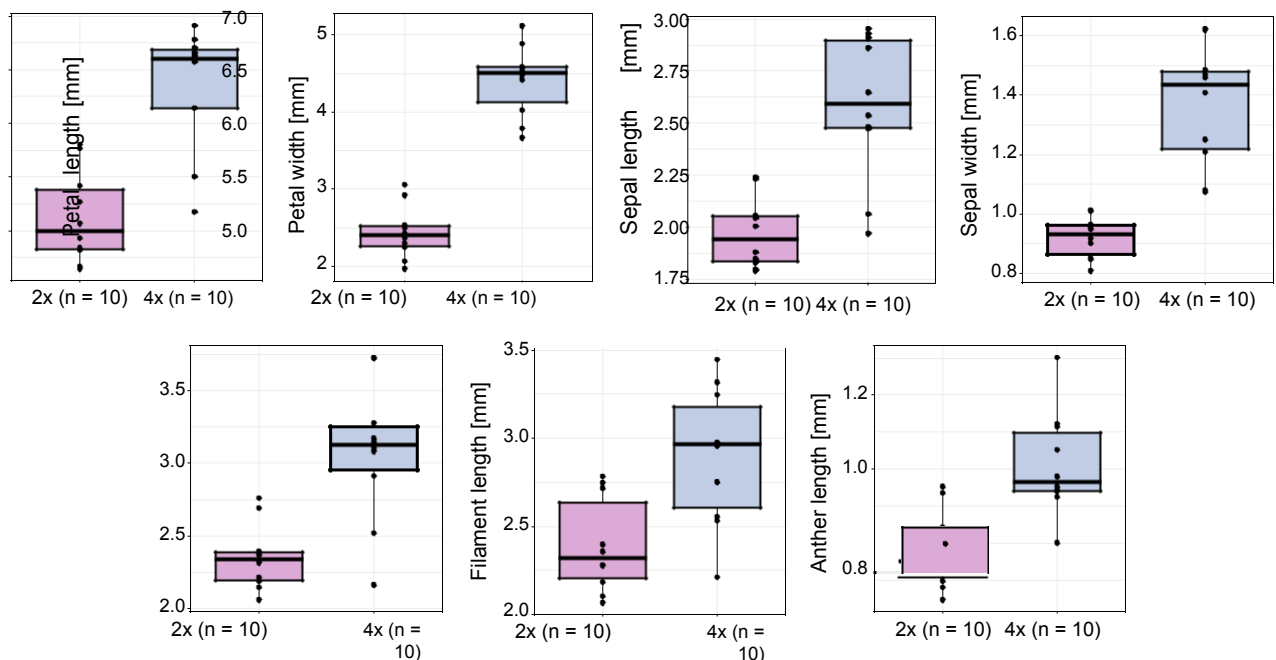


Figure 1. Box plots displaying the variability of the morphological traits in diploid and induced autotetraploid *Arabidopsis arenosa*. The number of individuals (n) is shown for each cytotype.

This resulted in induced autotetraploids showing a significantly higher relative fruit set (64.2%) compared to diploids (50.34%; $W = 76.5$, $p\text{-value} = 0.001482$, $n = 19\text{--}20$ per ploidy; Fig. 2).

Floral visitation to diploids and tetraploids

We found a diversity of insects visiting the plants, mostly species of bees and flies (Supporting information). We found no statistically significant difference in the amount of time that floral visitors spent on plants of different ploidy levels ($n = 162\text{--}169$ visitors per ploidy; Table 1, Fig. 3). On average, floral visitors spent 36.494 ± 46.437 s (mean $\pm$ SD) and 30.048 ± 33.613 s in diploid versus tetraploid plants, respectively. In contrast, plant size significantly influenced the time floral visitors spent on a plant, with larger plants receiving overall longer visits (Table 1).

Movement of floral visitors among plants was highly inconstant ($CI = -0.637$ and -0.710 for plots A and B, respectively), suggesting that visitors moved mostly between, rather than within, ploidy levels (Supporting information). Both χ^2 tests using either the frequency of ploidies in each plot ($\chi^2 = 107.04$ and 150.41 for plots A and B, respectively;

$df = 1$, $p < 0.001$), or the corrected expectation based on ploidy-specific distances to the nearest neighbor ($\chi^2 = 60.287$ and 93.568 for plots A and B, respectively; $df = 3$, $p < 0.001$) support visit sequences that are not independent of ploidy level.

Progeny of open-pollinated plants

Mature seeds from open-pollinated diploid and tetraploid plants were collected. The rate of normal-looking seeds, potentially indicating the presence of triploid seeds, was of 47% in average for diploid plants and 13% for tetraploid plants ($n = 5$ maternal plants, total 431 seeds for diploid plants and 409 seeds for tetraploids; Fig. 4a, Supporting information). This rate was significantly different between cytotypes ($W = 25$, $p\text{-value} = 0.01219$). The germination rate, however, was not significantly different between cytotypes, with a mean of 33% for diploid plants and 36% for tetraploid plants ($W = 12$, $p\text{-value} = 0.1$, $n = 5$ maternal plants, 315 seeds for diploids and 291 seeds for tetraploids; Fig. 4a, Supporting information). Strikingly, while only diploid seedlings were obtained from germinated seeds of diploid maternal plants, germinated seeds from tetraploid maternal plants

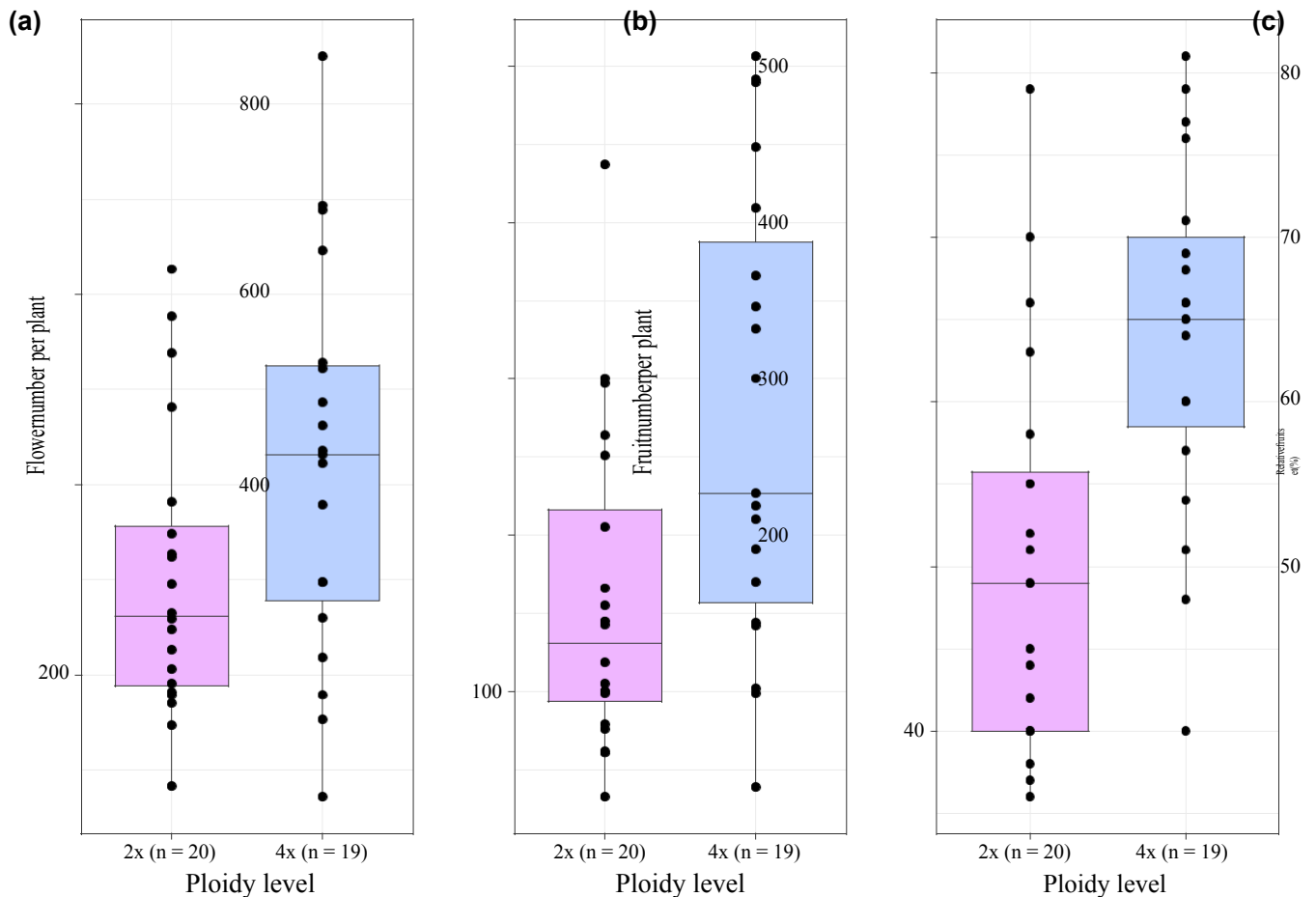


Figure 2. Reproductive success of diploid and induced autotetraploid *Arabidopsis arenosa*. The total number of flowers (a), the number of fruits per plant (b), and the relative fruit set per plant (c) is indicated for each cytotype, together with the number of individuals (n).

Table 1. Coefficient estimates, standard errors and statistical significance of main effects for pollinator visit durations.

	Estimate	SE	p
Intercept	3.113	0.093	
Ploidy			0.236
4x	−0.112	0.094	
Size			< 0.001
Large	0.319	0.160	
Medium	−0.314	0.104	
Small	−0.686	0.157	
Plot			0.940
B	0.007	0.097	

gave rise to a vast majority of triploid seedlings (55 out of 56 seedlings, $n = 5$ maternal plants; Fig. 4b).

Discussion

The early acting factors contributing to the establishment of new polyploids remain largely unknown. These factors may either influence the fitness of newly formed polyploid individuals and/or build reproductive isolation between cytotypes, leading to the emergence of genetically independent polyploid lineages. In this study, using synthetic *A. arenosa* tetraploids, we tested whether instant floral changes due to polyploidization could improve reproductive success and could lead to reproductive isolation between diploids and tetraploids, in the form of assortative mating mediated by changes in insect visitation.

Polyploidy-induced floral changes are not more attractive to visitors, but do improve mating success

As often reported in the literature, we found that synthetic polyploids had larger organs, including petals (Abdoli et al. 2013, Zahumenická et al. 2018, Meeus et al. 2020), an important trait for pollinator attraction (Moyroud and

Glover 2017). However, this did not translate into preferential visitation of tetraploids by insects in our study. This finding is consistent with the few available studies on this topic, yet in different plant families, in the species *A. mangium*, *C. angustifolium* and *T. pratense* (Nghiem et al. 2011, Husband et al. 2016, Porturas and Segraves 2020), which also did not find any impact of induced polyploidy on insect visitation. However, these studies did not test the quantitative preference of visitors (visit duration) between cytotypes but instead tested if assortative mating occurred, acting as an interploidy reproductive barrier. Here, we show that the floral changes induced by polyploidy in *A. arenosa* do not lead to differences in visit duration between diploids and tetraploids. These results, accompanied by previous results of the studies mentioned above, have not provided evidence to support the hypothesis that instant floral changes provoked by polyploidization contribute to reproductive isolation of new polyploids from their diploid progenitors. With this in mind, we must consider that synthetic polyploids may not perfectly emulate the formation of neopolyploids in nature. In particular, the formation of unreduced gametes in nature and subsequent outcrossing may capture more allelic diversity than with a somatic doubling induced by chemicals (De Storme and Geelen 2013). This may have consequences for the emergence of novel floral phenotypes in natural neopolyploids that may have stronger impact on visitor behavior which we cannot recapitulate in our study. For future research, working with species that are prone to unreduced gametes formation, such as *Cardamine* species (Kreiner et al. 2017), to produce neopolyploids may be a way to mimic better what happens in nature.

Despite no clear visitor preference, we found that the tetraploids set more fruits (both total number and relative fruit set) than the diploids, suggesting higher pollination rates. A higher pollination rate might still be explained by higher visitation rates, as visitations also happened outside our observation period. We performed the visitor observations only for 2.5 weeks, while the flowering period of *A. arenosa* is approximately two months. Therefore, the insect visitor community could have changed during the flowering period, and the tetraploids might have attracted specific visitors outside the time windows of the measurements. Also, we did not test for different foraging behavior between visitors as we did not identify pollinator species in detail, and it could be that specific visitor species were more attracted to tetraploids than others. Alternatively, the size of floral organs like the style has been shown to influence the efficiency of pollen transfer from the pollinator body to the stigma (Mazer et al. 2019), and it could be that the larger floral organs (including the pistil) of synthetic polyploids improved the pollen transfer and, consequently, increased fruit set. More generally, we studied visitors and not actual pollinators, and differences in fruit set between cytotypes could be due to cytotype-specific foraging behaviors/pollen transfer efficiency of certain insect visitors. Polyploidization could lead to the breakdown of self-incompatibility (SI; Ramsey and Schemske 1998, Mable 2004), and thus higher fruit set could be explained

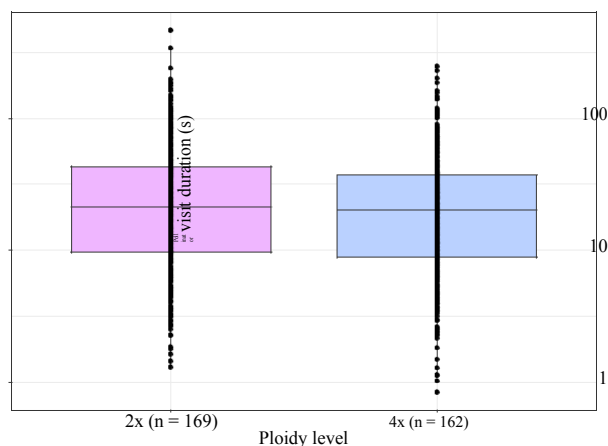


Figure 3. Box plots of visit duration of diploid and induced autotetraploid *Arabidopsis arenosa*. Duration values in seconds (s) are $\log_{10}$ -transformed for readability purposes. The total number of pollinator individuals (n) is indicated for each plant cytotype.

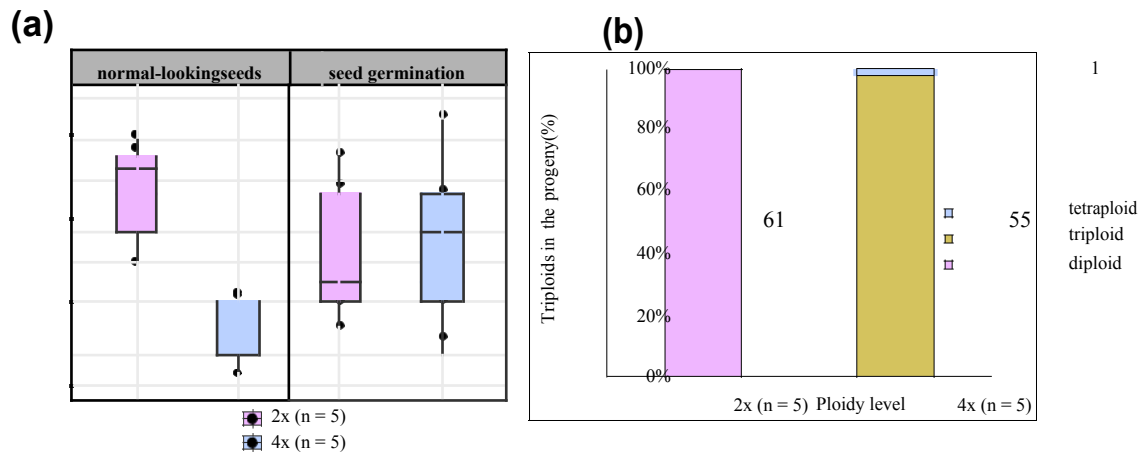


Figure 4. Progeny assessment of diploid and induced autotetraploid *Arabidopsis arenosa* mother plants. (a) Number of normal-looking seeds obtained from diploid (n=431 seeds) and induced autotetraploid (n=409 seeds) *A. arenosa* mother plants; seed germination rate among the diploid (n=315 seeds) and induced autotetraploid (n=291 seeds). Dots represent the individual mother plants (n=5 mother plants per cytotype). (b) Proportion of triploid seedlings in the progeny of diploid and tetraploid mother plants. The number of mother plants (n) is shown for each cytotype. The numbers inside the bars indicate the number of seedlings in the progeny matching a given ploidy level (diploid: pink; triploid: brown; tetraploid: blue).

by selfing. However, SI breakdown by polyploidy occurs in the gametophytic SI system and not for the sporophytic one (Mable 2004). Combined with the fact that *Arabidopsis* species have a sporophytic SI system (Kusaba et al. 2001), and that natural *A. arenosa* tetraploids are self-incompatible (Kiefer et al. 2023), selfing is unlikely to explain our results. In addition, the vast majority of the progeny in tetraploid plants were triploid, which is inconsistent with a selfing scenario. Independently of the mechanisms that led to higher fruit set in tetraploids as compared to diploids, the fitness of tetraploids may not increase as a result, as triploids may have reduced fertility as often reported (Farco and Dematteis 2014, Morgan et al. 2021). Further studies should be performed to understand the causes of higher fruit set in synthetic *A. arenosa* tetraploids.

Polyploidy-induced floral changes may promote, rather than prevent, interploidy gene flow

We also tested whether floral changes provoked by polyploidization could lead to assortative mating via pollinators preferentially visiting plant individuals of the same cytotype rather than between cytotypes. We did not find evidence for assortative mating, consistent with previous studies on synthetic polyploids and their diploid progenitors (Nghiem et al. 2011, Husband et al. 2016, Porturas and Segraves 2020). Several other studies, however, focusing on natural polyploids, found partially different pollinator preferences in diploids versus polyploids (Segraves and Thompson 1999, Husband and Schemske 2000a, Thompson and Merg 2008), suggesting that pollinator shift is not acting as an early reproductive barrier causing the genetic divergence between diploid and polyploid lineages, but rather coming into play at a later stage. Importantly, the chessboard design we used, while helpful to normalize the probability of visitor

visitation between cytotypes, is not representative of natural conditions. Tetraploids are likely to be grouped spatially in their early emergence and later, and the patchy distribution of cytotypes has been shown to influence between-cytotype pollinator visitation (Husband and Schemske 2000b). In general, the size of floral patches has been shown to influence the amount of visitation as well as the pollinator movement between patches (Santos et al. 2021). It would thus be interesting to carry out an experiment closer to natural conditions of the emergence of neopolyploids, with small patches of polyploids interspersed with larger patches of diploids, and to observe the impact of this design on pollinator behavior.

Although polyploidy-induced floral changes might eventually lead to a reproductive barrier between ploidy levels, our results suggest that they may alternatively promote inter-ploidy gene flow. Indeed, we observed that visitors preferentially moved between cytotypes rather than within cytotypes. While this could be attributed to the chessboard design, which places a heteroploid plant as the closest one, we still observed a preferential movement between cytotypes even after statistically correcting for this spatial bias. Although puzzling, this type of disassortative mating in mixed ploidy experimental plots with similar experimental design as ours has been observed before. A study in *C. angustifolium* using a different statistical approach to measure pollinator movements also found that pollinators were more likely to move between plants of different cytotypes rather than within the same cytotype (Husband et al. 2016). The biological explanation for this behavior remains obscure. However, taken at face value, our results and those of Husband et al. (2016) suggest that this kind of disassortative mating could, on one hand, hinder the establishment of neopolyploids through exacerbating the effects of minority cytotype exclusion, which disfavors the initially rare polyploid cytotype. On the other hand, if pollinators preferentially move between diploids and

neopolyploids, this would promote interploidy gene flow and the formation of a triploid bridge (Harlan and deWet 1975, Husband 2004), potentially helping neopolyploids to establish.

The scenario of a triploid bridge is somewhat supported by the vast majority of the germinated seeds from tetraploid plants being triploid, likely a consequence of the visitors preferentially switching between cytotypes. A very similar design to the present study was performed in the same location, growing diploid and natural tetraploid *A. arenosa* in a common garden in a chessboard manner (Morgan et al. 2021). The results were opposite to ours: less than 10% of seeds from natural tetra-ploid plants were triploids, the rest were tetraploid. This suggests that since the origin of tetraploids 11 000–30 000 or 226 000 generations ago (Arnold et al. 2015, Marburger et al. 2019) until now, additional reproductive barriers arose, preventing the formation of triploid seeds even when diploid and tetraploid plants are in direct contact. These barriers could be pollinator preferences and/or postmating prezygotic barriers such as pollen rejection. On the other hand, while we did not find triploids in the germinated progeny of diploid plants in the current study, it does not necessarily mean that no triploid seeds were formed: crosses performed between diploid maternal plants and tetraploid pollen donors in *A. arenosa* produced triploid seeds, but > 90% of these seeds were inviable (triploid seeds are mostly viable in the other cross direction, tetraploid mother × diploid father; Morgan et al. 2021). In our study, seeds from diploid plants germinated to a rate of 33%. It is likely that triploid seeds largely contributed to the remaining 67% that did not germinate.

In conclusion, our study suggests that floral changes upon polyploidization impact pollinator behavior and mating success in ways that are more complex than a mere increase in insect visitation or the rise of an interploidy reproductive barrier. Ploidy-induced floral changes increased visitation between cytotypes, suggesting the opposite of a reproductive barrier, and higher reproductive success of the induced tetraploids was found, implying a pervasive fitness effect of whole genome duplication. Future works may disentangle the mechanisms underlying reproductive success of newly formed polyploids.

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Data availability statement

Data are available from the Figshare Digital Repository: <https://doi.org/10.6084/m9.figshare.23302055.v1> (Schmickl et al. 2024).

Supporting information

The Supporting information associated with this article is available with the online version.

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Case study 3

ORIGINAL ARTICLE

Divergent evolutionary trajectories of male and female investment in the Brassicaceae

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Background and Aims Plants have colonised diverse habitats, with their ecological success depending on key adaptive traits. Among these, reproductive characteristics such as pollen and ovule production are critical. The allocation to sexual reproduction is likely shaped by diverse selective pressures and constraints. This study investigates the evolution of pollen and ovule production and its drivers in the Brassicaceae family.

Methods To achieve this, we analysed 158 Brassicaceae species, recorded pollen and ovule numbers, and used phylogenetic analyses to characterise the evolution of these traits. We then cultivated a subset of 66 species in a common garden and tested the association of pollen and ovule production with environmental conditions of original habitats.

Key Results & Conclusions Our results reveal that ovule production is strongly constrained by phylogeny, whereas pollen production is more evolutionarily flexible. Expectedly, recurrent mating system shifts played a role in this observation, as selfing species consistently produced fewer pollen grains but did not show significant changes in ovule number, supporting selection for reduced male investment in response to increased mating efficiency. In contrast, environmental associations with both traits were limited yet significant, with pollen and ovule production evolving according to different environmental pressures. Polyploidy did not influence the evolution of these traits. These phylogeny-informed findings refine our understanding of the evolutionary drivers of sexual allocation, demonstrating that male and female investment evolve under distinct selective pressures in hermaphroditic species.

Keywords: Sexual allocation, evolutionary response, phylogenetic constraints, mating system, ploidy level, pollinator abundance, Brassicaceae

INTRODUCTION

The success of an organism depends on how effectively it allocates limited resources for growth, survival, and reproduction, as defined by the resource allocation theory (Charnov, 1982; Gillet and Gregorius, 2020). Sexual reproduction, in particular, involves trade-offs between male and female investment, influencing species-specific reproductive strategies (Charnov, 1982; Delph, 1999). In plants, reproductive investment involves balancing pollen and ovule production costs, which require different amounts of energy. Ovules are thought to require greater maternal resources, so their production is more limited, whereas pollen is likely produced at a lower cost (Charnov, 1982). Consistently, simulations indicate that successful fertilisation requires greater reproductive investment in ovules than pollen, reinforcing that pollen is less costly to produce (Gillet & Gregorius, 2020). In addition, calorimetry in *Amaryllis vittata* showed that the energetic investment in total pollen and ovule per flower was comparable (Smith & Evenson, 1978), despite a greater number of pollen grains than ovules, and thus a lower energetic cost per pollen grain. Consequently, selection may favour different investment strategies for the male and female function. Species exhibit a wide range of ovule numbers, from uniovular taxa to those producing thousands per flower (Burd et al., 2009; Friedman and Barrett, 2011; Wetzstein et al., 2013). This variation reflects an evolutionary trade-off between offspring quantity and quality: producing more ovules reduces per-seed resource allocation, thereby influencing offspring fitness and survival (Baker, 1972; Greenway & Harder, 2007). Although pollen production is energetically cheaper, excessive investment leads to diminishing returns due to inefficient pollen transfer (Harder & Johnson, 2025), as only a small fraction reaches stigmas (Minnaar et al., 2019). Short-term plasticity studies suggest that pollen production, however cheap, is still limited by environmental resources. For

instance, Wilson Rankin et al. (2020) found that *Trifolium willdenovii* reduced pollen output under drought stress. Therefore, in resource-rich environments, selection may favour plants that invest more in pollen production, increasing pollination success and fertilisation rates. Conversely, in resource-limited environments, excessive pollen production may be disadvantageous, as its effect on mating success is limited due to low pollen transfer efficiency (Minnaar et al., 2019) and the energy cost would not compensate for the toll on survival. In these conditions, natural selection would favour plants that reduce pollen production, as lower reproductive costs improve survival and long-term reproductive success (Obeso, 2002; Edward and Chapman, 2011). Nevertheless, these expectations are contradicted by a recent modelling study showing that the evolutionarily optimal pollen production was not affected by the amount of resources available for reproduction (Bochynek & Burd, 2024), calling for additional investigation. Ovule production may follow a clearer trend, with greater investment in response to higher resource availability to maximise reproductive potential. Consistently, in *Embothrium coccineum* (Proteaceae), ovule production was positively associated with rainfall (Chalcoff and Aizen, 2016). The evolutionary pressures shaping pollen and ovule production may thus not be identical, as ovules require direct maternal investment in offspring (Yadegari & Drews, 2004; Phillips & Evans, 2020), while pollen disperses to achieve fertilisation (Harder & Johnson, 2008; Ollerton et al., 2011). This fundamental difference in parental investment suggests that both traits may respond to environmental constraints in different ways, but this remains to be tested.

These distinct trajectories are also reflected in the factors influencing male and female reproductive success. Bateman's principle predicts that male reproductive success is mating-limited, while female success is

resource-constrained (Bateman, 1948; Moore & Pannell, 2011; Collet et al., 2014). In plants, this means that increased pollen transfer enhances pollen transfer, as even a small pollen load can fertilise all ovules (Elle & Meagher, 2000; Collet et al., 2014). Since only ~5% of pollen reaches conspecific stigmas (Gong & Huang, 2014; Minnaar et al., 2019), selection may favour increased pollen production to compensate for inefficient transfer. Consistently, empirical studies have shown a negative relationship between pollen production and transfer efficiency (Gong & Huang, 2014). Another study indicates that pollen production is more responsive to pollinator availability (Harder & Johnson, 2023), whereas ovule investment may be constrained by maternal resource allocation. Thus, in pollinator-scarce habitats, plants may produce more pollen as a consequence of selection on the male function for increasing mating opportunities. Interestingly, these observations and expectations collide with a recent modelling study suggesting that in large panmictic populations at least, pollen production does not evolve to increase to compensate for low transfer efficiency even under pollen limitation (Bochynek & Burd, 2024), calling for further investigation.

Female reproductive success is often maximised with a single visit, making ovule production less responsive to pollinator availability. This is however different in the case of pollen limitation, and some researchers proposed ovule overproduction as a bet-hedging strategy against pollen receipt variability (Friedman and Barrett, 2009, 2011): in pollinator-scarce environments, uncertain pollen transfer increases reproductive failure risk, and thus producing more ovules may ensure seed production, particularly in fluctuating pollinator populations, seasonal climates, or high-elevation habitats. Empirical studies yield some evidence in this direction. Burd et al. (2009) found ovule number positively correlated with stochastic pollination, suggesting selection for increased ovule production in unpredictable environments.

mostly male reproductive success (Queller, 1984), whereas female reproductive success remains stable independently of. Similarly, Arroyo et al. (2019) found an increase in ovule number in central Chilean alpine plants, supporting the idea that plants in high-altitude environments, where pollinator visits are less predictable, may produce a high number of ovules as a bet-hedging strategy. These observations point towards pollen limitation as a cause of female reproductive investment strategies to prevent reduced seed output. As an alternative strategy, the transition to selfing may ensure reproductive assurance despite limited mating opportunities, such as in populations at species range edges (Griffin & Willi, 2014). Convergently across angiosperm species, the transition to selfing is accompanied by a reduction in pollen production (Cruden, 1977; Tsuchimatsu and Fujii, 2022), assumed to be a consequence of efficient pollen transfer and thus relaxed selection on pollen production. To the contrary, the transition to selfing does not seem to have an impact on ovule production (Harder & Johnson, 2023).

Altogether, the impact of resource availability and pollination uncertainty on the evolution of male and female investment remains poorly understood in plants beyond specific cases. While past studies suggest that pollen and ovule production respond to both abiotic and biotic selection pressures, the relative influence of these factors in shaping reproductive allocation evolution is unresolved. Moreover, the extent to which the evolution of pollen and ovule production responds differently to these factors remains to be explored. Finally, a phylogeny-informed approach to these questions could give a broader view on the evolution of these traits. To address these gaps, we focus on the Brassicaceae family, which shows variation in habitat preference, mating system and ploidy level, and a well-studied phylogeny (Nikolov, 2019; Al-Shehbaz, 2025). In this study, we investigate: 1) how constrained the evolution of pollen and ovule production is, 2) whether pollen and ovule production show an

evolutionary response to environmental resources, and 3) how pollination assurance (via pollinator availability and transition to selfing) influences the evolution of these traits. Overall, we investigate how evolutionary constraints shape trait diversity by examining the interactions between stabilising selection, ecological pressures, and phylogenetic history in influencing reproductive investment in plants.

MATERIAL AND METHODS

Species sampling

For the phylogenetic analyses, pollen and ovule numbers were retrieved for 158 species from the Brassicaceae, spanning 71 genera and 29 tribes. These data originated from two different sources: our measurements (see below) and the literature (Supplementary Dataset S1).

For the association between environment and reproductive traits, a total of 66 species were cultivated at the Heidelberg Botanic Garden. For each cultivated species, two populations were selected, with two or three individuals per population (Supplementary Dataset S2).

Plant material and measurements

For the 66 cultivated species, all plants were grown in outdoor greenhouses in Heidelberg University facilities (Germany) under non-limiting water and nutrient conditions. Perennial species were cultivated from spring 2018 and harvested for flower buds between September 2019 and March 2021. Annuals were cultivated for one year maximum prior to the same harvesting period. Flower buds close to anthesis were collected from the top part of the peduncle (three buds per individual), transferred to a fixing solution of ethanol:acetic acid (9:1), and kept at 4°C overnight. The fixed buds were then washed with 90% ethanol and stored in 70% ethanol at 4°C. In addition, buds from the same individuals were collected in teabags and dried, providing supplementary material for further analyses. Among the cultivated plants, 43

species, encompassing 25 genera and 14 tribes, had detailed locality data that were used for trait-environment associations (for a map showing the localities, see Supplementary Fig. S1).

For ovule number measurements, flower buds stored in 70% ethanol were transferred to a clearing solution (66.7% chloral hydrate, 25% double-distilled water, 8.3% glycerol) to render tissues transparent. The pistils were carefully dissected from the remaining floral parts (sepals, petals, anthers) and mounted on microscope slides. The ovules in each ovary were counted using a microscope equipped with differential interference contrast. Three buds per individual were measured to account for within-plant variation in ovule number.

For pollen measurements, dried flower buds (previously stored in teabags) were individually placed into 2 mL microtubes containing 95% sulfuric acid (Analytical Grade, PENTA CHEMICALS) and incubated at 37°C for ten days, dissolving all tissues except for pollen. After incubation, the tubes were centrifuged at 6000 rpm using an Eppendorf Centrifuge 5424 R (Eppendorf), and the supernatant was discarded. The pellet was resuspended in 1.5 mL of 0.01% (v/v) Triton™ X-100 (Serva Electrophoresis GmbH), and this washing step was repeated three times to remove any residual acid. The final pellet was then resuspended in 100 µL of Triton™ X-100 solution. After vortexing, 1 µL of the suspension was transferred to a BRAND® counting chamber, BLAUBRAND® Neubauer improved (Sigma-Aldrich). Under the microscope, pollen grains were counted in five 0.1-µL squares. The total pollen count per 100 µL was then calculated by multiplying by 10 the average number of pollen grains per square (to reach an average number of pollen in 1 µL) and then multiplying the result by 100 (to adjust from 1 µL to 100 µL). Three buds per individual were measured to account for technical variation in pollen number.

Abiotic and biotic environmental data

The geographic positions of species occurrences were extracted from the original collection records of the seeds from which the plants had been grown from the database of the Heidelberg Botanic Garden and Herbarium (HEID). Subsequently, the species occurrence data were complemented with altitude and climate data extracted from the WorldClim database version 2.1 (Fick and Hijmans, 2017). In particular, we included the data of mean annual temperature, annual precipitation, temperature seasonality, precipitation seasonality, and precipitation of the driest quarter, as these represent the main climatic gradients influencing many plant reproductive traits (Zhao et al., 2018; Huang et al., 2021; Dorji et al., 2020). We calculated two additional climatic metrics that are closely related to plant physiology and distribution (Title and Bemmels, 2018): growing degree days, which is the sum of mean monthly temperatures for months with a mean temperature $>5^{\circ}\text{C}$ multiplied by the number of days, and the climatic moisture index, a metric of relative wetness and aridity representing the difference between annual precipitation and potential evapotranspiration. Besides climatic variables, data on soil nutrient availability and rooting conditions were extracted from the Harmonised World Soil Database version 1.2 (Fischer et al., 2008). This database provides globally harmonised information on key soil properties such as organic carbon content, cation exchange capacity (critical for assessing nutrient availability), soil texture, bulk density, coarse fragments, and soil phases (influencing root penetration, soil depth, and soil volume). These factors play a fundamental role in shaping plant reproductive traits by influencing resource acquisition, water retention, and root development, ultimately affecting plant fitness and adaptation across diverse environments (Wieder et al., 2015; Batjes, 2016; Lehmann & Kleber, 2015).

Each species record was also matched with a potential number of pollinators. We used the relative species richness map of bees

(Anthophila) (Orr et al., 2021) to extract information on the potential number of pollinator species dwelling in a given spot. All abiotic and biotic characteristics were combined in a matrix of predictor variables. We screened the matrix for pairwise correlations and iteratively removed redundant variables from the dataset. In each pair of variables showing an absolute correlation $r > 0.65$, the more informative variable was retained until all pairwise correlations were less than the threshold. The subset used in the analyses involved altitude, temperature seasonality, precipitation seasonality, climatic moisture index, nutrient availability, rooting conditions, and number of pollinators.

Mating system, ploidy level and lifespan data

Mating system classifications, either self-fertilisation or outcrossing, were compiled from the literature and the FloraWeb database (German Federal Agency for Nature Conservation [BfN]; www.floraweb.de), with full details provided in the supplementary material (Dataset S1). We considered the categories “obligate selfer” and “facultative selfer” as selfing and “obligate outcrosser” and “facultative outcrosser” as outcrossing. Ploidy levels (coded as diploid and polyploid) were sourced from the *BrassiBase* database (Koch et al., 2012; Kiefer et al., 2014; Koch et al., 2018; <https://brassibase.cos.uni-heidelberg.de/bb2/>). Species classified as diploid or mesopolyploid were coded as diploids, whereas mesoneopolyploids and neopolyploids were coded as polyploids (Lysak and Koch, 2011). Additional ploidy data were obtained from the literature that estimated ploidy levels using the “Stebbins fraction”, a metric reflecting reproductive strategies (Román-Palacios et al., 2020). Moreover, the lifespan (annual, biennial, or perennial) information was extracted from *BrassiBase*.

Phylogenetic reconstruction

To investigate the evolutionary history of traits, we used a species-level dataset comprising pollen number and ovule number (see above).

For our phylogenetic comparative analyses, we employed a 1004-species tree of the Brassicaceae, including eight outgroup taxa from the order Brassicales. Taxonomic treatment follows *BrassiBase* version 2.0.1 (<https://brassibase.cos.uni-heidelberg.de/bb2/>). We compiled sequences of the internal transcribed spacers (ITS1 and ITS2) of nuclear ribosomal DNA, including the 5.8S gene, from *BrassiBase* and added sequences from the NCBI nucleotide database to increase our taxonomic sampling. We also collected sequences of the plastid *trnL/trnF* region and complete plastome sequences from the NCBI nucleotide database. ITS1, 5.8S, and ITS2 were aligned with MUSCLE5 (Edgar, 2022). For the alignment of the *trnL/trnF* region and complete plastome sequences, we used MAFFT version 7 (Katoh & Standley, 2013) to facilitate the alignment of the long plastome sequences. The alignments were trimmed using trimAL v1.4.rev15 (Capella-Gutiérrez et al., 2009), removing positions with gaps if $\geq 30\%$ of the accessions showed the gap. Additionally, a few adjustments were made manually in AliView version 1.30 (Larsson, 2014). In particular, in a few cases, unalignable sequence ends bordering indels were removed. Trimming completely removed the *trnL-trnF* intergenic spacer that comprised pseudogenes (Koch et al., 2005). To find the best substitution model, we used ModelFinder for both regions separately (Kalyaanamoorthy et al., 2017). Then, the two alignments were concatenated, and a maximum likelihood (ML) tree was constructed using IQ-TREE MPI multicore version 2.3.5.cmaple (Chernomor et al., 2016; Minh et al., 2020), inferring branch support using 1000 ultrafast bootstrapping (Hoang et al., 2018). The trimmed alignment for phylogenetic reconstruction and the ML tree are available on OSF (Haghighatnia et al. 2025). This tree was made ultrametric using penalised likelihood in treePL v1.0 (Smith & O'Meara, 2012) using one fossil (*Dressiantha bicarpellata* [93.6–89.3 million years (Ma)]; Gandolfo et al., 1998) and three secondary calibration points (crown

Brassicaceae [25.7–23.1 Ma], crown lineage III [15.42–13.5 Ma], and crown *Arabidopsis* genus [6.84–5.54 Ma]; 95% confidence intervals of the highest posterior density (HPD) values previously estimated for the respective crown ages are given; Hendriks et al., 2023). Molecular dating analyses were then conducted by integrating this primarily dated phylogeny as the starting tree, the *D. bicarpellata* fossil used to calibrate the stem node of the order Brassicales, and secondary calibration points of the crown Brassicaceae and lineage III under an optimised relaxed clock rate (ORC) in BEAST v2.7.7 (Bouckaert et al., 2019; Douglas et al., 2021; Müller & Bouckaert, 2020). The XML file of the run is available on OSF (Haghighatnia et al. 2025). Hereafter, we refer to this dated phylogeny as the main tree.

In parallel, we generated an alternative dated phylogeny incorporating genus-level relationships from the most recent Brassicaceae phylogeny reconstructed by target enrichment of a 1081-gene dataset (Hendriks et al., 2023). To do this, we matched species names between the Hendriks et al. (2023) phylogeny and our ITS-*trnL/trnF* phylogeny, and pruned the tree to retain only genera with at least one shared species. This resulted in a genus-level backbone phylogeny comprising 198 species. We then reconstructed a maximum likelihood phylogeny using IQ-TREE, applying the same substitution models, partition scheme, and branch support estimation framework as in our primary analysis. The constrained topology argument (-g) was used to fix genus-level relationships (based on pruned tree) while allowing phylogenetic placement of the remaining taxa to be optimized across the full dataset. This constrained tree was subsequently made ultrametric using the same penalised likelihood approach and fossil and secondary calibration points as before. We then performed molecular dating analyses in BEAST, maintaining the same criteria described previously. To preserve the genus-level relationships enforced during

tree inference, we incorporated a multi-monophyletic constraint distribution in the BEAST configuration file. This ensured that all genera defined in the constrained topology were treated as monophyletic during tree sampling, while allowing species to be freely placed within their respective constrained genera. Hereafter, we refer to this dated phylogeny as the constrained tree.

To run subsequent analyses, species absent from the dataset were pruned from the time-calibrated trees using the function *drop.tip* in the *ape* package v5.8 (Paradis & Schliep, 2019) in R to exclusively retain species in the phylogeny for which trait data were available (158 species).

Phylogenetic signal and trait evolution

Pollen and ovule numbers were log-transformed ($\log(x + 1)$) for all analyses to address potential skewness and heteroscedasticity and ensure the values remained positive. The phylogenetic signal was estimated using Blomberg's K (Blomberg et al., 2003) and Pagel's λ (Pagel, 1999) with the function *phylosig* implemented in the *phytools* package v2.3-0 (Revell, 2024) in R. To determine which continuous trait model best fits our datasets, we used the *Geiger* package v2.0.11 (Harmon et al., 2008) in R. Three continuous trait models were tested: Brownian Motion (BM), Ornstein–Uhlenbeck (OU), and Early Burst (EB). The BM model assumes that traits evolve randomly over time without constraints, representing neutral drift (Felsenstein, 1985). The OU process can be visualised as particles under Brownian motion experiencing friction. It assumes traits evolve toward an optimal value (adaptive peak), capturing selection signatures (Hansen, 1997). The EB model represents evolutionary scenarios where the rate of trait evolution is highest near the tree's root and decreases over time. This model is associated with adaptive radiation, where traits diversify rapidly early in the evolutionary history before stabilising (Harmon et al., 2010). Once the best-fit model was identified, we estimated the corresponding parameters (e.g., σ^2 , z_0 , and others) based on

the chosen model to better understand the traits' evolutionary dynamics. Ancestral state reconstructions were performed using ML estimation (Harmon, 2019). ML estimations with variances and 95% confidence intervals were conducted using the function *fastAnc* in the *phytools* package. The ancestral states mapping was accomplished by interpolating the estimated states along each edge using Felsenstein's equation [2] (Felsenstein, 1985). The reconstructed ancestral states were visualised on the tree using the function *contMap*.

Phylogenetically independent contrasts

Phylogenetically independent contrasts (PICs) were performed following the method of Felsenstein (1985) using the function *pic* in the *ape* package v5.8. Log-transformed values of ovule and pollen numbers ($\log(x + 1)$) were used to address skewness and ensure non-negative values. PICs were calculated separately for ovule and pollen numbers based on the Brownian motion model of evolution, representing independent differences between sister taxa scaled by branch lengths. A linear regression model was used to examine the relationship between the PIC-transformed pollen and ovule values, with pollen as the dependent variable. Pearson correlation was also calculated using the function *cor.test* to assess the phylogenetically adjusted relationship. A scatterplot was created using the *ggplot2* package (Wickham, 2016), showing the PIC-transformed values, a regression line with 95% confidence intervals, and the correlation coefficient (r) and p -value.

Statistical analysis of mating system and ploidy level

We analysed species with available data on these traits to determine whether ploidy level and mating system explain pollen and ovule number variation. Ploidy was categorised as diploid or polyploid, while mating systems were classified either as selfing or outcrossing. Due to the lower sample availability, species with a "Mixed" mating system were excluded (3

species). We conducted a Type II analysis of variance (ANOVA) using the function *Anova* from the *car* package v3.1-3 (Fox and Weisberg, 2019) separately for the ploidy level and mating system to assess their independent effects on pollen and ovule number. This approach accounts for potential imbalances in sample sizes and provides more robust estimates of factor significance (Langsrud, 2003). By this method, we evaluated whether either factor significantly influenced trait variation while ensuring a more reliable comparison between groups. Further, we investigated whether ploidy level and mating system interact in shaping pollen and ovule numbers, using a two-way Type II ANOVA from the *car* package. To compare the relative contributions of ploidy and mating system to variation in pollen and ovule count, we computed Eta-squared (η^2 partial) as a measure of effect size using the *effectsize* package (Ben-Shachar et al., 2020). This metric quantifies the proportion of variance explained by each predictor, with higher values indicating a stronger effect.

To account for the impact of ploidy level and mating system on ovule and pollen numbers while accounting for phylogenetic non-independence, we performed Phylogenetic Generalized Least Squares (PGLS) regression. We first created a correlation structure using the *corMartins* function in *ape* package v5.8. PGLS models were implemented using the function *gls* from the *nlme* package v3.1-164 (Pinheiro et al., 2011). Pollen and ovule numbers were assigned as dependent variables, while ploidy and mating systems were categorical predictors. To determine whether phylogenetic correlation was necessary, we also fitted the same models using Ordinary Least Squares (OLS) via the function *lm* from the *stats* package, which assumes species observations are independent. To compare PGLS and OLS, we computed Akaike Information Criterion (AIC) values for each model, where lower AIC values indicate better model fit.

Statistical analysis of environmental factors (abiotic and biotic)

We employed generalised additive mixed models (GAMMs) (Wood, 2014) to assess how pollinator availability and abiotic factors influence pollen and ovule production. Pollen and ovule numbers were modelled as Poisson-distributed responses with a logarithmic link function. Since the full models showed considerable overdispersion ($\phi \gg 1$), both GAMMs were refitted with observation-level random effects to account for unmodeled heterogeneity (Harrison, 2014). GAMMs were fitted using the *gam4* package (Wood, 2014), incorporating thin plate regression spline smooths (Wood, 2003) of pre-selected environmental predictors. Given our hierarchical sampling design, with multiple observations per individual nested within species, genera, and tribes, we accounted for non-independence in observations by including a nested phylogenetic random effect structure. To build the most parsimonious models, coefficients of variables with weak predictive performance were shrunk toward zero using double-penalty smoothing (Marra & Wood, 2011). Model diagnostics were assessed using residual plots, and no significant violations of the model assumptions were detected. We evaluated model fit using R^2 and adjusted intraclass correlation coefficients (ICC) following Nakagawa et al. (2017). Data visualisation was conducted using *ggplot2* (Wickham, 2016), and model performance metrics were assessed using the *performance* package (Lüdtke et al., 2021).

All analyses using different packages were performed in R v4.3.3 (R Core Team, 2023). AI assistant (*GPT-4o*, *OpenAI*) was used to debug R scripts, enhance code efficiency, modify scripts to improve figure aesthetics and select color-blind-friendly palettes. It also aided in tuning BEAST2 XML operator settings to improve MCMC mixing and reduce setup time. The explanation of the prompt that helped to

narrow down the number of operators to modify for the run is available on OSF (Haghighatnia et al. 2025).

RESULTS

Strong phylogenetic signal for ovule number compared to pollen number

Phylogenetic reconstruction inherently varies based on methodological and marker selection, potentially influencing the inferred evolutionary trajectories. Our main tree, estimated within a Bayesian framework, is based on the classical marker set of nuclear ITS and plastid *trnL/trnF* due to the unavailability of phylogenomic data for the majority of the species used in this study. Notably, the recent phylogenetic tree by Hendriks et al. (2023) included only 60 of the 158 species for which we had trait data, necessitating a *de novo* reconstruction to ensure full trait-phylogeny alignment. Node support in the main tree was moderate, with an average posterior probability of 0.78 (range: 0-1) for backbone nodes and 0.75 for shallower nodes. At the tribal level, the average posterior probability was 0.77, indicating that the node support is consistent across the tree. To complement this reconstruction and incorporate robust genus-level relationships, we also built a constrained tree using the tree topology by Hendriks et al. (2023) to enforce monophyly of genera while allowing species-level placements to vary freely within each genus.

Pollen and ovule production showed a substantial variation among the 158 studied Brassicaceae species. Pollen numbers ranged from 22 to 234,228 grains, with a mean ($\pm$ standard deviation) of $36,007 \pm 41,201$ grains and a high coefficient of variation of 114%. Ovule number showed lower variability, ranging from 1 to 153, with a mean of 25 ± 26 , and a CV of 104%.

We then aimed to determine the role of mere phylogeny in the interspecific variation for pollen and ovule numbers. Pagel's λ values indicated a strong phylogenetic signal for ovule

number ($\lambda = 0.95$) and a moderate signal for pollen number ($\lambda = 0.74$) in the main tree. The constrained tree yielded similar results ($\lambda = 0.94$ for ovule number and $\lambda = 0.75$ for pollen number), suggesting that ovule number is more evolutionarily constrained by lineage history. This pattern was consistent with Blomberg's K estimates, with 0.44 for pollen and 0.61 for ovule number in the main tree, and 0.52 and 0.66, respectively, in the constrained tree. The significance of these phylogenetic signals was confirmed through 1,000 randomisations ($p = 0.001$), reinforcing that while both traits exhibit phylogenetic structure, ovule number is more strongly conserved across evolutionary lineages than pollen number in both trees. A likelihood ratio (LR) test comparing models with estimated λ to models assuming no phylogenetic signal ($\lambda = 0$) confirmed the presence of a robust phylogenetic structure. For the main tree, the LR statistic was 57.95 for pollen number ($p = 2.68 \times 10^{-14}$) and 89.60 for ovule number ($p = 2.9 \times 10^{-21}$). Results from the constrained tree were consistent, with LR = 60.4 ($p = 7.8 \times 10^{-15}$) for pollen and LR = 85 ($p = 3.02 \times 10^{-20}$) for ovule. These results further confirm the presence of a robust phylogenetic structure in the data (Supplementary Table S1, S2). However, the relatively moderate K values for pollen and ovule numbers suggest that a substantial fraction of the observed variation deviates from the Brownian motion model's expectations, potentially reflecting the influence of adaptive evolution, environmental heterogeneity, or other non-phylogenetic factors.

Adaptive evolution explains the evolution of ovule and pollen numbers in the Brassicaceae

The evolution of ovule and pollen numbers in the Brassicaceae was analysed using comparative evolutionary models, with the OU model emerging as the best fit for both traits, and both trees (Supplementary Table S3, S4). This conclusion is supported by the lowest AIC values and highest AIC weights among the tested models. For ovule number, the AIC values for the BM, EB, and OU models were 406.8, 408.8, and 403.9, respectively, in the

main tree. Using the constrained tree, the corresponding AIC values were slightly higher: 409.13 (BM), 411.13 (EB), and 406.39 (OU). The OU model showed the strongest support with an AIC weight of 0.76, compared to 0.18 for BM and 0.07 for EB in the main tree. The constrained tree yielded slightly lower support for the OU model but retained it as the best fit, with AIC weights of 0.74 (OU), 0.19 (BM), and 0.07 (EB). While the preference for the OU model indicates that ovule number evolution is primarily driven by stabilising selection toward an adaptive optimum, the moderate support for the BM model suggests that random drift also shapes variation. Parameter estimates for the OU model revealed a relatively low rate of adaptation ($\alpha = 0.04$ for both trees) and a low rate of evolutionary variance ($\sigma^2 = 0.15$ for the main tree; 0.14 for the constrained tree) (Table S3, S4), indicating a slow evolutionary process predominantly shaped by stabilising selection (Fig. 1 and Supplementary Fig. S2, right).

For pollen number, the AIC values for the BM, EB, and OU models were 568.7, 570.7, and 543.8, respectively, in the main tree. Using the constrained tree, the corresponding values were 557.90 (BM), 560.06 (EB), and 538.28 (OU). In both cases, the OU model showed a high support, with an AIC weight of nearly 1. In contrast, the BM and EB models had negligible support, with AIC weights approaching zero in both trees (Table S3, S4). The dominance of the OU model for pollen number suggests that its evolution is driven almost entirely by stabilising selection, with minimal contribution from stochastic processes. The OU model parameters for pollen number showed a higher rate of adaptation ($\alpha = 0.12$ for the main tree; 0.10 for the constrained tree) and greater evolutionary variance ($\sigma^2 = 0.56$ for main tree; 0.46 for the constrained tree) compared to ovule number, reflecting a faster evolutionary process with a larger variation (Fig. 1 and Fig. S2, left). These findings indicate that both traits are shaped by stabilising selection, while their evolutionary

dynamics differ. This difference in evolutionary dynamics between pollen and ovule numbers was also supported by phylogenetically independent contrasts (PICs), which revealed a statistically significant but weak positive correlation between pollen and ovule numbers across the Brassicaceae (Supplementary Fig. S3, S4). The Pearson correlation coefficient for the PIC-transformed values in the main tree was $r = 0.21$ ($p = 0.009$), and $r = 0.22$ for the constrained tree ($p = 0.0067$), indicating a modest but statistically significant evolutionary association between these traits after accounting for phylogenetic relatedness.

Environmental conditions are weekly associated with pollen and ovule number evolution

We then investigated the role of ecological conditions in shaping the evolution of pollen and ovule numbers. For this purpose, we measured pollen and ovule numbers in a subset of species cultivated in a common garden (see Methods, Dataset S2) testing associations between environmental conditions of origin and pollen or ovule number. The environmental association with ovule number was relatively weak ($R^2 = 3.3\%$), while the phylogenetic signal had the strongest impact. Genus played the most pronounced role (ICC = 67.0%) followed by species (22.5%) and tribe (10.5%). Regarding the influence of the native environment, the most parsimonious model involved a significant impact of precipitation seasonality (GAMM: effective degrees of freedom [edf] = 0.84, $\chi^2 = 4.57$, $p = 0.0171$) and climatic moisture index (edf = 0.83.0, $\chi^2 = 4.54$, $p = 0.0184$). Other environmental characteristics showed no significant association (Supplementary Table S5). Both variables were positively associated with ovule number (Fig. 2), implying that plants in moist environments with higher seasonality evolved to produce more ovules.

The environmental association with pollen number was comparable to that for ovules ($R^2 =$

3.8%), and the parsimonious GAMM involved significant effects of altitude ($\text{edf} = 1.06$, $\chi^2 = 22.3$, $p < 0.0001$), temperature seasonality ($\text{edf} = 0.93$, $\chi^2 = 8.44$, $p = 0.0017$), and rooting conditions ($\text{edf} = 1.73$, $\chi^2 = 11.7$, $p = 0.0009$) (Table S5). The highest number of pollen grains is expected in conditions of high altitude, low-temperature seasonality, and favourable rooting conditions (Fig. 3). The phylogenetic signal was strong, with tribe and species accounting for the most variability in pollen number ($\text{ICC} = 17.7\%$ and 76.2% , respectively), while genus variation was less important ($<0.1\%$).

For both traits, no significant association was found with lifespan (annual vs perennial) and pollinator richness (Table S5).

Mating system drives the evolution of pollen production but does not affect ovule number variation

Since phylogeny had a moderate impact on pollen and ovule number evolution, we investigated the role of species traits such as mating system and ploidy level as evolutionary drivers of these traits. In the two-way ANOVA, when both ploidy and the mating system were analysed together, the mating system was the primary factor influencing pollen production ($F(1, 70) = 24.11$, $p = 0.0001$, $\eta^2 = 0.26$), with selfing species producing significantly fewer pollen grains than outcrossers (Fig. 4). Ploidy did not have a significant effect ($F(1, 70) = 2.73$, $p = 0.10$, $\eta^2 = 0.04$) (Supplementary Table S6). For ovule count, neither the mating system ($F(1, 70) = 0.0002$, $p = 0.98$, $\eta^2 = 0$) nor ploidy ($F(1, 70) = 0.1728$, $p = 0.68$, $\eta^2 = 0.002$) showed a significant effect. The interaction between ploidy and mating system was not significant for either trait, affecting neither pollen ($F(1, 70) = 0.91$, $p = 0.34$, $\eta^2 = 0.01$) nor ovule number ($F(1, 70) = 0.77$, $p = 0.383$, $\eta^2 = 0.01$) (Table S6).

The PGLS analysis, accounting for phylogenetic correlation, revealed that phylogenetic correction improved model fit, with the PGLS model ($\text{AIC} = 1789.12$ for the main tree;

1792.34 for the constrained tree) performing better than the OLS model ($\text{AIC} = 1793.99$) (Supplementary Table S7, S8). For the main tree, the mating system was a significant predictor of pollen number (PGLS: $\beta = 43,006.68$, $\text{SE} = 14,351.05$, $t = 3.00$, $p = 0.0038$; OLS: $p = 0.001$), with outcrossers producing significantly more pollen than selfers. In contrast, ploidy ($\beta = 7,349.18$, $\text{SE} = 12,463.33$, $p = 0.557$) and the interaction between ploidy and mating system ($\beta = -14,636.33$, $\text{SE} = 17,612.06$, $p = 0.409$) were not significant. The residual standard error was higher in the PGLS model ($57,216.07$) than the OLS model ($42,740$), suggesting that, while phylogenetic correction accounts for evolutionary relatedness, considerable variability in pollen number remains, likely influenced by additional factors. For ovule number, the PGLS model ($\text{AIC} = 705.19$) outperformed the OLS model ($\text{AIC} = 716.59$), indicating that phylogenetic correction improved model fit. None of the tested predictors were significant: ploidy ($\beta = 5.27$, $\text{SE} = 8.22$, $t = 0.61$, $p = 0.524$), mating system ($\beta = -3.47$, $\text{SE} = 9.47$, $t = -0.37$, $p = 0.715$), or their interaction ($\beta = 1.04$, $\text{SE} = 11.62$, $t = 0.09$, $p = 0.929$) (Table S7). The constrained tree yielded similar results for both traits (see Table S8). This suggests that ovule number is largely conserved across evolutionary lineages, shaped more by historical constraints than by recent reproductive shifts or whole genome duplication. In contrast, pollen number is more evolutionarily flexible and strongly influenced by the mating system.

DISCUSSION

Reproductive investment strategies are crucial for plant fitness, influencing mating success, fecundity, and adaptation. Pollen and ovule production, as key reproductive traits, vary widely across species, yet the relative influence of evolutionary constraints, environmental adaptation, and reproductive strategies in shaping these traits remains unresolved. This study addresses this question, examining the phylogenetic constraint of pollen and ovule production, how abiotic factors (e.g., moisture,

altitude) and biotic factors (pollinator availability) influence both traits, and the extent to which mating system shifts and ploidy affect their evolution.

Using phylogenetic comparative analyses of 158 Brassicaceae species and common garden cultivation to control for phenotypic plasticity, we found that ovule number is strongly constrained by phylogeny, while pollen number is more evolutionarily flexible, largely due to mating system shifts—selfing species producing significantly less pollen. The association between environment and both traits were limited yet significant. Ovule number increased in wetter environments, while pollen number was influenced by altitude and temperature seasonality. Whole-genome duplication had no significant impact on the evolution of either trait.

Distinct evolutionary patterns for pollen and ovule production

Understanding evolutionary rate variation helps explain how morphological diversity arises and persists (Foote 1997; Sidlauskas 2008). Traits evolve at different rates; some change rapidly under strong selection, while others are constrained by development, functional trade-offs, or phylogenetic inertia (Pagel, 1999; O'Meara et al., 2006; Cooper & Purvis, 2010; Uyeda et al., 2011). Our results show distinct selective pressures on pollen and ovule numbers: ovule number was more conserved, while pollen number was more flexible across the phylogeny (Fig. 1 and Fig. S2). This pattern is consistent with other families like Lamiaceae and Boraginaceae exhibiting deep phylogenetic conservatism in ovule number, while others, such as Rosaceae, show greater variation (Potter et al., 2007; Burd et al., 2009). The strong phylogenetic signal observed in our study (Blomberg's $K = 0.61$, Pagel's $\lambda = 0.95$ for the main tree; 0.66 and 0.94, respectively, for the constrained tree) and the weak association with environmental factors ($R^2 = 3.3\%$) suggest that

ovule number is primarily constrained by historical lineage-specific factors in Brassicaceae rather than being highly responsive to other factors such as ecology. Ovule production was also unaffected by mating system shifts and ploidy. Stabilising selection models predict that traits crucial to fitness remain within an optimal range to minimise deleterious variation (Waddington, 1942; Debat & David, 2001; Flatt, 2005). Our findings align with this framework, with the OU model indicating that ovule number evolves under strong selective constraints, reflected in its slow evolutionary rate ($\alpha = 0.04$, $\sigma^2 = 0.15$ for the main tree; 0.04 and 0.14, respectively, for the constrained tree).

A likely explanation for this constraint lies in the fundamental trade-off between reproductive output and offspring survival. Maternal resources are finite, and plants must allocate them strategically to balance ovule production with seed provisioning. Producing more ovules can come at the cost of per-seed investment, potentially reducing overall reproductive success (Haig & Westoby, 1988; Greenway & Harder, 2007). This limitation can impose an evolutionary restriction, as excessive ovule production without sufficient resource provisioning could reduce seedling fitness and long-term reproductive success. With this in mind, whether investment per seed is fully correlated to ovule number remains to be investigated, and it is likely that the evolution of maternal investment follows its own trajectory to a certain extent. Genetic constraints may also limit the evolutionary flexibility of ovule number, as developmental regulation often restricts trait variability (Hallgrímsson et al., 2002). In *Arabidopsis thaliana*, genes involved in ovule development also play essential roles in broader reproductive processes, including gametophyte formation and carpel development, with mutations often leading to reduced fertility and structural defects (Pinyopich et al., 2003). Moreover, ovule number and gynoecium size

are interconnected traits governed by shared hormonal and genetic pathways, meaning that changes in ovule production can have cascading effects on floral architecture and seed yield (Cucinotta et al., 2020). Because of these pleiotropic effects, strong selection likely acts to maintain ovule number within an optimal range, ensuring reproductive success while preserving developmental stability.

Compared to ovule production, pollen number was more evolutionarily flexible, exhibiting a moderate phylogenetic signal (Blomberg's $K = 0.44$, Pagel's $\lambda = 0.74$ for the main tree; 0.52 and 0.75, respectively, for the constrained tree) and higher evolutionary rate ($\alpha = 0.12$, $\sigma^2 = 0.56$ for the main tree; 0.10 and 0.46, respectively, for the constrained tree). This pattern contrasts previous studies reporting strong phylogenetic constraints for pollen number in several plant communities (Arroyo et al., 2019; Nepal et al., 2023). This evolutionary flexibility largely arises from the well known influence of mating systems on pollen production (Cruden, 1977; Tsuchimatsu and Fujii, 2022). Specifically, selfing species produced significantly fewer pollen grains than outcrossers (Fig. 4), confirmed by phylogenetically corrected PGLS, indicating that reproductive strategies are major drivers of pollen number evolution. The different impact of mating system shifts on ovule and pollen production, similarly found in other species (Harder & Johnson, 2023), may reflect the different factors limiting the reproductive success of male and female functions—mating opportunities and maternal resources, respectively, according to Bateman's principle. This is consistent with the idea that outcrossing species face higher competition among pollen grains for fertilisation success, favouring increased pollen output to maximise pollination opportunities and reproductive success (Cruden, 1977; Harder and Johnson, 2008). In contrast, selfing species experience reduced pollen competition and guaranteed fertilisation, allowing selection to favour lower pollen production as a resource-conserving strategy (Barrett, 2002; Sicard & Lenhard,

2011). Another factor potentially explaining the relative evolutionary flexibility of pollen production compared to ovule production might be different genetic constraints. Indeed, the effect of genes influencing pollen production might be more modular and less restrictive than ovule-regulating genes. The *REDUCED POLLEN NUMBER1 (RDPI)* gene exemplifies this flexibility; mutations in this gene reduce pollen number but do not seem to influence other traits, highlighting weaker pleiotropic constraints (Kakui et al., 2022). This is the case of one gene, however, and whether the genetic architecture underlying pollen production follows this trend remains to be explored.

Although both traits are influenced by phylogenetic factors, their evolutionary paths are largely independent, as suggested by the weak but significant phylogenetic correlation ($r = 0.21$, $p = 0.009$ for the main tree; 0.22 and 0.0067, respectively, for the constrained tree) between pollen and ovule numbers (Fig. S2, S3). Besides mating systems, the differing environmental associations with ovule and pollen numbers (see next section) further support the idea that these traits follow distinct evolutionary pathways. Together, these patterns suggest that while both traits respond to phylogenetic and ecological factors, they do so through largely independent mechanisms, shaped by different environmental constraints and adaptive pressures.

Is pollen production “cheap”?

Pollen is often viewed as a low-cost male investment due to its small size and excess production compared to ovules (Charnov, 1982; Harder & Routley, 2006). However, this overlooks ecological and evolutionary factors influencing pollen production. Hermaphroditic plant species must balance resource allocation between pollen and ovules, following resource allocation theory, which predicts optimal investment to maximise fitness under constraints (Charnov, 1982; Lloyd & Bawa, 1984; Obeso, 2002). Pollen costs depend on quantity, viability, dispersal efficiency, and

gamete success, requiring carbon and nitrogen inputs (Ashman and Schoen, 1994; Delph & Ashman, 2006; Nepi & Pacini, 1993; Franchi et al., 2011). The assumption that pollen is inherently “cheap” is largely based on Bateman’s principle, which itself is an assumption based on the relative number and size of pollen grains versus ovules rather than direct cost comparison (Cruden, 1977; Charnov, 1982).

To examine this assumption, we tested how pollen and ovule numbers evolve in response to selective pressures. The association between environmental factors and both traits was weak and similar, with ovule number showing an R^2 of 3.3% and pollen number an R^2 of 3.8%. Ovule production appeared primarily shaped by water availability, as indicated by its association with precipitation seasonality and climatic moisture levels (Fig. 2). This suggests that plants may adjust ovule investment in response to water resources, potentially under resource (water) constraints). In contrast, pollen number was associated with a broader range of environmental factors, including altitude, temperature fluctuations, and rooting conditions (Fig. 3). Altitude and rooting conditions also suggest resource constraints on the evolution of pollen production, to a similar extent as ovule production according to R^2 values.

Beyond environmental associations, the reduction in pollen number in selfing species across Brassicaceae supports a model of resource allocation optimisation (Stearns, 1998; Roff, 2002), where selection would favour lower pollen production as a resource-conserving strategy in selfers (Barrett, 2002; Sicard & Lenhard, 2011). Further supporting this, in the selfer *Arabidopsis thaliana*, a mutation that reduces pollen production was found to be under positive selection (Tsuchimatsu et al., 2020). This indicates that rather than relaxed selection, lower pollen production confers a selective advantage, likely

by reducing the reproductive cost.

Overall, these findings challenge the traditional view that pollen is a negligible investment. The evolution of pollen production did not appear less resource-constrained than for ovule production, and the consistent reduction in pollen number in selfers may be adaptive due to its significant cost reduction.

Pollinator availability may explain the evolution of pollen production

Pollinators are vital for plant reproduction, and it has been suggested that pollen and ovule production evolve in response to pollinator fluctuations (Burd et al., 2009; Ollerton, 2017; Tong et al., 2023). In particular, increased ovule production in pollination-limited environments could maximise reproductive opportunities if pollination occurs, while pollen production might be more directly shaped by selection to enhance male reproductive success under pollinator scarcity (Harder & Johnson, 2023).

Our results suggest that pollinator availability is not a primary driver of pollen investment in the Brassicaceae. Despite the role of pollen production in male competition (Snow & Spira, 1991; Lankinen and Skogsmyr, 2002), its variation was not linked to pollinator richness in our study. Nevertheless, pollen production increased with altitude, supporting the hypothesis that in pollinator-scarce environments, such as higher elevations, plants may produce more pollen to counteract reduced mating opportunities (Harder & Johnson, 2023). Alternatively, a short growing season in higher altitudes may select for more pollen-rich flowers, which would maximize each visit by pollinators and thereby improve male fitness. In contrast, ovule production remained unaffected by pollinator richness or altitude, challenging the hypothesis that plants compensate for stochastic pollination by increasing ovule production (Burd et al., 2009). This finding also contrasts with a study reporting higher ovule

numbers at high elevations, potentially as a response to reduced pollination opportunities (Arroyo et al., 2019). Overall, our results are partially consistent with empirical studies suggesting that pollen production is more responsive to pollinator availability than ovule production (Harder & Johnson, 2023).

However, one clear limitation of our study is the reliance on bee species richness as a proxy for pollinator availability (Orr et al., 2021), which overlooks factors like pollinator abundance, visitation frequency, or floral fidelity—key predictors of effective pollination and plant reproductive success (Vázquez et al., 2005; Klein et al., 2007; Winfree et al., 2009; Brosi & Briggs, 2013; Goulson et al., 2015). Thus, the absence of a relationship between pollinator richness and reproductive traits may reflect the limitations of our proxy rather than a true lack of pollination effects on reproductive allocation. Similarly, altitude is only an indirect proxy for pollinator abundance. Future research using direct pollination observations and pollen deposition analyses could refine our understanding of selective pressures, challenging current evolutionary frameworks for plant reproductive strategies.

CONCLUSION

Our findings reveal that reproductive trait evolution reflects phylogenetic constraints and adaptive responses. Ovule number is evolutionarily conserved, showing weak environmental association and no significant effect of mating system, suggesting strong stabilising selection. In contrast, pollen number is more evolutionarily flexible and primarily shaped by mating system shifts rather than environmental variation. The consistent reduction in pollen production in selfing species, along with evidence for positive selection on pollen-reducing mutations in *A. thaliana*, challenges the assumption that pollen is a low-cost investment. Instead, pollen production carries a cost, which plants minimise when selection favours increased efficiency under self-fertilisation. While we found no

direct link between pollinator richness and reproductive investment, the observed increase in pollen production at high elevations suggests that pollen output may adjust to pollination limitations in certain environments. Given the limitations of our proxy for pollinator availability, future studies incorporating direct measures of pollination success will be crucial for refining theories on male reproductive allocation in plants.

SUPPLEMENTARY DATA

Supplementary data are available at *Annals of Botany* online and consist of the following:

Fig. S1. Geographic distribution of species based on population-level sampling.

Fig. S2. Evolutionary patterns of pollen and ovule numbers in the Brassicaceae based on the constrained tree show similar trends to those observed in the main tree.

Fig. S3. Phylogenetic independent contrasts (PICs) between pollen and ovule numbers show a statistically significant but weak positive correlation in the main tree.

Fig. S4. Phylogenetic independent contrasts (PICs) between pollen and ovule numbers show a statistically significant but weak positive correlation in the constrained tree.

Table S1. Phylogenetic signal analysis of pollen and ovule numbers based on the main tree.

Table S2. Phylogenetic signal analysis of pollen and ovule numbers based on the constrained tree.

Table S3. Comparative phylogenetic model evolution for pollen and ovule numbers based on the main tree.

Table S4. Comparative phylogenetic model evolution for pollen and ovule numbers based on the constrained tree.

Table S5. Results of Generalized Additive Mixed (GAM) models testing the effects of biotic and abiotic variables on pollen and ovule production.

Table S6: ANOVA results and effect sizes for pollen and ovule numbers.

Table S7. Summary of PGLS (OU) and OLS results with AIC comparisons based on the main tree.

Table S8. Summary of PGLS (OU) and OLS results with AIC comparisons based on the constrained tree.

Dataset S1: The complete dataset of 158 Brassicaceae species, with the information on the mating system and ploidy level used for all phylogenetic and traits analyses.

Dataset S2: The complete dataset of cultivated species with abiotic and biotic measurements, and GPS coordinates used to examine the relationship between reproductive traits and environmental factors.

DATA AVAILABILITY

The final alignment for building the phylogeny and the maximum likelihood tree (IQ-TREE) is available online on OSF (<https://doi.org/10.17605/OSF.IO/XMREJ>).

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CONFLICTS OF INTEREST

The authors have no conflict of interest to declare.

AUTHOR CONTRIBUTIONS

M. H. and C. L. P. designed the study. M. H. and M. S. performed the analyses. M. H., M. S. and C. L. P. wrote the manuscript. M. A. K. provided the plant material and contributed with cytogenetic data, M. A. K. and R. S. provided

advice on the design and analyses and edited the manuscript.

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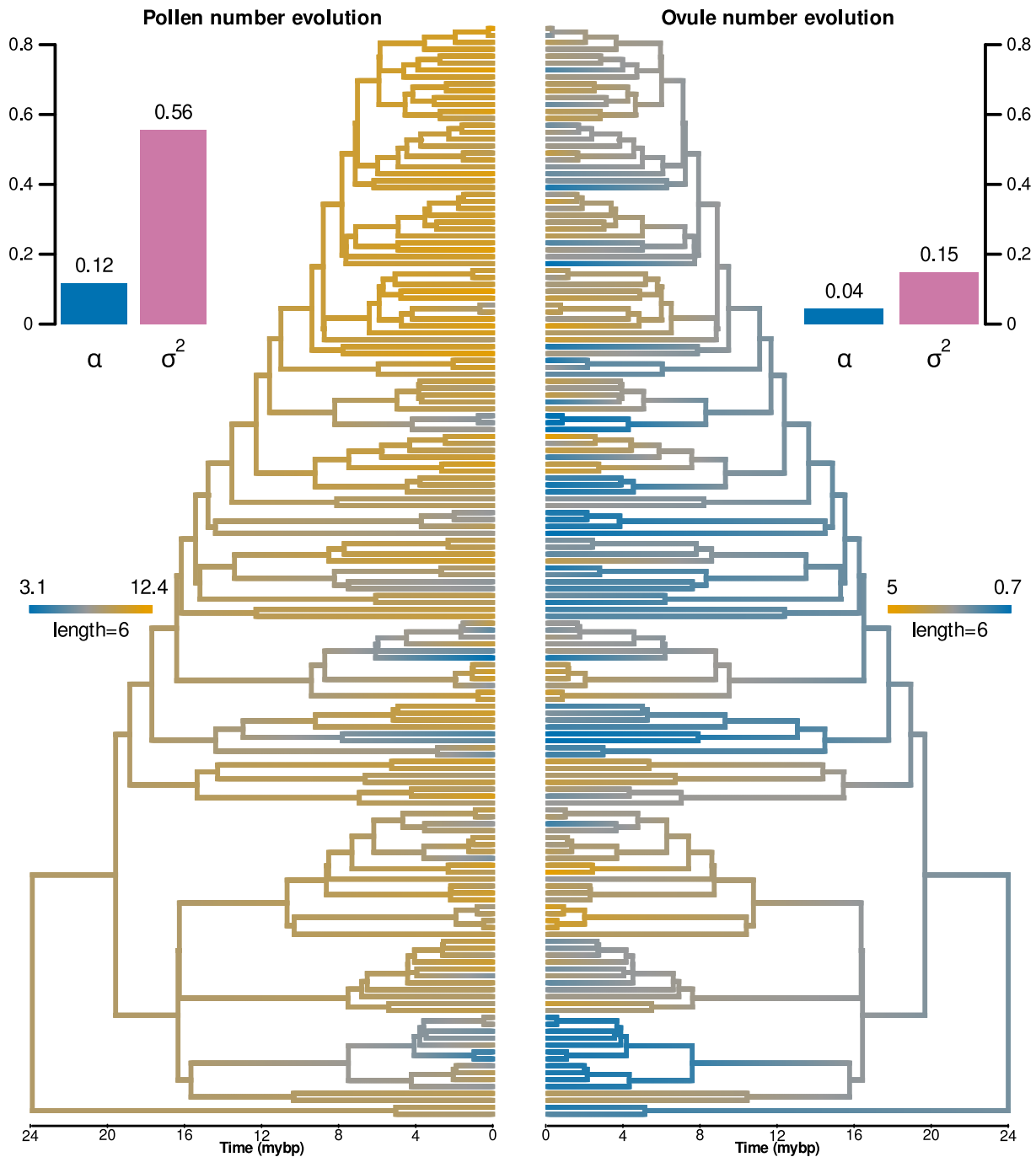


Fig. 1. Evolutionary patterns of pollen and ovule numbers in the Brassicaceae based on the main tree. Continuous trait maps depicting the evolution of (left) log-transformed pollen number and (right) log-transformed ovule number. Trait values are represented using a gradient that transitions from blue (low values) through grey (medium values) and gold (high values). Time is scaled in millions of years before the present (mybp). Bar plots adjacent to each phylogeny display the estimated parameters from the best-fitting Ornstein-Uhlenbeck (OU) model: the strength of selection (α) and evolutionary variance (σ^2) for pollen and ovule numbers.

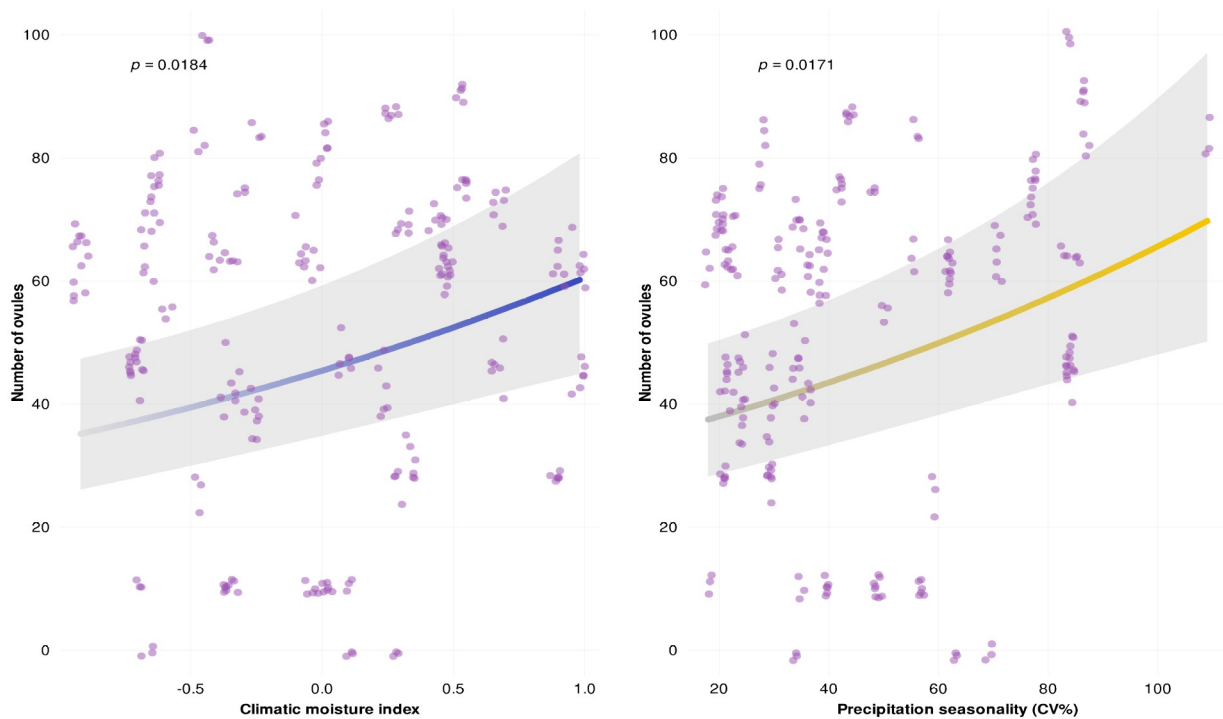


Fig. 2. Significant relationships between number of ovules, climatic moisture index (left), and precipitation seasonality (right). Model predictions (lines) and their 95% confidence intervals (grey bands) are displayed. Note that negative values of the climatic moisture index indicate moisture deficits, values close to zero (Light grey) represent a balance between moisture supply and demand, while positive values indicate wetter conditions (Blues). For precipitation seasonality, the grey and gold are representative of stable and extreme seasonality, respectively. The points represent raw measured values for each individual, ranging up to ~100 ovules. Points were slightly jittered in both horizontal and vertical directions to reduce overlap and improve visual clarity.

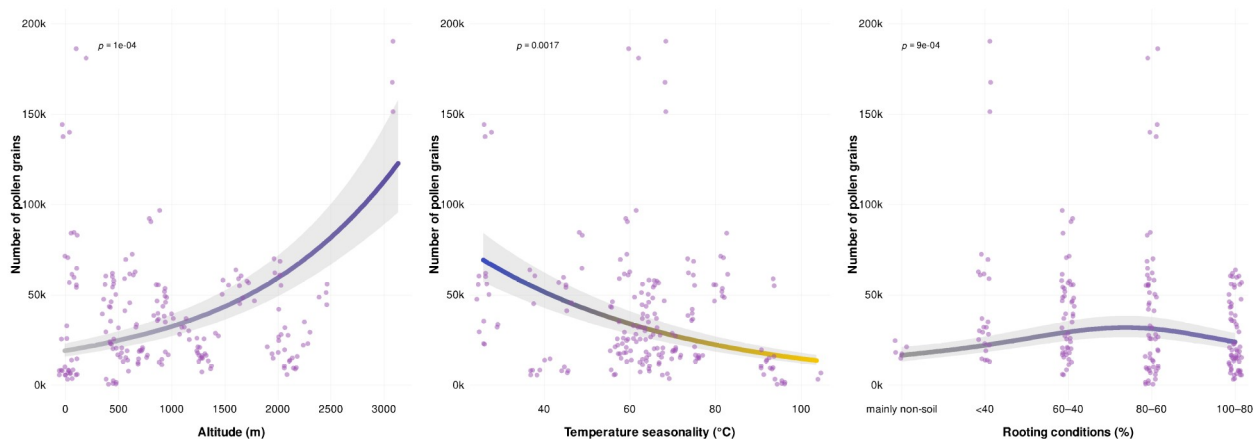


Fig. 3. Significant relationships between the number of pollen grains, altitude (left), temperature seasonality (middle), and rooting conditions (right). Model predictions (lines) and their 95% confidence intervals (grey bands) are displayed. Rooting conditions represent the percentage of plant growth potential. The points represent raw measured values for each individual, ranging up to ~188,000 grains. Points were slightly jittered in both horizontal and vertical directions to reduce overlap and enhance the visual distinction of individual observations.

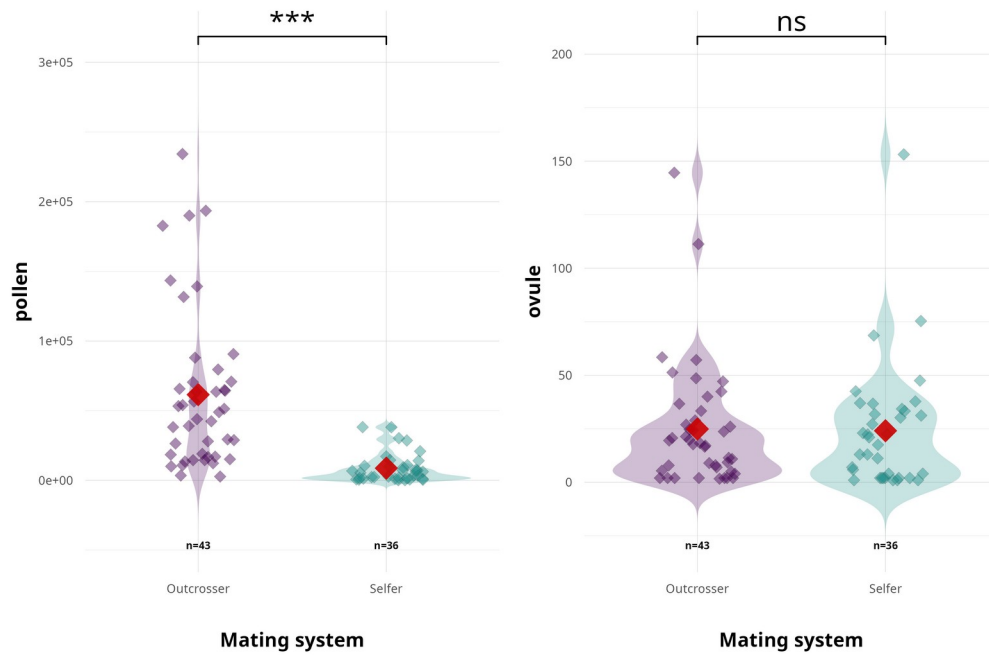


Fig. 4. Effects of the mating system on pollen and ovule production in the Brassicaceae. Violin plots with individual data points showing the distribution of (left) log-transformed pollen numbers and (right) ovule numbers in species with different mating systems: outcrossers and selfers. Red diamond shows the mean values of pollen and ovule numbers in different mating systems. Pollen numbers are significantly higher in outcrossers compared to selfers (Type II ANOVA, $p = 0.001$), indicating a strong influence of the mating system on male reproductive investment. In contrast, ovule numbers do not differ significantly between mating systems (ns, not significant), suggesting ovule production is independent of mating strategy.

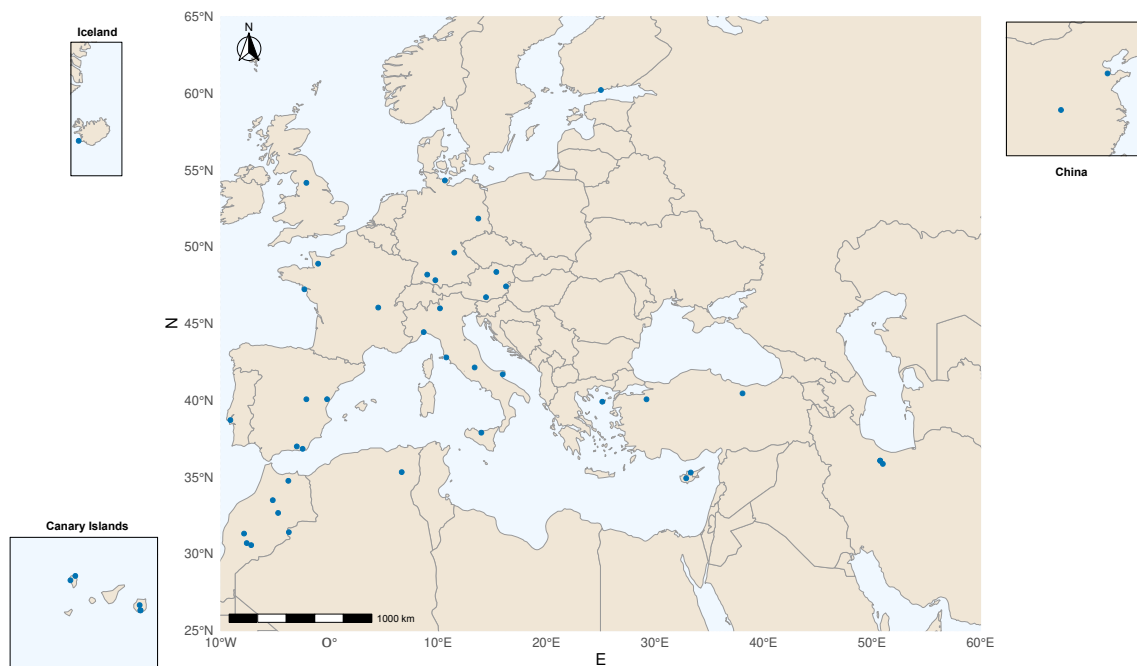


Fig. S1. Geographic distribution of species based on population-level sampling. Each point

represents a unique population used for ovule and pollen trait measurements. Insets (not to scale or geographic position) highlight peripheral regions: the Canary Islands, China, and Iceland.

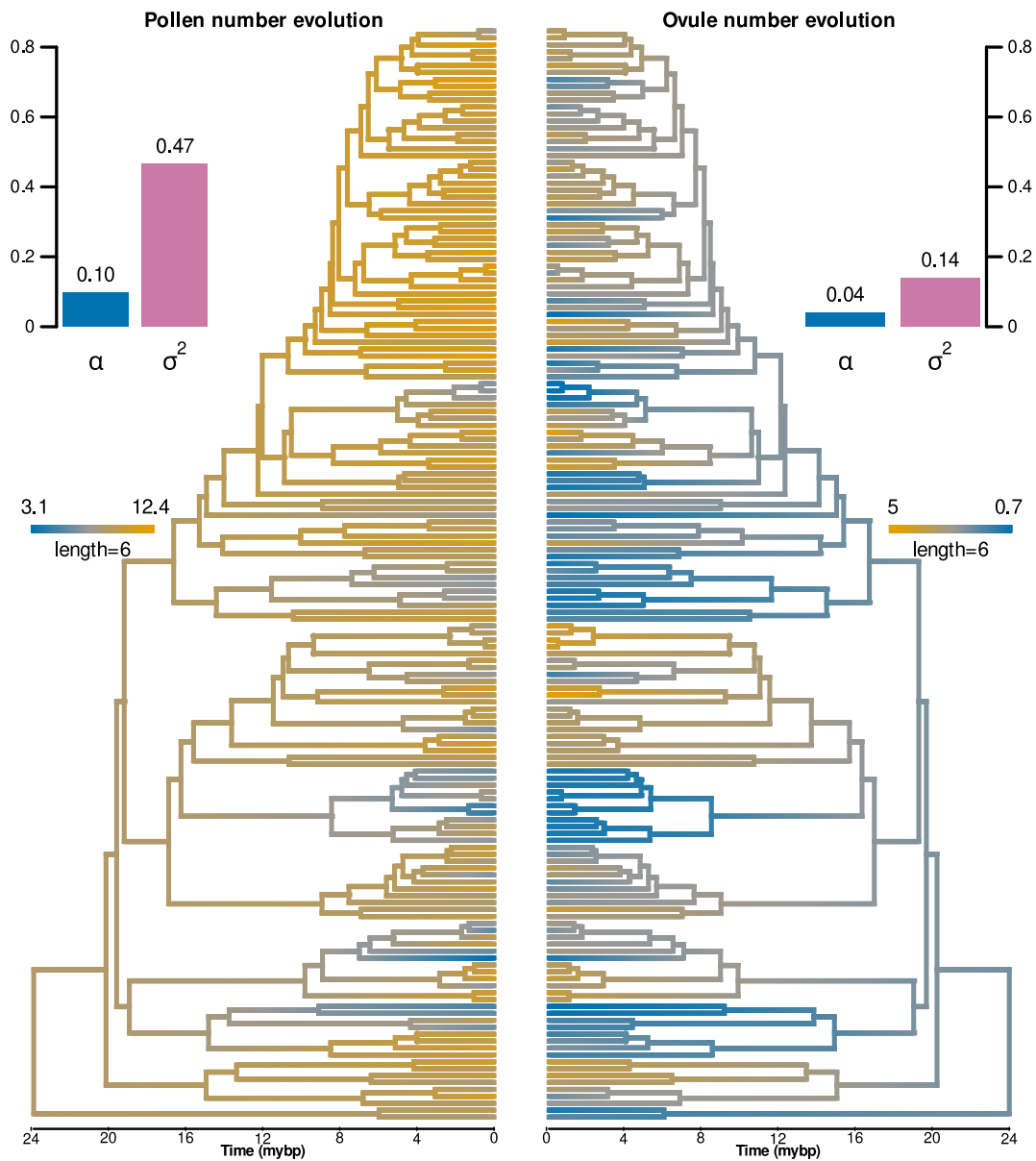


Fig. S2. Evolutionary patterns of pollen and ovule numbers in the Brassicaceae based on the constrained tree. Continuous trait maps depicting the evolution of (left) log-transformed pollen number and (right) log-transformed ovule number. Trait values are represented using a gradient that transitions from blue (low values) through grey (medium values) and gold (high values). Time is scaled in millions of years before the present (mybp). Bar plots adjacent to each phylogeny display the estimated parameters from the best-fitting Ornstein-Uhlenbeck (OU) model: the strength of selection (α) and evolutionary variance (σ^2) for pollen and ovule numbers.

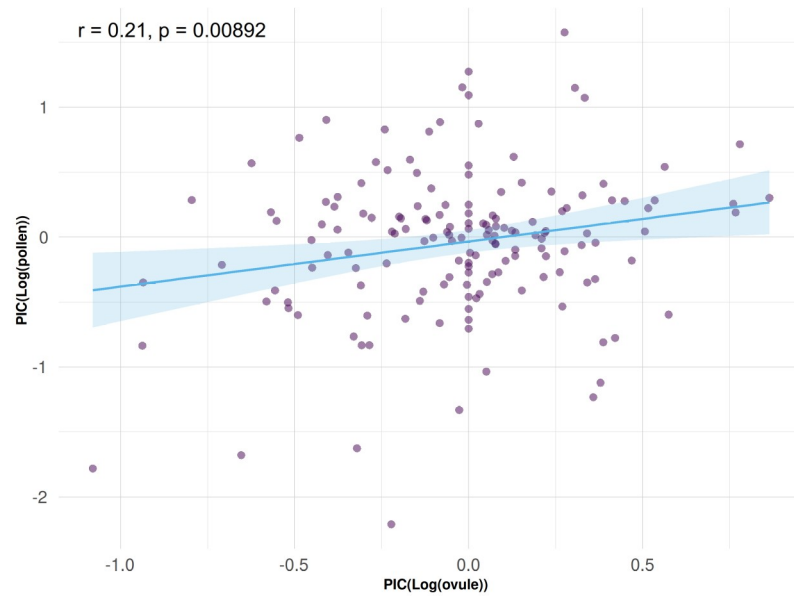


Fig. S3. Phylogenetically corrected relationship between ovule and pollen numbers in the Brassicaceae based on the main tree. Scatter plot showing the relationship between phylogenetically independent contrasts (PICs) of log-transformed ovule and pollen numbers across species. Each point represents a PIC value for a species, accounting for shared evolutionary history. The blue line represents the linear regression fit, with the shaded area indicating the 95% confidence interval. A positive correlation was observed (Pearson's $r = 0.21$, $p = 0.009$), suggesting a weak but significant association between ovule and pollen numbers after controlling for phylogenetic relatedness.

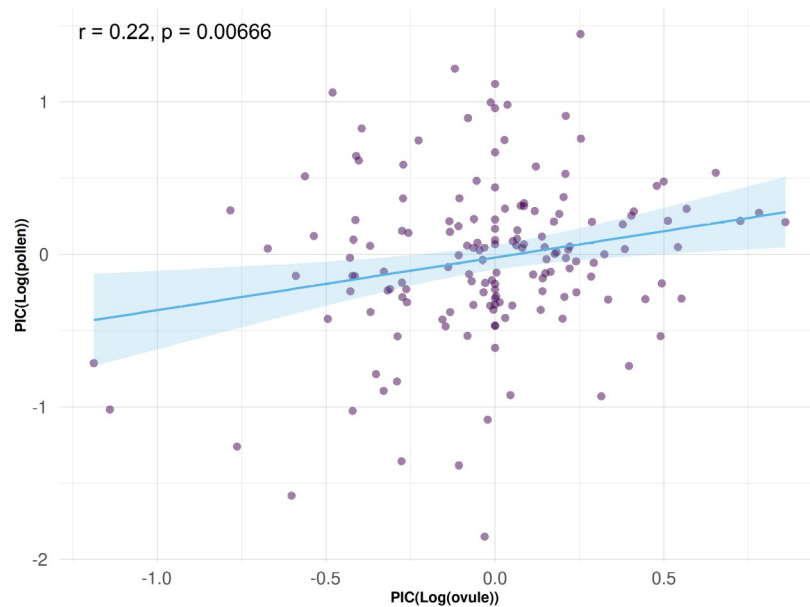


Fig. S4. Phylogenetically corrected relationship between ovule and pollen numbers in the Brassicaceae based on the constrained tree. Scatter plot showing the relationship between phylogenetically independent contrasts (PICs) of log-transformed ovule and pollen numbers across species. Each point represents a PIC value for a species, accounting for shared evolutionary history. The blue line represents the linear regression fit, with the shaded area indicating the 95%

confidence interval. A positive correlation was observed (Pearson's $r = 0.22$, $p = 0.0067$), suggesting a weak but significant association between ovule and pollen numbers after controlling for phylogenetic relatedness.

Table S1. Phylogenetic signal analysis of pollen and ovule numbers based on the main tree. The signals are estimated by two methods: Blomberg's K with p-value (K) and Pagel's Lambda with logL and likelihood ratio (LR) and LR p-values. Significant p-values are shown with strike and bold font.

Table S2. Phylogenetic signal analysis of pollen and ovule numbers based on the constrained tree. The signals are estimated by two methods: Blomberg's K with p-value (K) and Pagel's Lambda with logL and likelihood ratio (LR) and LR p-values. Significant p-values are shown with strike and bold font.

S1	Metric	Pollen	Ovule
	Blomberg's K	0.44	0.61
	P-value (K)	0.001*	0.001*
	Pagel's Lambda	0.74	0.95
	logL (Lambda)	-268.11	-775.07
	LR (Lambda=0)	57.95	89.61
	P-value (LR Test)	2.68E-14*	2.9E-21*

S2	Metric	Pollen	Ovule
	Blomberg's K	0.52	0.66
	P-value (K)	0.001*	0.001*
	Pagel's Lambda	0.75	0.94
	logL (Lambda)	-266.57	-199.4
	LR (Lambda=0)	60.4	85.00
	P-value (LR Test)	7.8E-15*	3.02E-20*

Table S3. Model evolution of pollen and ovule numbers based on the main tree. All the models tested for their evolution are shown, including Brownian Motion (BM), Early Burst (EB), and Ornstein-Uhlenbeck (OU). Akaike information criterion (AIC), corrected AIC (AICc), AIC weight (AICw), estimated evolutionary variance (σ^2), and ancestral state (z_0) are shown for all models. The strength of selection (α) is estimated for the OU model. The best-fitting models, identified by the lowest AICc values, are shown in bold.

Model	log-likelihood	AIC	AICc	σ^2	α (OU)	z_0	AICw
BM_pollen	-282.33	568.66	568.74	0.31	-	9.26	0.00
EB_pollen	-282.33	570.66	570.82	0.31	-	9.26	0.00
OU_pollen	-268.89	543.79	543.95	0.56	0.12	9.35	1.00
BM_ovule	-201.39	406.77	406.85	0.11	-	2.37	0.18
EB_ovule	-201.39	408.77	408.93	0.11	-	2.37	0.07
OU_ovule	-198.95	403.90	404.05	0.15	0.04	2.45	0.76

Table S4. Model evolution of pollen and ovule numbers based on the constrained tree. All the

Haghighatnia et al. – The evolution of pollen and ovule production in the Brassicaceae

models tested for their evolution are shown, including Brownian Motion (BM), Early Burst (EB), and Ornstein-Uhlenbeck (OU). Akaike information criterion (AIC), corrected AIC (AICc), AIC weight (AICw), estimated evolutionary variance (σ^2), and ancestral state (z_0) are shown for all models. The strength of selection (α) is estimated for the OU model. The best-fitting models, identified by the lowest AICc values, are shown in bold.

Model	log-likelihood	AIC	AICc	σ^2	α (OU)	z_0	AICw
BM_pollen	-276.95	557.90	557.98	0.27	-	9.21	0.00
EB_pollen	-276.95	559.90	560.06	0.27	-	9.21	0.00
OU_pollen	-266.05	538.12	538.27	0.46	0.10	9.30	1.00
BM_ovule	-202.57	409.13	409.13	0.11	-	2.37	0.19
EB_ovule	-202.57	411.13	411.13	0.11	-	2.37	0.07
OU_ovule	-200.19	406.39	406.54	0.14	0.04	2.45	0.74

Table S5. Results of Generalized additive mixed models testing the effects of biotic and abiotic variables on pollen and ovule production. Test statistics (χ^2), associated probabilities (p), and estimated degrees of freedom (df) are shown for each variable. Note that smooth terms with low predictive performance were penalised toward zero during model fitting. Significant effects are shown with bold font.

Variable	Number of ovules			Number of pollen grains		
	(e)df	χ^2	p	(e)df	χ^2	p
Lifespan	1.00	0.71	0.3980	1.00	0.02	0.8860
Altitude	<0.01	<0.01	1.0000	1.06	22.32	<0.0001
Temperature seasonality	<0.01	<0.01	1.0000	0.93	8.44	0.0017
Precipitation seasonality	0.84	4.57	0.0171	<0.01	<0.01	1.0000
Climatic moisture index	0.83	4.54	0.0184	<0.01	<0.01	1.0000
Number of pollinators	<0.01	<0.01	1.0000	<0.01	<0.01	1.0000
Nutrient availability	<0.01	<0.01	1.0000	0.46	0.85	0.1662
Rooting conditions	0.71	2.30	0.0684	1.73	11.72	0.0009

Table S6. ANOVA results and effect sizes for pollen and ovule numbers. For each factor, we estimated the sum of squares (Sum_Sq), degrees of freedom (Df), F-statistics (F_value), p values, effect size (Eta_squared), and 95 % confidence interval (CI_95). Significant p-values are shown with strike and bold font.

Trait	Factor	Sum_Sq	Df	F_value	p_value	Eta_squared	CI_95
Pollen	ploidy	4989192300.58466	1	2.7314	0.1029	0.0376	0.95

Pollen	mating	44031251187.4489	1	24.1055	0.000005747*	0.2562	0.95
Pollen	ploidy:mating	1660640032.03896	1	0.9091	0.3436	0.0128	0.95
Pollen	Residuals	127862509390.471	70	-	-	-	-
Ovule	ploidy	150.023470862579	1	0.1728	0.6789	0.0025	0.95
Ovule	mating	0.205228211067151	1	0.0002	0.9878	0	0.95
Ovule	ploidy:mating	669.039256451957	1	0.7707	0.383	0.0109	0.95
Ovule	Residuals	60765.0852201855	70	-	-	-	-

Table S7. Summary of PGLS (OU) and OLS results with AIC comparisons based on the main tree. The table includes AIC values, intercept estimates, coefficients (β) for ploidy (Ploidy_Coeff), and mating system (Mating_Coeff), their associated p-values (Ploidy_p, Mating_p), and interaction effects (Interaction_Coeff, Interaction_p). Significant p-values are shown with strike and bold font.

Model	Trait	AIC	Intercept	Ploidy_Coeff	Ploidy_p	Mating_Coeff	Mating_p	Interaction_Coeff	Interaction_p
PGLS	Pollen	1789.12	4649.69	7349.18	0.56	43006.68	0.00*	-14636.33	0.41
PGLS	Ovule	705.19	20.69	5.27	0.52	-3.47	0.71	1.04	0.93
OLS	Pollen	1793.99	12836.18	-5057.89	0.76	63652.20	0.00*	-20430.89	0.34
OLS	Ovule	716.60	33.38	-10.80	0.35	-8.08	0.49	12.97	0.38

Table S8. Summary of PGLS (OU) and OLS results with AIC comparisons based on the constrained tree. The table includes AIC values, intercept estimates, coefficients (β) for ploidy (Ploidy_Coeff), and mating system (Mating_Coeff), their associated p-values (Ploidy_p, Mating_p), and interaction effects (Interaction_Coeff, Interaction_p). Significant p-values are shown with strike and bold font.

Model	Trait	AIC	Intercept	Ploidy_Coeff	Ploidy_p	Mating_Coeff	Mating_p	Interaction_Coeff	Interaction_p
PGLS	Pollen	1792.34	2276.87	80.96.54	0.55	50459.64	0.00*	-20038.79	0.3
PGLS	Ovule	707.28	22.17	2.82	0.74	-3.72	0.71	2.95	0.81
OLS	Pollen	1793.99	12836.18	-5057.89	0.76	63652.20	0.00*	-20430.89	0.34
OLS	Ovule	716.60	33.38	-10.80	0.35	-8.08	0.49	12.97	0.38

Dataset S1. Information on pollen and ovule numbers, mating system and ploidy level for the 158 Brassicaceae species used for phylogenetic and trait evolution analyses. Columns prefixed with ‘ref_’ indicate the source of the corresponding data value.

Dataset S2. Information on pollen and ovule numbers, abiotic and biotic data, and GPS coordinates for common garden species used to examine the relationship between reproductive traits and environmental factors. It consists of two sheets: Pollen_data and Ovule_data. Both sheets include detailed information at the individual level, including species identity (genus and tribe), individual identifiers, and lifespan. Species were assigned to tribes according to the classification by Hendriks et al., 2023. GPS coordinates correspond to population locations from which given individuals were collected and grown in the common garden. Each row corresponds to a replicate for pollen and ovule measurements (1-3 per individual). The Pollen_data sheet contains pollen number measurements, while the Ovule_data sheet contains ovule number measurements. Additionally, both sheets include a comprehensive set of environmental variables, such as geographic coordinates (latitude and longitude), climate indices (growing degree days [GDD] and climatic moisture index [CMI]), elevation (ALT), soil nutrient levels, rooting conditions, bee richness, and bioclimatic variables (B1, B4, B12, and B15).

Case study 4

How do newly arisen polyploids beat their initial disadvantage? A meta-analysis of the consequences of whole genome duplication on reproductive traits

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ABSTRACT

Polyploidy is considered to be one of the main diversification drivers, representing a widespread evolutionary phenomenon among angiosperms. Newly formed polyploids (neopolyploids) face multiple challenges to their establishment and yet, the high frequency of polyploid lineages in nature suggests that these challenges are successfully overcome. Here, we test the hypothesis that instant polyploidy-associated reproductive changes foster the establishment of polyploids amongst their diploid progenitors in two ways: i) higher reproductive fitness of polyploids, allowing them to survive and prevail despite their rarity, and ii) instant reproductive isolation, allowing the emergence of genetically independent polyploid lineages. In both scenarios, reproductive traits are the key candidates undergoing polyploidy-associated changes. We conducted a literature review supported by a meta-analysis to test this hypothesis, comparing diploids with their derived synthetic autopolyploids. Focusing on synthetic polyploids allowed us to look at early stages of polyploid formation. We found that 11 out of 15 reproductive traits undergo instant changes during polyploidization, with neotetraploids having increased size of flowers and multiple floral parts, delayed flowering time, increased pollen and seed size, but decreased pollen and seed germination as well as fewer seeds. We further discuss the functional implication of these results both for polyploid reproductive success and isolation from their diploid progenitors.

INTRODUCTION

Polyploidy, the presence of more than two sets of chromosomes, is “an evolutionary and ecological force in stressful times” (Van de Peer et al., 2021). The formation of unreduced gametes, which is considered to be the most common mechanism of polyploid formation (Harlan & de Wet, 1975; Mason & Pires, 2015), has been estimated to generally occur at low rates, especially in undisturbed environments (Kreiner et al., 2017). This suggests that polyploidization is a rare event. Therefore, polyploids form at lower frequency than diploids and, with random cross-pollination, newly formed polyploids (neopolyploids) face a high risk of extinction due to a frequency-dependent mating disadvantage known as minority cytotype exclusion (Levin, 1975). Moreover, nascent polyploids face challenges that are associated with their genetic instability, increased occurrence of meiotic aberrations, and other phenomena decreasing their fitness (Porturas et al., 2019; Clo & Kolář, 2021). Nevertheless, polyploidy is frequent in nature, as suggested by the reports of the proportion of 35% or 24% of polyploids among large datasets of extant plant species (Wood et al., 2009; Barker et al., 2016, respectively). This implies that neopolyploids are able to establish and persist despite their initial low frequency in early stages of their appearance.

The establishment of a new polyploid lineage likely relies on two early-acting factors: the direct impact of polyploidization itself on reproductive fitness, and the establishment of instant reproductive barriers from its diploid ancestor(s). The impact on reproductive fitness is likely to influence the survival and success of new polyploid individuals, while reproductive isolation may contribute to the emergence of a genetically independent polyploid lineage. Both the effect on reproductive fitness and reproductive isolation may be caused by immediate phenotypic changes often accompanying polyploidization. The increased genomic content (i.e. nucleotypic effect; Doyle & Coate, 2019) entails increased cell and organ

size (i.e. gigas effect; Müntzing, 1936), and altered gene dosage affects gene expression, overall leading to differences in morphology and floral phenology as well as changes in physiology and ecological tolerance (Ramsey & Ramsey, 2014; Doyle & Coate, 2019; Bomblies, 2020). These instant changes have been suggested to play an important role in polyploid establishment, although many of them are deleterious, leading to an initial decrease in fitness, which is only later reverted by subsequent evolutionary processes, among which likely selection (Clo & Kolář, 2021). Thus, the role of instant phenotypic changes due to polyploidization remains ambivalent.

A new, independent polyploid lineage may only emerge if reproductive barriers arise, reducing gene flow with its diploid ancestor(s) (Husband, 2000). On the contrary, complete reproductive isolation of a neopolyploid early in its establishment may directly contribute to its exclusion due to an unavailability of mating partners. Reproductive isolation is the consequence of different pre- and postzygotic barriers (Rieseberg & Willis, 2007). Interestingly, prezygotic isolation shows higher effect in gene flow reduction between ploidy levels compared to postzygotic isolation (Ramsey et al., 2003; Martin & Willis, 2007). One well known immediate reproductive barrier established by polyploidization is the triploid block, i.e. the inviability of interploidy seeds due to failure of the hybrid endosperm (Scott et al., 1998). While other instant reproductive barriers after polyploidization have been rarely studied, they are likely to exist. In particular, immediate phenotypic changes associated with polyploidy, such as increased flower size and delayed flowering time, possibly leading to a pollinator shift, may contribute to a reduction in gene flow between cytotypes (Segraves & Anneberg, 2016). Among all phenotypic changes instantly induced by polyploidization, those affecting reproductive traits are likely to play a crucial role in polyploid establishment, because they can both affect polyploid reproductive fitness and establish reproductive barriers.

Instant polyploidy-induced changes affecting floral traits are substantial and include increased flower size but decreased reproductive outputs (Porturas et al., 2019), suggesting an ambivalent impact of polyploidization on reproductive fitness. Regarding the immediate consequences on interploidy reproductive barriers, some studies found a shift in floral phenology in newly formed polyploids (e.g. Joshi & Verma, 2004; Dibyendu, 2010), while other studies did not (e.g. Husband et al., 2016; Innes et al., 2021), suggesting an ambiguous role of polyploidy-induced floral changes as an instant prezygotic barrier. Similarly, the few studies focusing on the instant impact of polyploidy-induced floral changes on pollinator behavior did not find any sign of pollinator shift between cytotypes (e.g. Porturas & Segraves, 2020; Schmickl et al., 2024).

Aspiring to address the evolutionary significance of polyploidization, many experimental studies aimed to evaluate the phenotypic and ecological consequences of genome doubling using diploids and natural polyploid populations (e.g. Husband & Schemske, 2000; Nuismer & Cunningham, 2005). However, such comparisons do not allow for clear differentiating between the effects caused by polyploidization itself, possible gene flow, and evolutionary processes acting later, such as selection or genetic drift (Bomblies, 2020). Instead, synthetically induced polyploids may be used to simulate natural autopolyploids in the earliest phase after their emergence, thereby being invaluable for examining the initial stage of whole genome duplication (WGD) and the effects of WGD on phenotypic traits and the genotype (Porturas et al., 2019). To induce polyploidy, antimitotic agents, which are compounds interfering with mitosis, are frequently used, both in horticulture (Eng & Ho, 2019) and scientific research (Westermann et al., 2024). Importantly, the phenotypic impact of polyploid induction seems to be lineage- or species-specific, possibly even genotype-specific (Anneberg et al., 2023). Thus, comparisons across different plant groups are

necessary when one aims to infer the processes behind the establishment of new polyploid lineages. At the same time, it needs to be taken into consideration that antimitotic agents are toxic compounds which might have an impact on the morphology and performance of plants that undergo polyploidy induction treatment. However, there exist methodological approaches that decrease the probability that the phenotype of the analyzed plants will be influenced by the antimitotic agent.

In this article, we review studies that used diploids and derived synthetic tetraploids of the same genetic background, using a meta-analysis to compare their differences in selected traits associated with sexual reproduction, such as floral organ size and flower number, floral phenology, and pollen and seed characteristics. Consequently, we focus only on studies on angiosperms, and we do not include studies on synthetic allopolyploids. This enables us to distinguish between effects of genome duplication and effects associated with the merging of subgenomes as a result of hybridization. Porturas et al. (2019) conducted a similar type of review article, including also studies on allopolyploids, while Clo & Kolář (2021) focused on autopolyploids only, but included studies on established polyploids (i.e. polyploids from natural populations). Our meta-analysis applies a different approach. Focusing exclusively on diploids and their synthetic autotetraploid derivatives, we aim to explore the effects of polyploidization on specific reproductive traits that have been evaluated in these previous two review articles only within broader trait categories. This allows us to investigate the functional implications of polyploidy-associated phenotypic changes and their potential impact on neopolyploid reproductive fitness and reproductive isolation (see Figure 1 for our working hypotheses). We cover several new traits and tens of articles that have not been studied in the previous meta-analyses by Porturas et al. (2019) and Clo & Kolář (2021). More specifically, our objectives are to 1) identify reproductive traits reported to

change after induced polyploidization; 2) discuss how reproductive trait changes may positively or negatively affect the reproductive fitness of neopolyploids, and how they could be linked to the formation of reproductive isolation mechanisms; 3) summarize the methodologies applied in the reviewed studies, mainly regarding the specifics of plant treatment with antimitotic agents; and 4) summarize the knowledge gaps and advocate perspectives in future research in this field.

META-ANALYSIS

Literature search

In a first step, a thorough literature search was employed to select the studies to be suitable for fulfilling the objectives of the review. We performed a search in databases of Web of Science and Google Scholar between June 2021 and March 2024, using combinations of the keywords “induc*”, “synthe*”, “*polyploid*”, “colchicine”, “oryzalin”, “trifluralin”, “plant*”, “flower*”, “floral”, and “pollen”. A study was considered to be relevant and was included in the meta-analysis, if it met all of the following criteria: 1) It involved the production of synthetic autotetraploids from diploids. The taxa used in the experiments must have been specified, and only studies on angiosperms were taken into consideration; 2) At least one of the reproductive traits analyzed in the present study must have been measured and compared between the diploid and its synthetic autotetraploid derivative. In advance, we prepared a list of 25 traits (Table S1) which we considered to have a potential impact on reproductive success; 3) Numeric values of the focal traits and sample sizes must have been specified for both cytotypes; 4) Methods of either chromosome counting or flow cytometry must have been used to confirm the tetraploid status of the plants treated with antimitotic agents at least once in the process. Some studies utilized indirect methods of induced polyploidy detection by using morphological or anatomical features as markers, but we did not consider these studies reliable.

For each article that met the above mentioned criteria (Table S2), the following information was recorded: reference, name of the species used and its taxonomic classification into family, agent used for polyploidy induction, information if the authors analyzed the traits in diploid plants that also underwent treatment with the polyploidy-inducing agent and stayed unconverted, and information if the authors analyzed plants grown from tissue directly treated by the agent or plants from descendant seed generations derived from the treated individuals. Iannicelli et al. (2016) reported measurements of both untreated and treated (unconverted) diploid plants, and Münzbergová (2017) compared measurements of untreated diploids and diploid offspring emerged from synthetic tetraploids with measurements of induced neotetraploids. We used the values of untreated plants in these cases to remain consistent with most remaining studies that only made this kind of comparison. Some studies provided trait measurements for multiple accessions or genotypes within a single species. In these cases, we calculated mean values for each cytotype to avoid pseudoreplication. When studies reported data from different species, intraspecific varieties, cultivars, or other entities that were expected to show a significant differentiation in reproductive traits, we treated them independently, stated as standalone experiments in separate rows of the table. The dataset compiled from the literature survey included observations of 106 experiments extracted from 88 studies (Table S2). Noteworthy, 22 studies were found during the literature search that reported comparisons of traits focal to our research aims, but they could not be included in our meta-analysis, as the authors did not include sample sizes for the specific measurements (Table S3).

STATISTICAL ANALYSIS AND RESULTS

Among the 25 focal traits (Table S1), we selected and further analyzed 15 traits that have been reported in at least five experiments. In the case of seven studies, the trait values were not

directly specified in the text or tables but were extracted from figures included in the article, using the library *metaDigitise* (Pick et al., 2019) in R (R Core Team, 2022). Whenever possible, the mean value, standard deviation (SD), and sample size (n) were recorded for each of the focal traits and both cytotypes. In 27 studies, the authors provided the standard error (SE) instead of the standard deviation. In those cases, the standard deviation was recalculated using the formula ($SD = SE \times \sqrt{n}$). Studies that did not report SDs or other measures of variability that can be transformed into SDs were retained, and their SD values were estimated using log-log regression of SDs on means (Higgins et al., 2019). To predict SDs, we used a linear mixed model (Pinheiro & Bates, 2000) with plant species treated as random intercepts. The model fitted the data reasonably well ($F(1, 425) = 1826$, $p < 0.0001$, marginal $R^2 = 0.75$) and showed good predictive performance in cross-validation ($r = 0.92$). Since this imputation technique makes assumptions about an unknown relationship, we performed a sensitivity analysis to assess the impact of data imputation on the results of the meta-analysis. For all traits except pollen germination and style length, the same general conclusions were reached regardless of whether the datasets contained imputed SDs or not. Therefore, we retained the estimated SDs in the models except for those of pollen germination and style length. For pollen germination, the dataset with missing values was used. As the dataset for style length included only three studies with full information, we excluded this trait from the meta-analysis.

To quantify effect sizes, we calculated log-transformed response ratios (Hedges et al., 1999) as the average trait value of synthetic tetraploids over the average value of diploids for each paired measurement. Thus, a positive value of the response ratio means that tetraploids reached larger values than diploids and vice versa.

The independence of individual effect sizes is a fundamental assumption of a meta-analysis (Borenstein et al., 2009). Given that the structure of our data was inherently hierarchical, with multiple experiments per study and repeated observations of the same taxa, the independence assumption is clearly violated. Therefore, we performed a multilevel random-effects meta-analysis that allowed us to account for non-independence and to quantify the different variance components (Nakagawa & Santos, 2012; Fernández-Castilla et al., 2020). Specifically, the four-level random-effects model was used to quantify the sampling variance (level 1), the variance between experiments within studies (level 2), the variance between studies within species (level 3), and the variance between species (level 4). Since nested grouping levels were completely confounded in some traits (e.g. each species investigated in a single study or studies involving one experiment only), we reported the variance for higher hierarchical levels in those cases. The multilevel models were fitted using restricted maximum-likelihood estimation (Raudenbush 2009). Variance estimates (τ^2) and heterogeneity indices (I^2) were used to quantify variability among nested levels (Nakagawa & Santos, 2012), and Cochran's Q-tests were employed to evaluate residual heterogeneity (Hedges & Olkin, 1985). We reported mean effect sizes with t-tests of no differences between diploids and neotetraploids. Ninety-five percent confidence and prediction intervals were displayed in orchard plots to visualize the differences between ploidy levels. Since all models we fitted showed very high significant heterogeneity ($Q = 24 - 40454$, $I^2 = 74 - 100\%$), we followed the recommendation of Harrer et al. (2022) and completely avoided publication bias analyses, as those can be misleading. The analyses were performed in R using the libraries *lme4* (Bates et al., 2015), *lmerTest* (Kuznetsova et al., 2017), *metafor* (Viechtbauer, 2010), and *orchaRd* (Nakagawa et al., 2021).

Studies that were used in the meta-analysis (Table S2) were done on 73 genera representing

42 angiosperm families, while the genera from Asteraceae, Fabaceae, Brassicaceae, and Rosaceae were the subjects of the highest number of articles. In the majority of studies (81%), colchicine was used as the antimitotic agent, while in the remaining ones other agents were applied (e.g. oryzalin, nitrous oxide gas), or the authors reported results of experiments that involved multiple agents. In 10 studies, the authors reported the traits of diploid plants that underwent the same treatment as the induced polyploids but stayed unconverted. In 19 studies, traits were analyzed from at least the first seed generation derived from the treated plants. Flower or petal size represented the trait that was measured in the largest number of studies (59%), while four out of 25 of the preselected traits were not addressed in any study that met our criteria (Table S1). Fifteen traits that have been reported in at least five experiments were selected and included in the meta-analysis (Table S1).

The meta-analysis revealed significant differences between ploidy levels in 11 out of 15 studied traits (Figure 2; Table S4). Synthetic autotetraploids showed significantly larger flowers or petals, longer stamens, greater anther diameter, and larger pollen grains, ovaries, and seeds (or larger seed weight) as well as delayed flowering time (i.e. days to onset of anthesis), while diploids displayed significantly higher pollen and seed germination, and larger seed set (measured in seed set/fruit and total seed yield). In contrast, flower number (per inflorescence or per individual plant), flowering period (i.e. days of anthesis), and pollen viability were comparable between both cytotypes.

All models showed very high heterogeneity as indicated by highly significant results of Cochran's Q-tests, heterogeneity indices, and corresponding variance components (Table S4). The high heterogeneity is also demonstrated by broad prediction intervals of the effect sizes that include zeroes (Figure 2). The prediction intervals did not include zero for stamen length, pollen size, and pollen germination, suggesting

that also future studies on cytotype differences will likely reveal increased stamen length and pollen size, accompanied by lower pollen germination in neotetraploids. Interspecific differences were the major source of variation in the studied traits (Figure S1).

Synthesis

Before discussing the results of our meta-analysis and specific traits, we highlight the inconsistency in the applied methodologies found among the analyzed studies and their potential impact on the interpretation of our results. Even though it is generally acknowledged that the agents used for induction of polyploidy (mostly colchicine) are toxic compounds, there is little known about its actual effects on specific phenotypic traits, especially fitness-related traits, including those targeted in our meta-analysis. Münzbergová (2017) warns that the phenotypic changes observed in plants surviving the application of antimitotic agents may be associated with various effects other than genome duplication, for example by its possible impact on the production of plant hormones, or by the strong selection pressure that is exerted during the treatment. We attempted to review the nuances in the methodological approaches applied in the selected studies, focusing on the two approaches explained below in more detail. However, the methodological descriptions of the inspected studies were sometimes unclear in this regard, and it was thus impossible to unambiguously deduce which plants were used for the specific measurements and analyses. For example, it was quite often reported by the authors that they compared induced neopolyploids with "diploid control plants", but it was not clear if these "control plants" refer to diploids that underwent the same treatment and remained diploid, or those that were not treated. Hence, we only report here the cases in which we were able to state the use of the methodological approach. One option to account for the toxicity of the polyploidy-inducing agent is to compare induced neopolyploids with diploids that underwent the same treatment but remained

unconverted, as mentioned above. Only 10 studies of those included in our meta-analysis used treated (unconverted) diploids for the measurements (Table S2). To uncover the actual effects of the antimitotic agent, treated and untreated diploids should therefore be compared in the same experiment. However, this procedure was performed very rarely. Husband et al. (2016) found significant differences in all three analyzed traits between treated (unconverted) and untreated diploids of *Chamerion angustifolium* in the directly exposed generation, while the differences completely diminished in the first seed (F1) generation. On the other hand, Wei et al. (2020) studying different (non-reproductive) traits in *Fragaria* found the effect of colchicine negligible when comparing the treated unconverted and untreated diploid plants. Another option that is expected to significantly diminish the effects of the polyploidy-inducing agent *per se* is to analyze the plants from the derived seed generation instead of the plants that have been directly treated by the agent at some stage of its development. Oswald & Nuismer (2011) observed reduced growth rates of the plants treated with colchicine, and therefore did not make measurements on the exposed generation but instead produced the first seed generation for further analysis. Such an approach was applied in 19 (22%) of the inspected studies (Table S2). Münzbergová (2017) addressed this problematics in the most complex way: she reported that the effects of the colchicine application on the plant phenotype are apparent not only in the treated maternal generation, but also in the F1 generation, and suggested that it would be better to use the F2 generation for future comparisons. The combination of these two options, using treated (unconverted) diploids as well as the F1 generation for analysis, could be considered an ideal methodology to account for the toxicity of the induction agents. We found only four studies that applied both options (Husband et al., 2016; Münzbergová, 2017; Siopa et al., 2020; Schmickl et al., 2024). Focusing only on these few methodologically ideal studies would not

have allowed us to draw any conclusions, and we instead considered the numerous studies that did not take all possible measures to avoid a potential toxicity effect of the antimitotic agent.

PART 1: Higher reproductive fitness of polyploids compensating for their rarity?

1.1 Premating phase: flower size, flower number and flowering period

The finding of bigger flowers in neopolyploids has previously been interpreted to increase flower visitation, which could aid nascent polyploids in their successful establishment, since it would improve their reproductive fitness (Vamosi et al., 2007). Consistently, an increase in the size of petals (e.g. Ye et al., 2010; Huang et al., 2015) or whole flowers (e.g. Abdoli et al., 2013; Zahumenická et al., 2018) in induced polyploids was observed in tens of studies included in our meta-analysis, thereby constituting a significant difference between neopolyploids and their diploid progenitors (Figure 2; Table S4). Induction of polyploidy was associated with an enlargement of additional floral parts in our meta-analysis, in particular stamen length as well as anther diameter and ovary size (Figure 2; Table S4). However, as these traits play no obvious role in pollinator attraction, they likely do not impact flower visitation. Overall, if an increase in floral organ size as a result of a WGD event may provide an instant advantage in reproductive fitness remains largely untested, as the few studies focusing on pollinator behavior in synthetic polyploids mostly looked at pollinator movement between cytotypes as a reproductive barrier (e.g. Porturas & Segraves, 2020; Schmickl et al., 2024) instead of the potentially increased attractiveness of neopolyploids *per se*. While the size of single flowers or its parts may have an influence on pollinator attraction, the overall floral display (i.e. number of open flowers) may also be important. The number of flowers per inflorescence or the number of flowers per individual plant may affect the attractivity for pollinators, by attracting inexperienced pollinators with large displays or

by alluring experienced pollinators with the promise of greater rewards (Makino & Sakai, 2007). Our meta-analysis revealed that overall diploids and synthetic autopolyploids do not substantially differ in flower number per inflorescence or individual plant (Figure 2; Table S4). Some studies showed that the number of flowers per inflorescence in synthetic polyploids was lower in comparison to diploids (e.g. Ye et al., 2010; Hannweg et al., 2013), but the opposite was also found (e.g. Zahedi et al., 2014; Chambhare & Nikam, 2022).

Flower color and scent could also influence the attractivity of a flower to flower visitors. These traits are attributable to the content of secondary metabolites. For example, the higher concentration of a particular pigment can lead to higher reflectance and thus a more vivid flower color, which could contribute to higher flower visibility and attractiveness (Trunschke et al., 2021). As many pollinators are scent-oriented, this may also apply to specific odor (Gfrerer et al., 2021) or the amount of pollinator reward, such as nectar (Barberis et al., 2024). Evidence on changes in floral scent- or color-related traits in induced polyploids is scarce and obscure, and we did not include these traits in our meta-analysis due to the scarcity of relevant studies (Table S1).

Finally, for successful reproduction it may be beneficial for a plant to enhance the floral display not only in space but also in time, as a longer flowering period allows for more pollinator visits. However, our meta-analysis did not show a difference between diploids and neopolyploids (Figure 2; Table S4). The flowering period was either shorter or longer compared to the diploids, depending on the species (e.g. Oswald & Nuismer, 2011; Husband et al., 2016).

In conclusion, the direct impact of polyploidization on floral changes and their consequences on reproductive fitness of neopolyploids remains ambiguous and requires more investigation.

1.2 Postmating pre-fertilization phase: pollen size and mating success

When pollen is deposited on the stigma, the number of pollen often exceeds the number of ovules, which leads to competition between pollen grains to fertilize ovules (Moore & Pannell, 2011). Various pollen traits such as pollen protein content, pollen size, production, viability, germination, and pollen tube growth rate (Haghighatnia et al., 2023; Russo et al., 2025) might contribute to the male reproductive success in the face of pollen competition for mating.

We found significantly enlarged pollen size after polyploidy induction in our meta-analysis (Figure 2; Table S4). The increase in pollen size associated with cell and genome enlargement as a consequence of WGD is one of the well documented changes linked to polyploidization (Tsukaya, 2013). Pollen size has been associated with the amount and type of nutrients stored (Baker & Baker, 1979), and larger pollen size has been associated with higher siring success (McCallum & Chang, 2016). Thus, bigger pollen grains produced by polyploids may provide more energy, increasing the speed of pollen germination and tube growth, and consequently they may increase polyploid reproductive fitness. At least theoretically, the bigger pollen from polyploids may occupy a greater surface on the stigma and reduce the chance for pollen from diploids to be deposited there as well, providing polyploids with an additional reproductive fitness advantage. On the other hand, an increase in pollen size after polyploidization is associated with an increase in the amount of energy that pollen grains require for germination and pollen tube growth (Williams & Oliveira, 2020). Our meta-analysis suggested that pollen germination associated with polyploidization significantly decreases (Figure 2; Table S4). For example, pollen germination in induced tetraploid *Raphanus sativus* was lower compared to the diploids (Limera et al., 2016), evidencing decreased male reproductive fitness in newly formed

polyploids. A contradicting finding was recently published: in *Arabidopsis arenosa*, Westermann et al. (2024) did not reveal a significant difference in the in vitro pollen germination rate between synthetic tetraploids and their diploid progenitors.

We could not include the trait pollen number in our meta-analysis because of very few available studies (Table S1), and these studies are contradicting each other. Shao et al. (2003) showed in *Punica granatum* that the number of pollen grains per anther in synthetic polyploids is higher than in diploids, while Husband et al. (2016) observed the opposite in *Chamerion angustifolium*. Without additional studies, it is therefore impossible to predict whether neopolyploids show higher male reproductive success or not via an instant change in pollen production.

Finally, there is only a single, recent study comparing the pollen tube growth rate (PTGR) between diploids and synthetic polyploids, even though it represents a crucial factor in pollen reproductive success in the face of pollen competition for ovule fertilization. Westermann et al. (2024) compared the PTGR of diploids with that of induced tetraploids and established tetraploids of *Arabidopsis arenosa*, and found a slower pollen tube growth of neotetraploids compared to the diploids but also established polyploids. A comprehensive study on natural polyploids of 451 seed plants by Reese & Williams (2019) showed that the PTGR of very young autopolyploids tends to be lower, but most of the established polyploids demonstrate fast PTGRs which are equal or outperforming those of their diploid counterparts. Hence, it becomes evident that WGD leads to a decrease in PTGR, and this may be compensated by evolutionary processes in later stages of polyploid establishment (Reese & Williams, 2019).

1.3 Post-fertilization phase: seed size, number and viability

Seeds that arose from induced polyploids turned out to be significantly larger or heavier compared to their diploid progenitors in our meta-analysis (Figure 2; Table S4). It is expected that polyploidy-induced seed size increase may directly affect the fitness of the plants, when considering the effect of seed size or weight changes on seed germination, seedling establishment, and seed dispersal. In general, larger seeds have more resources for the embryo, and tend to germinate earlier (Moles et al., 2005), produce larger seedlings compared to smaller seeds (Moles & Westoby, 2004), and impact adulthood fecundity (Halpern, 2005). On the contrary, the advantage of small seeds on germination ability and plant growth rate has also been shown in several studies (Turnbull et al., 2012). Thus, the correlation between seed size, seed germination, and seedling performance might depend on seed characteristics, but also on the specific metrics, for example of the growth rate (Simpson et al., 2021). In addition, seed size has been shown to influence seed dispersal, which can have an impact on the establishment of a new population, geographic isolation, and adaptive divergence (Leslie et al., 2017). Thus, the increased seed size of neopolyploids may strongly affect the establishment of new polyploid populations and the divergence from their diploid progenitors.

However, our meta-analysis revealed that seed germination rates of induced polyploids are significantly lower compared to seeds of their diploid counterparts (Figure 2; Table S4). Based on this result, it is apparent that the larger seed size of neopolyploids does not promote higher germination rates, and the increased seed size may be due to the gigas effect, without significant changes in the resource content of the seed. Increased seed size in neopolyploids may nevertheless affect dispersal, in a positive or negative way depending on the mode of dispersal (e.g. large seeds may aid in zoochory, small seeds in anemochory), and this may help or hinder neopolyploids to establish and colonize a potentially new environment.

Finally, in our meta-analysis we found a significant decrease in seed set per fruit and total seed yield of induced polyploids when compared with their diploid progenitors (Figure 2; Table S4). This may partially be explained by decreased pollen germination in neopolyploids, which has been shown by our meta-analysis and discussed in section 1.2. Interestingly, the recent findings of Westermann et al. (2024) indicated slower pollen tube growth of synthetic polyploids, which would result in decreased seed set under a pollen tube growth competition scenario involving diploids. Alternatively, a decrease in seed set may be explained by ovule sterility in neopolyploids, which was shown to occur in synthetic *Chamerion angustifolium* (Husband et al., 2008). Also, a lower number of ovules in neopolyploids could explain a lower number of seeds, but no difference was found by Husband et al. (2008).

In summary, it appears that neopolyploids show lower seed germination rates compared to diploids despite larger seeds, and they also show a decrease in the amount of seeds they produce, suggesting that their reproductive fitness is impaired. These results suggest that neopolyploids show a decreased reproductive output, and this is likely to be an obstacle to their establishment.

PART 2: Instant polyploid reproductive isolation promoting cytotype differentiation?

2.1 Premating phase: floral phenology and pollinator shift

The manifestation of a neopolyploid lineage may eventually require the formation of reproductive barriers. Premating barriers between neopolyploids and their diploid progenitors limit interploidy gene flow, which is crucial for escaping the extinction risk due to minority cytotype exclusion.

Changes in phenological traits, such as the first day of anthesis or the flowering period, were frequently observed in neopolyploids. Our meta-analysis supported this, showing significantly delayed flowering in synthetic

polyploids (Figure 2; Table S4). One plausible explanation relies on the fact that genome duplication is associated with the enlargement of cells, and therefore cells are expected to take longer to divide (Doyle & Coate, 2019). Consequently, this might slow down the overall development of the plant, potentially delaying its flowering time (Ramsey & Schemske, 2002). Interestingly, this trend was not statistically supported by the previous meta-analyses (Porturas et al., 2019; Clo & Kolář, 2021) which found no differences in floral phenology between diploids and neopolyploids. This discrepancy with our meta-analysis may be explained by our focus on autopolyploids, while Porturas et al. (2019) also included allopolyploids, likely adding hybridization as a confounding factor. Also, we analyzed more studies compared to Porturas et al. (2019) and Clo & Kolář (2021), which gave us a stronger statistical power. Changes in floral phenology caused by polyploidization may serve as an immediate premating reproductive barrier. It is, however, unlikely to be a complete barrier, as the flowering time and period of diploids and neopolyploids still overlap.

As previously mentioned (see section 1.1), the abundance and composition of pollinators visiting plants of a nascent polyploid may be influenced by changes in its floral morphology or structure. Studies on the behavior and composition of pollinators on induced polyploids and their diploid progenitors are very scarce, and consequently we could not include them in our meta-analysis (Table S1). The general trend seems to be a lack of cytotype preference of pollinators in comparisons of diploids and induced tetraploids, and also established tetraploids (Nghiem et al., 2011; Husband et al., 2016; Porturas & Segraves, 2020). This suggests that instant floral changes induced by polyploidization are not enough to lead to a pollinator shift and a reproductive barrier between diploids and neopolyploids. Even further, a recently published study indicated nonrandom preference of flower visitors for switching between diploids and

synthetic tetraploids (Schmickl et al., 2024), potentially preventing the establishment of a reproductive barrier.

In contrast, established polyploids have been shown to be able to attract partially different groups of pollinators, which could potentially lead to their reproductive isolation from the diploid progenitors (Segraves & Anneberg, 2016), even though at this later stage in polyploid establishment, the pollinator spectrum was rarely unique for particular cytotypes (but see Roccaforte et al., 2015). This suggests that pollinator shift might not play a major role in the early or late divergence between diploids and polyploids. Segraves (2017) points out to the complexity in plant-pollinator interactions and advocates that attention should be paid to studies on pollinator behavior in a community context, focusing on multiple guilds of pollinators associated with given diploid-polyploid cytotypes. The apparent lack of a clear pollinator shift in early or established polyploids might thus be due to technical artifacts.

2.2 Postmating phase: pollen competition and triploid block

Pollen competition on the stigma or style can act as a hybridization barrier in the case of foreign pollen competing with pollen from the same lineage as the maternal plant. As a result of pollen-pollen and/or pollen-pistil interaction, ‘local’ pollen may be favored for ovule fertilization to the detriment of foreign pollen, preventing hybridization (Rieseberg et al., 1995). When it comes to neopolyploids, to our knowledge, there is no study that crossed diploids and synthetic polyploids to test whether maternal plants would discriminate in favor of pollen of the same cytotype. Alternatively, a hybridization barrier may arise if the pollen of one lineage shows higher performance (higher viability, germination, or PTGR), leading to this pollen outcompeting the other one and siring more seeds independently of the maternal plant. While there is no experiment involving mixed pollen load to test for diploid-neopolyploid

pollen competition, in diploids and natural tetraploids of *Chamerion angustifolium* the pollen grains from tetraploids were found to consistently sire more seeds in both diploid and tetraploid maternal plants (Husband et al., 2002). The authors interpreted this result as a siring advantage of tetraploids over diploids during pollen competition. This may foster tetraploid establishment in sympatric populations by maximizing tetraploid success. However, the slower pollen tube growth of synthetic tetraploids of *Arabidopsis arenosa* (Westermann et al., 2024) indicates lower siring success of neopolyploids at their earliest stage, counteracting their establishment.

Ultimately, the fate of neopolyploids may be threatened by their inability to produce viable progeny when mating with their more abundant diploid progenitors. The fusion of gametes produced by a diploid and a tetraploid may result in inviable triploid seed, referred to as the triploid block, resulting from a maternal and paternal ploidy disbalance having a negative impact on the formation of the endosperm (Martinez et al., 2018). Several studies have found large seed inviability in crosses between diploids and synthetic polyploids (e.g. Valentine & Woodell, 1960; Scott et al., 1998). The triploid block has been extensively studied and is not the direct focus of this article, and therefore we invite the reader to consult previously published reviews on this topic (e.g. Köhler et al., 2021).

Conclusions and perspectives

We investigated whether instant changes in reproductive traits due to polyploidization could lead to a significant increase in reproductive fitness and the establishment of reproductive barriers between newly formed polyploids and their diploid progenitors. These changes act both at the pre- and postmating stage and involve organs associated with sexual reproduction: the flower, pollen, and seed. Changes in reproductive traits potentially affect the reproductive fitness of the initially rare

polyploids in the diploid progenitor populations, by increasing pollinator attractivity, pollen and seed(ing) performance. Changes possibly establishing reproductive barriers between cytotypes are discussed to be phenological shifts, pollinator shifts due to changes in floral morphology, pollen competition, and certainly the triploid block, affecting endosperm development in the seed. Several of these changes in reproductive traits are in fact of dualistic nature: they affect polyploid reproductive fitness while at the same time impacting reproductive barriers between polyploids and their diploid progenitors.

While our meta-analysis identified clear effects on certain reproductive traits (see Figure 3 for a schematic summary), these changes did not indicate any increase in fitness or rise of hybridization barriers that would be significant enough to balance the obstacle of minority cytotype exclusion. The reasons for that may be multiple. First, studies on synthetic polyploids often use different methodologies (type of antimutagenic agent, controls), hindering direct comparisons and calling for a standardised approach in the future (Münzbergová, 2017). Then, the instant phenotypic changes upon polyploidization are barely studied for some reproductive traits, such as pollen biology or pollinator behavior, making it difficult to elaborate firm conclusions about instant potential fitness gain or interploidy reproductive barriers. Third, neopolyploids may not be fitter than their diploid progenitors (Clo & Kolář, 2021) or only in subsequent generations (Anneberg et al., 2023). Consistently, we often found that synthetic polyploids likely suffered from decreased fitness rather than the other way around, as our meta-analysis showed, for example, decreased seed set and seed germination of neopolyploids. The early establishment of polyploids may thus not rely on better fitness. A recent theoretical study actually proposed that polyploids may emerge by genetic drift, such as in small marginal populations, even if they are less fit than diploids (Clo et al., 2022). Environmental

factors may further impact the fixation of newly formed polyploids in a population, although the simulated environmental change in Cheng et al. (2024) needed to be rather strong for polyploid fixation. Finally, the rise of reproductive barriers between diploids and neopolyploids may be irrelevant for the establishment of a polyploid lineage. It may even hinder it, if happening too early, as it would lead to the limitation of mating partners at the initial stage of polyploid formation. In other words, the first polyploid individuals within an otherwise diploid population may rely on initial gene flow with diploids, as observed by Schmickl et al. (2024), while in natural polyploid populations with predominantly sexual reproduction intermediate cytotypes are rare to absent (e.g. Husband and Sabara, 2004; Pegoraro et al., 2016), indicating the establishment of pre- and postzygotic barriers, possibly even within a relatively short time after polyploid formation (Simón-Porcar et al., 2017).

Box 1. Bullet point summary

- Most reproductive traits in plants undergo instant changes during polyploidization.
- These immediate changes in reproductive traits potentially affect the reproductive fitness of the initially rare polyploids, by increasing pollinator attractivity, pollen and seed(ing) performance.
- Changes establishing reproductive barriers between cytotypes are phenological shifts, pollinator shifts, pollen competition, and the triploid block.
- Several of these instant changes in reproductive traits are of dualistic nature: they affect polyploid reproductive fitness while impacting reproductive barriers between cytotypes.
- Despite these clear effects on certain reproductive traits, these changes are not significant enough for polyploids to overcome minority cytotype exclusion.

Box 2. Open questions

- Do we need a standardized approach for the application of antimutagenic agents in studying synthetic polyploids?
- Which effects do polyploidization-induced, instant changes in reproductive traits have on pollinator behavior?
- Are neopolyploids maybe not fitter than their diploid progenitors?
- Is the rise of reproductive barriers between diploids and neopolyploids irrelevant for the establishment of a polyploid lineage or may it even hinder it, if happening too early?
- How do instant changes in reproductive traits during polyploidization translate into large-scale diversification?

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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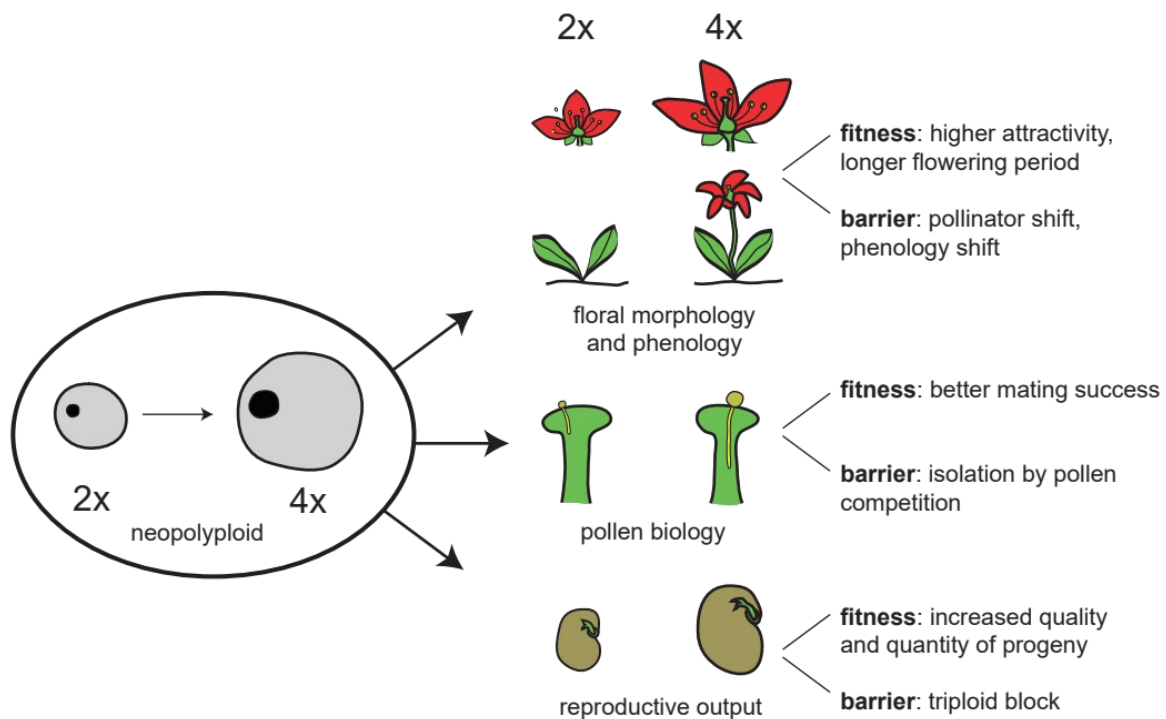


Figure 1. Working hypotheses of this article. The left part shows the categories of reproductive traits affected by polyploidy induction (floral morphology and phenology, pollen biology, reproductive output). In the right part, the theoretical consequences of polyploidy-associated changes in reproductive traits are listed, which are expected to play a role in the establishment of neopolyploids, either by having a direct impact on their reproductive fitness, or by promoting the establishment of reproductive barriers.

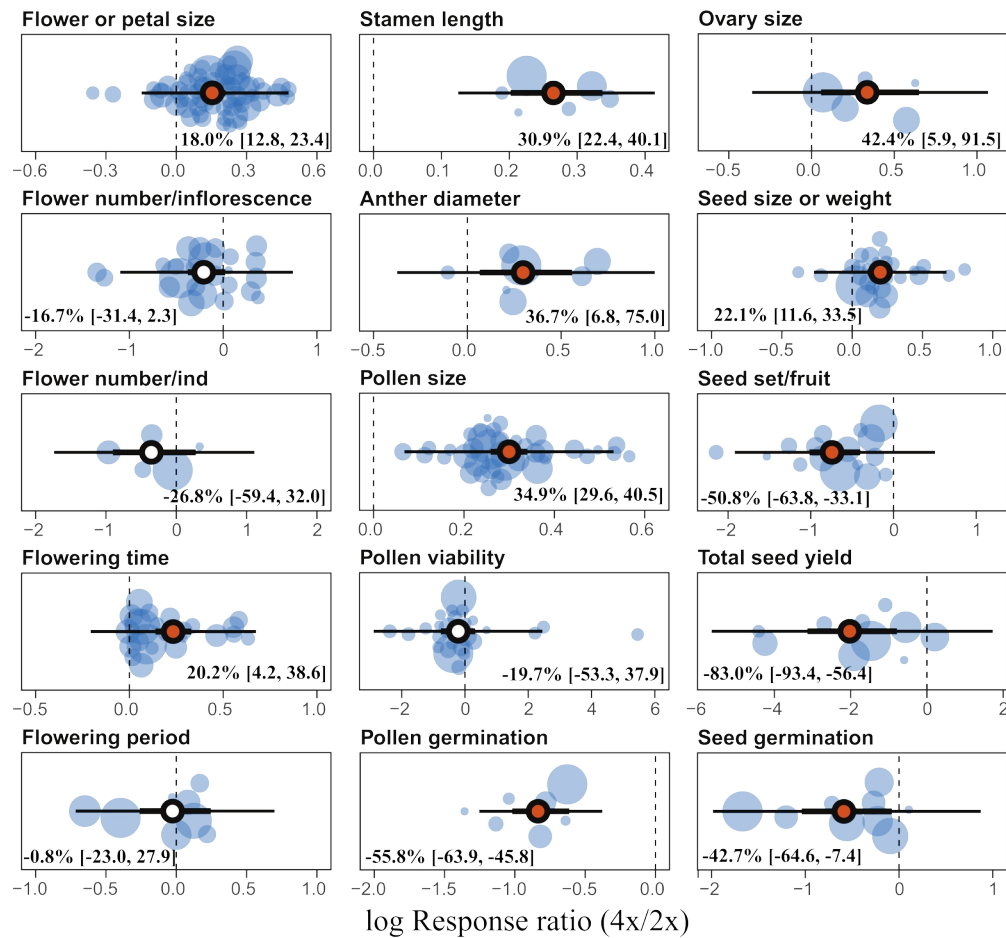


Figure 2. Orchard plots of the 15 investigated plant reproductive traits showing estimated log response ratios (black circles), their 95% confidence intervals (thick error bars) and prediction intervals (thin error bars). Semi-transparent circles depict effect sizes of individual observations; their size is proportional to their weights in the analysis. Back-transformed response ratios, converted to percentages, and their 95% confidence intervals (in square brackets) are provided in italics. Significant differences between diploids and neotetraploids ($p < 0.05$) are highlighted by the red color of the estimated effect sizes. Details on the studies involved in each model are given in Figure S1.

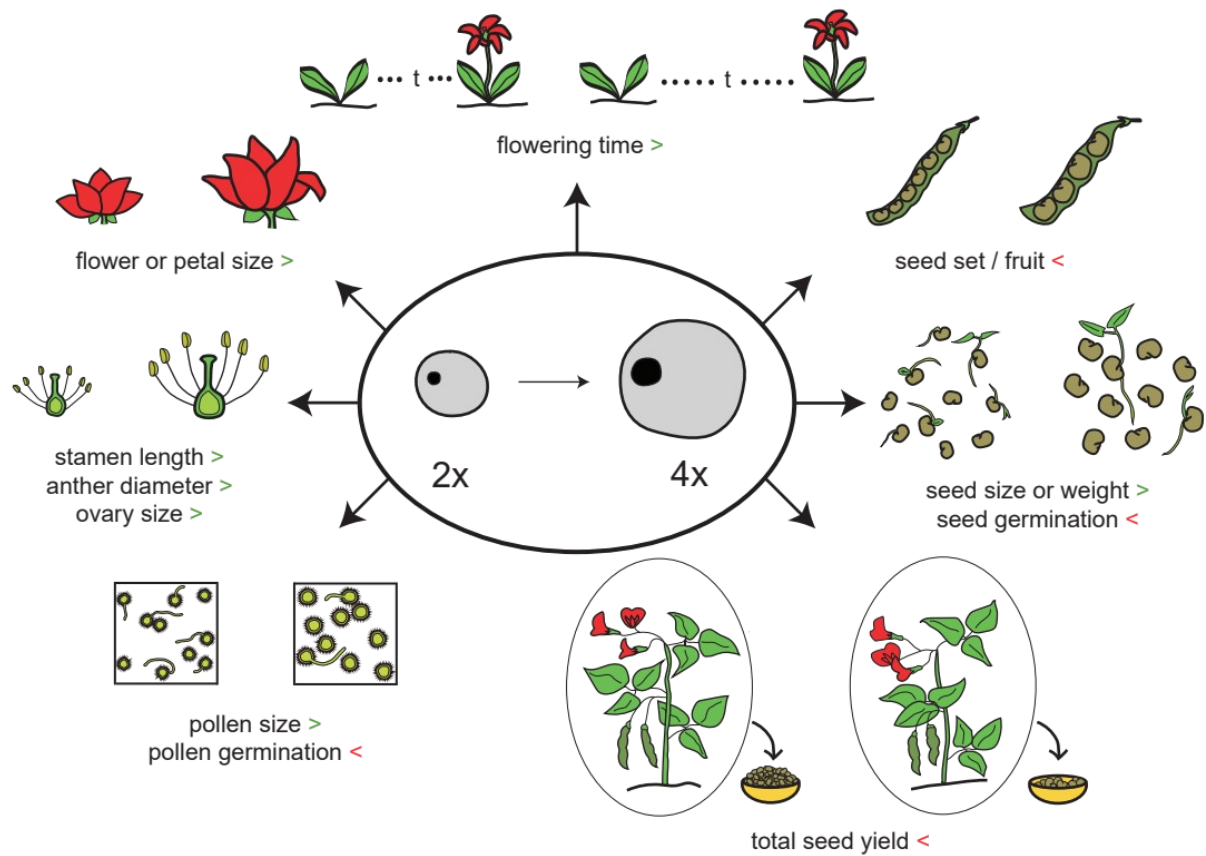


Figure 3. Schematic summary of the statistically significant effects of polyploidization on plant reproductive traits. An increase in trait values is depicted with > while a decrease is shown with <.

Case study 5

From micro- to macroevolution: sexual selection shapes plant speciation across a 1000-species Brassicaceae phylogeny by generating trait divergence

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Abstract

The role of sexual selection in shaping speciation remains contested—especially in plants, where hermaphroditism, the absence of behavioral mate choice and the interplay with mating systems obscure its signature. Here, we test whether sexual selection drives macroevolutionary diversification in plants by analysing the evolution of floral traits linked to male–male competition and female choice across a 1,000-species phylogeny of Brassicaceae. The mating system emerged as a major source of confounding, influencing both the expression of sexual traits and speciation rates. Notably, selfing lineages consistently exhibited lower net diversification than outcrossing ones. Setting confounding effects aside, trait magnitudes weakly predicted speciation rates. However, faster evolutionary divergence in sexually selected traits showed a moderate positive correlation with higher speciation, particularly among outcrossers. Directional analyses reveal divergent evolutionary trajectories between mating systems, with upward shifts in floral traits linked to increased speciation in outcrossers but not selfers. Our results demonstrate that sexual selection leaves a detectable macroevolutionary signal in plants—mediated not by trait elaboration, but by the tempo and direction of trait evolution, conditioned by mating system.

Main

Sexual selection is widely invoked in animals to explain the elaboration of ornaments, courtship displays, and behaviors that increase mating success (Darwin, 1871; Kokko & Jennions, 2008). Yet despite extensive theoretical development, its role in driving speciation remains contested. Some studies report positive associations between sexual selection intensity and species richness (Janicke et al., 2018), while others found no supportive evidence (Gomes et al., 2016).

In plants, these uncertainties are even more pronounced. The prevalence of hermaphroditism and the absence of behavioral mate choice have led to the historical neglect of sexual selection in botanical systems (Willson, 1979; Moore & Pannell, 2011). However, recent advances reveal striking parallels between plants and animals in how sexual selection operates—particularly through male–male competition and female choice (Tonnabel et al., 2021; Haghighatnia et al., 2023). Whether these processes consistently lead to reproductive isolation and macroevolutionary diversification remains unresolved.

Darwin’s distinction between intrasexual selection (competition within one sex) and intersexual selection (mate choice) applies broadly—even in hermaphroditic species (Darwin, 1871; Beekman et al., 2016). In angiosperms, pollen transfer to the stigma (hereafter pollination) is analogous to mating in animals, and traits enhancing pollen delivery to conspecific stigmas—such as petal size, floral biomass, and color—can evolve under pre-pollination sexual selection (Willson, 1979; Queller, 1997; Delph & Ashman, 2006; Paterno et al., 2020). These traits are assumed to particularly influence the male reproductive success, as the latter is often limited by the number of mating events (Collet et al., 2014; Gong & Huang, 2014), consistent with Bateman’s principle (Bateman, 1948). In addition, the ratio of pollen to ovule (P:O ratio) often negatively correlates with pollen transfer

efficiency (Gong & Huang, 2014), and can thus be regarded as a proxy for the male mating success probability, i.e. pre-pollination sexual selection. Following pollination, postmating sexual selection may act on pollen traits such as germination rate, tube growth speed, and chemical compatibility (Mulcahy, 1979; Willson, 1994; Moore & Pannell, 2011). Faster-growing pollen gains a competitive edge, as shown in *Hibiscus moscheutos* and *Viola tricolor* (Snow & Spira, 1991a,b, 1996; Skogsmyr & Lankinen, 1999). Female choice may be mediated by style length, stigmatic surface area and/or molecular interactions (Lankinen & Skogsmyr, 2002; Mazer et al., 2016; Tonnabel et al., 2021; Haghighatnia et al., 2023). Longer styles may prolong pollen–pistil interactions, enhancing selectivity (Mulcahy & Mulcahy, 1975; Ramesha et al., 2011). Yet a meta-analysis found stronger non-random mating in species with shorter styles (Ruane, 2009), likely reflecting different mechanisms of postmating sexual selection—where germination speed, rather than tube growth, drives male advantage in short-styled species (Travers & Shea, 2001).

Sexual selection may promote reproductive isolation by driving divergence in reproductive traits (Reviewed in Haghighatnia et al., 2023). Premating barriers can arise when floral divergence alters pollinator behavior and reduces gene flow, as in *Mimulus lewisii* and *M. cardinalis*, where differences in petal color and nectar traits correspond with pollinator shifts and reduced hybridization (Schemske & Bradshaw, 1999; Ramsey et al., 2003). Postmating barriers may stem from mismatches in co-evolved sexual traits, such as pollen tube growth and style length (Arnold et al., 1993; Hodges & Arnold, 1995; Klips, 1999; Travers & Shea, 2001). Divergent sexual selection may also underlie unilateral incompatibility—as seen in *Solanum* and *Arabidopsis*—where pollen from self-compatible species is rejected by self-incompatible (SI) pistils (Covey et al., 2010; Baek et al., 2015; Li et al., 2018). This “weak inbreeder/strong outbreeder” pattern may reflect

asymmetric female choice and male performance (Brandvain & Haig, 2005), though the SI recognition system itself may be partly responsible for interspecies pollen rejection, introducing a confounding factor. Empirical demonstrations of traits mediating both non-random mating and reproductive isolation remain rare in plants (Tonnabel et al., 2021).

At macroevolutionary scales, the role of sexual selection in diversification remains speculative. Traits such as bilateral symmetry, nectar spurs, and SI have been associated with elevated diversification (Dodd et al., 1999; Kay et al., 2006; Goldberg et al., 2010), but causality is difficult to infer. Many studies rely on categorical traits (e.g. pollination mode, SI), which mask the complexity of reproductive processes and introduce confounding factors, complicating interpretations (Igic et al., 2008; Cutter, 2019). The use of sexual selection proxy traits based on informed hypotheses, similarly to animal studies, may solve this issue.

Here, we tested whether sexual selection promotes speciation in flowering plants using a trait-based approach across a densely sampled phylogeny of Brassicaceae. We ask whether floral traits linked to male competition and female choice predict lineage diversification, or their evolutionary rates better capture the macroevolutionary imprint of sexual selection. We accounted for mating system and ploidy as potential confounders.

Results

Mating system contributes to variation in reproductive traits and diversification

We reconstructed a new phylogeny because existing Brassicaceae trees included too few species to support macroevolutionary analysis, covering less than half of our trait-sampled taxa. To address this, we built a *de novo* tree spanning ~25% of the family's described diversity using nuclear and plastid markers. Given the known incongruence between nuclear and plastome phylogenies due to hybridisation, we also

constructed a constrained tree enforcing genus-level relationships from Hendriks et al. (2023). This framework enabled estimation of tip-specific speciation rates, trait evolutionary rates, and trait-dependent diversification using state-dependent models. Speciation rates estimated using ClaDS were highly concordant between unconstrained and constrained phylogenies (Spearman's $\rho = 0.83$; Supplementary Fig. 1), providing justification for conducting all downstream analyses on the unconstrained phylogeny. The unconstrained tree revealed substantial heterogeneity in speciation rates across Brassicaceae, with marked differences among tribes; rates were highest in Erysimeae, Boechereae, and Arabideae, and lowest in Aphragmeae and Stevenieae (Fig. 1a).

To test the impact of sexual selection on the variable speciation rates across the family, and based on the literature, we considered petal dimensions and P:O ratio as proxies for male pre-mating sexual selection; we used style length as a proxy for post-mating sexual selection (female choice). To account for the link between style length and general flower size, we also used a style-to-petal ratio (S:P), corresponding to the relative floral investment in female selection. All these reproductive traits varied extensively across Brassicaceae. Taking into account phylogenetic relationships, mating system explained much of the variation in style length, petal dimensions, and P:O ratio (all $p < 0.05$; Fig. 1b), while S:P ratio showed a marginal effect ($p = 0.07$). We also tested ploidy as potential confounding effect, but it had no significant effect on trait values, and only modestly interacted with mating system for P:O ratio ($p = 0.02$; Supplementary Table 1). In outcrossers, all male proxy traits showed a significant bias towards upward evolution (evolution towards higher values) while style length showed no trend and the S:P ratio showed a downward evolution (Fig. 1c); all traits except for the S:P ratio showed a downward evolution in selfers (Fig. 1c).

Since mating system appears as a clear confounding effect for sexually selected traits, we further disentangled its impact by a) testing the impact of mating system on macroevolutionary diversification and b) in subsequent trait-dependent analyses, we separate selfers and outcrossers. Across all HiSSE models, selfing lineages consistently exhibited lower net diversification than outcrossers. The best-supported model for mating system (Supplementary Table 2), showed that outcrossers had the highest net diversification (e.g., $r = 1.93$ in hidden state A), while selfers had negative rates in both hidden states ($r = -0.11$ and -0.96 in A and B, respectively; Supplementary Table 3 and Fig. 1d). Although ploidy had limited influence on sexually selected traits, we assessed its association with diversification for completeness. The best-supported model (Supplementary Table 4) showed higher net diversification in diploids (e.g. $r = 0.54$ in A), while polyploids had near-zero or negative rates ($r \approx 0.005$ in B; $r = -0.51$ in A; Supplementary Table 5 and Supplementary Fig. 2).

Sexual selection proxies show inconsistent associations with speciation across mating systems

Next, we tested whether variation in the intensity of sexual selection, as measured by proxy traits, could account for variation in speciation rates across the phylogenetic tree. To take into account the confounding effect of mating systems, we treated selfers and outcrossers separately. Bayesian multilevel models (BMLMs) revealed variable trait–speciation associations across mating systems (Supplementary Tables 6, 7). Male-male competition traits showed weak or inconsistent associations (Fig. 2a,b,c). The P:O ratio was negatively associated with speciation in outcrossers ($\beta = -0.26$; $P(\beta < 0) = 97.2\%$), while selfers showed a smaller, non-credible negative estimate ($\beta = -0.13$; $P(\beta < 0) = 85.9\%$). Petal width and length showed non-significant slopes in both groups (e.g. petal width in selfers: $P(\beta > 0) = 93\%$). Female-related traits showed

similarly weak patterns (Fig. 2d,e). Style length was not associated with speciation (e.g. outcrossers: $\beta = -0.006$; $P(\beta < 0) = 52.3\%$). The S:P ratio was negatively associated with speciation in selfers ($\beta = -0.25$; $P(\beta < 0) = 97.4\%$) but not outcrossers ($\beta = -0.02$; $P(\beta > 0) = 55\%$). ICCs indicated that most trait variance was residual (61.8–85.7%), with moderate genus-level effects (11.2–35.2%) and minor tribe effects ($\leq 9.8\%$).

Faster and directional evolution of sexual selection traits predict speciation, with divergent effects between mating systems

Finally, instead of the general intensity of sexual selection, we tested whether the pace of evolution of sexually selected trait, i.e. how fast variation is generated, could account for speciation rates. Our assumption here is that with faster evolutionary rates, more variation is generated, this variation being the basis for reproductive incompatibilities and thus, speciation. BMLMs revealed consistent positive associations between trait evolutionary rates and speciation for both mating systems ($P(\beta > 0) > 98\%$ for all traits; Supplementary Tables 8, 9). Male-associated traits showed broadly similar effects across groups (Fig. 3a,b,c). For the P:O ratio, outcrossers had slightly steeper slopes ($\beta = 0.35$) than selfers ($\beta = 0.28$). Petal length showed comparable estimates ($\beta = 0.28$ in outcrossers; $\beta = 0.26$ in selfers), while petal width had weaker effects ($\beta = 0.07$ – 0.08). Female-associated traits showed stronger effects in outcrossers (Fig. 3d,e). The S:P ratio had $\beta = 0.42$ in outcrossers versus $\beta = 0.23$ in selfers. Style length also showed a steeper slope in outcrossers ($\beta = 0.35$ vs. 0.24). ICCs indicated low-to-moderate phylogenetic structure (residual: 40.9–74.3%; genus: 10.8–33.5%; tribe: 4.1–26.8%).

To further examine how sexual selection traits influence speciation, we considered not only the pace but also the *direction* of trait evolution. While BAMM captures variation in evolutionary rate, it does not distinguish between *increasing* and *decreasing* trends. In sexually selected traits, this distinction is

meaningful: increases in traits like style length or P:O ratio may indicate intensified pollen competition or female choice, while decreases may reflect relaxed selection or adaptation to selfing. Using signed rates from RRphylo, we tested whether the *direction* of trait change predicts speciation differently across mating systems. We expected outcrossers—subject to stronger sexual selection—to show higher speciation when traits evolve *upward*, while selfers might exhibit contrasting or inverse patterns. Building on this rationale, we used directional RRphylo rates to test our predictions and found more nuanced, mating system-specific associations (Supplementary Tables 10, 11). Male traits showed divergent patterns between selfers and outcrossers (Fig. 4a,b,c). Faster increase in P:O ratio was positively correlated with higher speciation in outcrossers ($\beta = 0.07$; $P(\beta > 0) = 99.1\%$) but showed no effect in selfers. Petal width increased in outcrossers ($\beta = 0.02$; $P(\beta > 0) = 96.4\%$) but declined in selfers ($\beta = -0.02$; $P(\beta < 0) = 96.7\%$). Female traits diverged more strongly between mating systems (Fig. 4d,e). Faster style length evolution towards high values was associated with higher speciation in outcrossers ($\beta = 0.02$; $P(\beta > 0) = 97\%$) while faster style length decline was associated with higher speciation in selfers ($\beta = -0.02$; $P(\beta < 0) = 96.9\%$), with non-overlapping posterior distributions. The S:P ratio showed no credible directional association in either group. Trait rate variation in RRphylo models was overwhelmingly residual ($ICC \geq 98\%$).

Discussion

Sexual selection is increasingly recognized as a potent evolutionary force beyond the animal kingdom, yet its role in shaping reproductive isolation and speciation in plants remains enigmatic (Willson, 1979; Moore & Pannell, 2011; Tonnabel et al., 2021; Haghighatnia et al., 2023). Here, we tested whether floral traits involved in intra- and intersexual selection promote speciation in Brassicaceae, and whether these effects are conditioned by mating system and ploidy. Using a densely sampled

phylogeny and trait dataset, we combined diversification modeling and trait evolution analyses to test whether sexual selection in plants leaves a macroevolutionary signal.

Mating system emerged as a central factor shaping both reproductive trait expression and macroevolutionary dynamics. Selfing lineages consistently evolved towards shorter styles, lower P:O ratios, and significantly smaller petals (Fig. 1b)—a suite of traits commonly associated with the selfing syndrome and the relaxation of sexual selection in plants (Sicard & Lenhard, 2011; Gutiérrez-Valencia et al., 2022; İltaş et al., 2025). On the other hand, in outcrossers petal traits and P:O ratio showed disproportionately upward evolutionary trajectories (Fig. 1c). This contrast may indicate that directional sexual selection actively drives the evolution of traits influencing male reproductive success through pollinator attraction and pollen competition (Haghighatnia et al., 2023). Style length, however, which we expected to show an upward evolution in outcrossers if under a runaway scenario, did not show such trend, suggesting that either male-female coevolution does not involve style length, or this trait may be constrained by morphological, developmental, or functional limitations. In parallel, macroevolutionary modeling revealed that selfing lineages exhibited consistently lower net diversification rates across hidden states, driven primarily by elevated extinction (Fig. 1d). This pattern mirrors findings by Goldberg et al. (2010), who showed that SC lineages in Solanaceae—those that had lost the molecular mechanism of SI—diversified more slowly than their SI, typically outcrossing, counterparts due to elevated extinction. However, while Goldberg’s study focused on SI as a genetically defined mechanism, our analyses are based on empirically observed mating strategies (selfing vs. outcrossing), offering a complementary and more functionally grounded perspective. Taken together, these findings suggest that mating system conditions both the tempo and direction of sexually selected trait evolution, and that

these dynamics cascade into long-term patterns of speciation.

After removing mating systems as confounding factor, we found that sexually selected trait magnitudes alone were poor predictors of speciation. We observed only weak and often negative correlations between trait values—such as P:O ratio, and style length—and speciation rates (Fig. 2a,e). These results suggest that higher trait investment, a proxy for stronger sexual selection, does not necessarily lead to increased diversification. In some cases, elevated values may reflect stabilizing selection or ecological trade-offs rather than drivers of reproductive isolation (Panhuis et al., 2001; Ritchie, 2007; Janicke et al., 2018; Mendelson & Safran, 2021).

By contrast, the evolutionary tempo of these traits—how quickly they change across lineages—was consistently associated with increased speciation (Fig. 3), with P:O ratio and style length showing particularly strong effects. This pattern held across all traits and both mating systems, but was especially pronounced in outcrossers (Fig. 3a,d,e; see posterior slope separation and low overlap for P:O ratio, style length, and S:P ratio). These results suggest that speciation is more closely tied to divergence in sexual traits than to their static expression. As sexual traits diverge, they may be associated with mismatches in pollinator preference, pollen–pistil interactions, or floral cues that reduce gene flow between populations, thereby contributing to reproductive isolation. Over time, this may facilitate speciation (Schemske & Bradshaw, 1999; Ramsey et al., 2003). The directional trait evolution analyses support the idea that the divergence in sexual traits happen under directional selection towards higher value in outcrossers, potentially due to sexual selection. Our results align with the idea that sexual selection can promote speciation by accelerating divergence in mating traits—particularly in outcrossing lineages where competition and choice remain strong (Gutiérrez-Valencia et al., 2022; İltaş et al.,

2025). Yet this relationship appears contingent on trait evolution, not trait expression *per se*. Fast-changing traits may be more likely to contribute to reproductive barriers, even when their absolute values are modest.

It is important to note that ecological factors likely interact with sexual selection but were not modelled explicitly in this study. As emphasized in theoretical and empirical syntheses, ecological divergence can amplify or constrain trait evolution when abiotic stress (e.g., drought) and biotic interactions (e.g., pollinator assemblages) impose opposing or reinforcing selection pressures on floral traits (Rundle & Nosil, 2005; Scordato et al., 2014; Servedio & Boughman, 2017; Irwin, 2020; Mendelson & Safran, 2021). This interplay is especially evident in traits such as petal size, nectar production, and floral display architecture, which evolve under both ecological constraints and pollinator-mediated sexual selection (Safran et al., 2013; García et al., 2023; Alquichire-Rojas et al., 2024). Accounting for such environmental heterogeneity will be essential to determine whether ecological gradients facilitate or constrain sexual trait divergence and speciation in plants.

Taken together, our findings suggest that the evolutionary rate and direction of sexually selected traits—rather than their magnitude—predict speciation in flowering plants. To our knowledge, this is the first study to directly test whether sexually selected trait evolution leaves a macroevolutionary signal in plants by jointly analysing trait evolutionary rates, directionality, and diversification patterns across mating systems. This integrated framework opens the door to future work that unpacks how sexual selection interacts with ecological pressures to shape speciation. Mating system shapes the context in which sexual selection operates, but it is divergence in traits under competition and choice that links sexual selection to macroevolutionary outcomes. Integrating ecological gradients into this framework remains a crucial next step toward a more

synthetic understanding of how mating biology, environmental pressures, and trait evolution together drive speciation.

Methods

Plant material

To achieve broad phylogenetic coverage across Brassicaceae, we compiled trait data for four reproductive traits—P:O ratio (684 species), style length (753), petal length (992), and petal width (652)—representing 350 genera and 58 tribes. Style length and P:O data were obtained from both our measurements and published sources (Supplementary Dataset 1). We cultivated 92 species at the Heidelberg Botanic Garden, sampling 2–3 individuals from 2–3 populations per species (Supplementary Dataset 2). Plants were grown under well-watered, nutrient-rich conditions in outdoor greenhouses. Perennials were established in spring 2018; annuals were grown for up to one year. Floral material was collected between September 2019 and March 2021.

Floral buds near anthesis were harvested from upper peduncles (three per individual), fixed in ethanol:acetic acid (9:1), rinsed in 90% ethanol, and stored in 70% ethanol at 4°C. Additional buds were dried in teabags for pollen quantification. Ovule counts were performed on cleared, dissected buds mounted in chloral hydrate–glycerol and examined via DIC microscopy (three buds per individual). For pollen counts, dried buds were incubated in 95% sulfuric acid at 37°C for 10 days, washed with 0.01% Triton™ X-100, and resuspended. A 1 µL aliquot was counted using a Neubauer chamber across five 0.1 µL squares. Estimates were scaled to 100 µL and averaged across three buds per individual. Species-level P:O ratios were calculated as the mean number of pollen grains divided by mean ovule number per species. Style length was measured from dissected pistils photographed under a macroscope and analyzed in ImageJ v1.54g. Petal traits were obtained exclusively from external sources; when multiple values were

available, we used the species mean. Moreover, S:P ratios were computed as style length divided by petal length.

To expand species coverage in the P:O dataset, we implemented a machine learning–based imputation pipeline. Candidate predictors—including style length, petal length, petal width, and sepal dimensions—were evaluated across several models: linear regression, generalized additive models (*mgcv* v1.9.3), support vector regression (*e1071* v1.7-16), *BART* v2.9.9, *randomforest* v4.7-1.2, *quantregForest* v1.3-7.1, and *XGBoost* v1.7.11.1. Models were tuned using grid or default optimization and assessed with 10-fold cross-validation (*caret*). A non-regularized XGBoost model trained on log-transformed P:O ratios using style length and petal width achieved the highest performance (cross-validated $R^2 = 0.78$), outperforming a regularized variant. This model minimized collinearity and was interpretable. Robustness was confirmed through bootstrap uncertainty (300 replicates), SHAP analysis, partial dependence plots, calibration curves, and residual diagnostics. Excluding ~7–9% extreme outliers marginally improved fit ($R^2 = 0.79$), indicating model stability. We applied the final model to species lacking P:O data but with complete predictors. Predicted values were back-transformed and labeled as ‘imputed’; original and imputed values were retained for all downstream analyses.

Mating system and ploidy level data

Ploidy was classified as diploid or polyploid based on BrassiBase (Kiefer et al., 2025, unpublished), following Lysak and Koch (2011): diploids and mesopolyploids were coded as diploid; mesoneopolyploids and neopolyploids as polyploid. For 24 species, additional estimates from Stebbins fraction data (Román-Palacios et al., 2020) were incorporated. Alternative inferences from ChromEvol/Chromploidy were excluded due to frequent misclassification of mesopolyploids.

Mating system (selfing vs. outcrossing) was compiled from published sources and FloraWeb (Bundesamt für Naturschutz, n.d.), with references listed in Supplementary Dataset 1. “Obligate” and “facultative” selfers were grouped, as were outcrossing categories. To expand coverage, we implemented a supervised classification framework using biologically relevant floral traits: P:O ratio, style length, petal length, petal width, and sepal dimensions. Binary classifiers (XGBoost and Bayesian logistic regression in *brms* v2.22.0) were trained and evaluated via 10-fold cross-validation.

Two imputation model sets were developed. The first, *StrongBioLogical*, focused on species with P:O data and included three XGBoost models: (1) P:O + petal width ($R^2 = 0.59$), (2) P:O + petal length ($R^2 = 0.55$), and (3) P:O alone ($R^2 = 0.34$). This set was prioritized given the well-established link between P:O ratio and mating strategy (Cruden, 1977). The second, *FlowerSyndrom*, used petal length or width alone in XGBoost and logistic regression models; while logistic regression with petal length showed the highest accuracy ($R^2 = 0.84$), these models were considered secondary due to weaker functional relevance.

Imputed mating system states were accepted only under strict consensus across all models within a set. Species with insufficient data for *StrongBioLogical* were evaluated under *FlowerSyndrom*. Final predictions were assigned as ‘Selfer’, ‘Outcrosser’, or ‘Potential Mix’ based on probabilistic thresholds (>0.6 , <0.4 , and intermediate). All imputed values were flagged and retained alongside observed data. Model interpretations were supported by SHAP analyses, logistic regression coefficients, and bootstrap confidence intervals (1,000 replicates), with repeated model refits confirming coefficient stability.

Phylogenetic reconstruction

We constructed a 1004-species Brassicaceae phylogeny including eight Brassicales outgroups (*Bretschneidera sinensis*, *Capparis decidua*, *Cleome pallida*, *Forchhammeria*

macrocarpa, *Synsepalum dulcificum*, *Tovaria pendula*, *Tropaeolum pentaphyllum*, *Vasconcellea quercifolia*). Taxonomy followed BrassiBase v2.0.1 (Kiefer et al., 2025, unpublished). Nuclear ITS (ITS1, 5.8S, ITS2) and plastid *trnL/trnF* sequences were compiled from BrassiBase and NCBI, supplemented with complete plastomes to maximize sampling. ITS regions were aligned using MUSCLE5 (Edgar, 2022), and plastid regions using MAFFT v7 (Katoh & Standley, 2013). Alignments were trimmed with trimAL v1.4.rev15 (Capella-Gutiérrez et al., 2009), removing sites with $\geq 30\%$ gaps. Manual adjustments were made where needed. Trimming also removed pseudogene-rich *trnL-trnF* intergenic spacers (Koch et al., 2005). Substitution models were selected using ModelFinder in IQ-TREE v2.3.5.cmaple (Kalyaanamoorthy et al., 2017; Minh et al., 2020), with BIC criteria applied separately to ITS and plastid datasets. SYM+FQ+R8 and TVM+F+R4 were selected, respectively.

Alignments were concatenated and analysed under a partitioned maximum likelihood (ML) framework in the MPI version of IQ-TREE. Each partition was assigned its own model and rate class to capture locus-specific evolutionary patterns. Node support was assessed with 1000 SH-like approximate likelihood ratio test replicates and 1000 ultrafast bootstraps, with nearest-neighbor interchange optimization to reduce model violation artifacts. Partition-aware resampling was used to improve multi-locus support estimation (Seo et al., 2005; Gadagkar et al., 2005). The ML tree was made ultrametric using penalized likelihood in treePL v1.0 (Smith & O’Meara, 2012), calibrated with one fossil constraint—*Dressiantha bicarpellata* (93.6–89.3 Ma; Gandolfo et al., 1998)—and three secondary calibrations: the crown Brassicaceae (25.7–23.1 Ma), crown Lineage III (15.42–13.5 Ma), and crown *Arabidopsis* (6.84–5.54 Ma) based on Hendriks et al. (2023).

Molecular dating was refined by integrating this tree as a starting topology in BEAST v2.7.7

(Bouckaert et al., 2019), using the fossil *D. bicarpellata* to calibrate the stem Brassicales, and secondary calibrations for crown Brassicaceae and Lineage III. We employed the optimized relaxed clock model under the Coupled Markov chain Monte Carlo (CoupledMCMC) framework, with one long chain (50 million generations, four coupled chains) and five shorter chains (10 million generations each). The first 20% of samples were discarded as burn-in. Outputs were combined in LogCombiner, yielding 80 million post-burn-in states. Convergence was assessed in Tracer v1.7.2 (Rambaut et al., 2018), with all parameters showing effective sample sizes (ESS) > 400 and stable trace behavior. The final time-calibrated tree was summarized as a maximum clade credibility (MCC) tree with 95% highest posterior density intervals using TreeAnnotator v2.7.7. In parallel, we generated a constrained genus-level tree incorporating relationships from Hendriks et al. (2023). A constraint topology was built using shared species to anchor genus-level placements (198 tips), ensuring each genus was represented. A new ML tree was inferred in IQ-TREE using the same models, partitioning, and support metrics. The constraint topology fixed genus-level relationships, while allowing remaining taxa to be placed freely. This tree was made ultrametric via treePL and dated in BEAST using the same calibrations, priors, and CoupledMCMC scheme. A multi-monophyletic constraint distribution was used to enforce genus monophyly. Post-processing (burn-in removal, MCMC merging, MCC tree extraction) followed the same protocol as the unconstrained tree.

The alignment, ML tree, and both dated trees (constrained and unconstrained) are available on GitHub (LINK). Following generation of the

initial dated tree, we reviewed all species names against the updated Brassicaceae checklist, pruning 10 duplicates and renaming 25 species. These updates were applied post hoc without re-running BEAST. The final curated tree reflecting these revisions is also available at the same repository.

Statistical analysis of mating system and ploidy level

We analysed whether ploidy level (diploid vs. polyploid) and mating system (selfing vs. outcrossing) explain variation in five floral traits: P:O ratio, style length, petal length, petal width, and S:P ratio. Species with mixed mating systems were excluded due to low representation. We fit Phylogenetic Generalized Least Squares (PGLS) models using the *caper* v1.0.3, with pruned phylogenies matched to each trait dataset to account for shared evolutionary history. Traits were transformed to improve residual normality (log for P:O; Yeo-Johnson for all others via *bestNormalize* v1.9.1). Each model included ploidy, mating system, and their interaction as predictors. Residual diagnostics were performed using Shapiro-Wilk tests and Q-Q plots. For each model, we report effect sizes, likelihood-based R^2 , and p-values.

Trait-dependent diversification with HiSSE

We used the Hidden-State Speciation and Extinction (HiSSE) framework (Beaulieu and O'Meara, 2016) to test whether mating system influences diversification dynamics in Brassicaceae. Mating system was coded as a binary trait (0 = selfing, 1 = outcrossing) for 688 species. The phylogeny was pruned to include only ingroup taxa, made ultrametric using *phytools*, and matched to species with complete trait data. A uniform sampling fraction was applied across mating system states.

We fit 11 irreversible models using the *hisse* function, including BiSSE-like models, HiSSE models with two hidden states, and a character-independent diversification (CID-2) model. Models differed in the constraints applied to turnover, extinction, and transition rates across observed and hidden states. All models were run in parallel using *furrr* and ranked by AIC. The best-supported model ($\Delta\text{AIC} > 2$) was used for inference of state-dependent diversification. *SupportRegionHiSSE* function was used to generate 10,000 samples around the MLE, yielding posterior-like distributions of speciation (λ), extinction (μ), and net diversification (r) by mating system and hidden state. Trait-dependent patterns were visualized using kernel density plots. The full procedure was repeated using ploidy as the binary trait (0 = polyploid, 1 = diploid), with 550 species. All preprocessing, model fitting, and inference steps followed the same protocol.

Estimation of speciation rates

All phylogenetic trees used for diversification analyses were pruned to exclude outgroup taxa, yielding a consistent ingroup set ($n = 1004$ species) across both constrained and unconstrained topologies. To meet ClaDS2 model requirements, trees were made fully ultrametric using branch extension in *phytools*.

Speciation rates (λ) were estimated using the ClaDS2 model (Maliot & Morlon, 2022), which infers lineage-specific rates under a framework allowing frequent, small shifts in diversification. ClaDS2 models speciation and extinction as continuous processes, enabling high-resolution inference of rate dynamics across lineages and time. Daughter lineages inherit speciation rates from a lognormal distribution governed by a deterministic trend (α) and stochasticity (σ), while extinction is modelled via a constant

turnover rate (ϵ). To account for incomplete sampling, genus-level sampling fractions were assigned based on species richness data from BrassiBase v2. ClaDS2 was implemented in PANDA.jl v0.0.8 in Julia v1.10.2, which uses a data augmentation framework to improve MCMC efficiency. Each tree was analysed in triplicate for 100,000 iterations with a thinning interval of 5 and 25% burn-in. Posterior samples were exported to R, cleaned, and assessed for convergence using Gelman–Rubin statistics (R hat), ESS, autocorrelation, and trace/density plots. R hat values were monitored across full chains and iteration windows, and compared against standard benchmarks (e.g. R hat < 1.01, 1.02, 1.05).

Tip- and branch-specific speciation rates were estimated as maximum a posteriori values from the augmented data, then averaged across chains. These were visualized on phylogenies using color mapping to explore variation across species and through time.

Estimation of trait evolutionary rates

We estimated rates of trait evolution using the Bayesian Analysis of Macroevolutionary Mixtures (BAMM) in BAMM v2.5.0 (Rabosky, 2014) and Phylogenetic Ridge Regression in *RRphylo* v3.0.0 (Castiglione et al. 2018). BAMM captures discrete shifts in macroevolutionary regimes, while *RRphylo* detects continuous, lineage-specific variation in both the speed and direction of phenotypic change.

BAMM was applied to log-transformed values of five traits (P:O ratio, petal length, petal width, style length, and S:P ratio) using unconstrained ultrametric phylogenies. Trait evolution was modelled under a time-dependent Brownian motion model, allowing rate shifts across lineages and through time. Three independent MCMC chains were run per trait (40–100 million generations depending on trait). Prior sensitivity was assessed by varying the

expected number of shifts (1, 500, and 1000) using *setBAMMpriors* from *BAMMtools* v2.1.12. Lineage-specific evolutionary rates were also estimated using RRphylo, applied separately to each log-transformed trait. For each species, we extracted the terminal branch rate; both the absolute value (rate magnitude) and the sign (rate direction) were retained for downstream analyses.

Comparative analysis

We used Bayesian multilevel models (BMLMs; Bürkner, 2017) to evaluate associations between floral traits, their evolutionary dynamics, and speciation rates across angiosperms. All models were implemented in the *brms* package (v2.22.0) with Gaussian error distributions, default weakly informative priors, and 4,000 iterations across four chains. Unless otherwise specified, all trait and rate variables were natural log-transformed prior to analysis. Models included log-transformed speciation rate as a fixed effect, mating system-specific slopes (selfing vs. outcrossing), and random intercepts for genus and tribe to account for phylogenetic hierarchy. Convergence and fit were evaluated using standard diagnostics and posterior predictive checks.

We first modelled speciation rate as a function of trait values related to prepollination male–male competition (e.g. P:O ratio), pollinator attraction (e.g. petal dimensions), and female choice (e.g. style length), estimating mating-specific slopes in each case. Second, we modelled species-specific mean rates of trait evolution, estimated via BAMM, as a function of speciation rate. Third, we used directional evolutionary rates from RRphylo (untransformed) to classify species into upward or downward trait evolution. Differences in polarity frequencies across mating systems were assessed with Fisher’s exact tests. Finally, we modelled directional RRphylo rates as a function of log-transformed speciation rates using BMLMs with mating-specific slopes.

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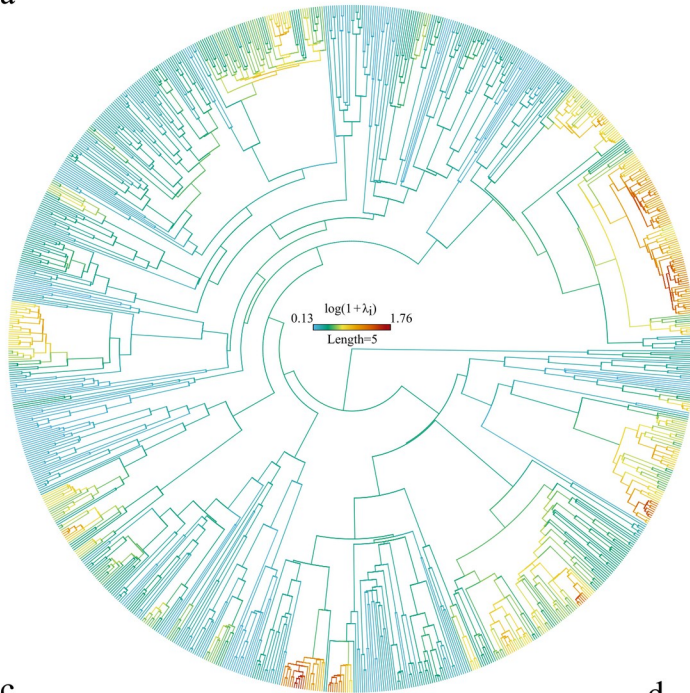
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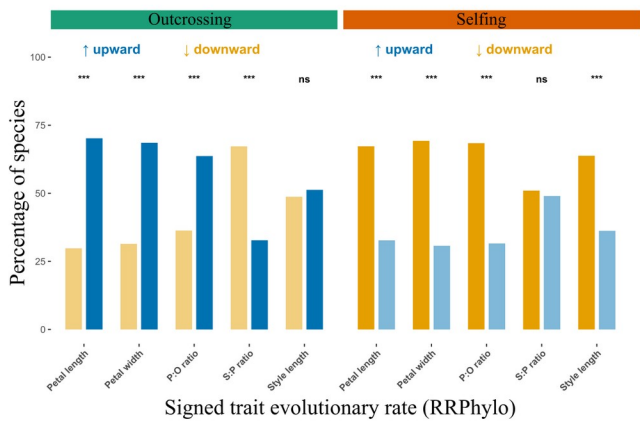
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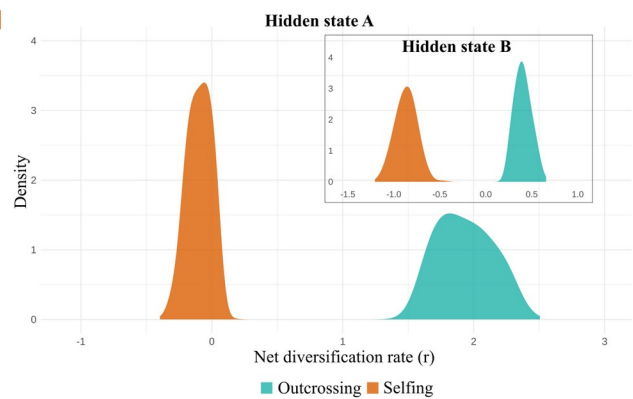


Figure 1 a. Speciation rates (λ) estimated with ClaDS for 1004 Brassicaceae species. The circular phylogeny shows log-transformed rates from blue (slow) to red (fast). Color intensity reflects variation across branches. The scale bar indicates 5 million years. b. Differences in trait values between mating systems across traits. Half plots represent the distribution of transformed trait values for outcrossers and selfers. Asterisks denote significance levels from PGLS models testing trait differences by mating system (ns = non-significant). c. Polarity of trait evolution by mating system. Bars show the percentage of species exhibiting upward (blue) or downward (gold) trait evolutionary rates, grouped by trait and mating system. Asterisks indicate binomial tests of whether upward vs. downward shifts differ from a 50:50 expectation within each mating system. d. Net diversification rates (r) for selfing and outcrossing lineages under the best-fitting HiSSE model. Distributions are based on adaptive support region sampling ($n = 10,000$) around the ML solution, shown for hidden state A (main) and hidden state B (inset).

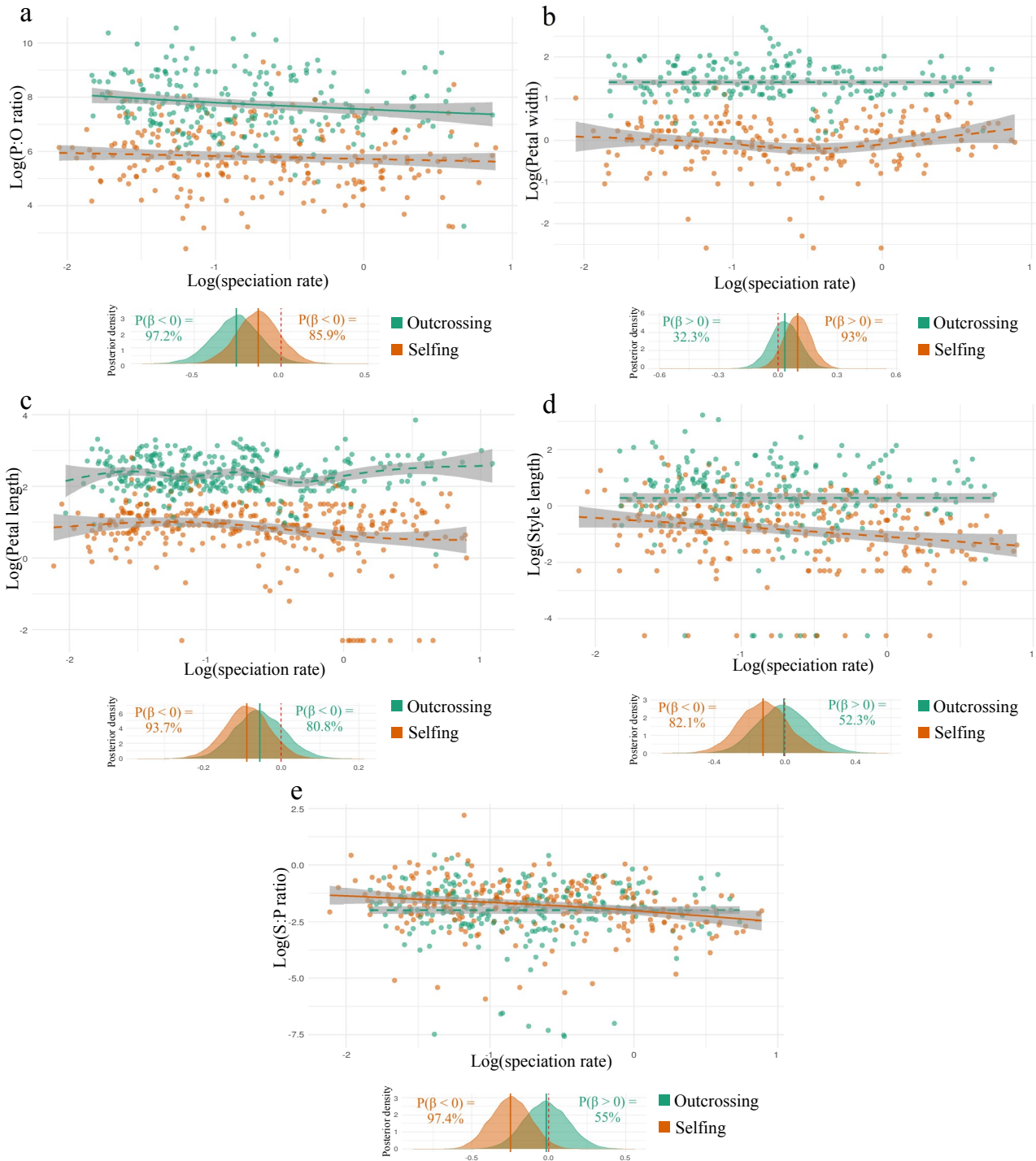


Figure 2 Relationship between log-transformed sexual selection proxies and log-transformed speciation rates for a. P:O ratio, b. petal width, c. petal length, d. Style length, and e. S:P ratio across mating systems. Scatterplots display species-level relationships, with points coloured by mating system. Solid lines indicate slope with strong posterior support ($P(\beta > 0) > 0.95$); dashed lines indicate no support. Posterior slope distributions (below each scatterplot) from BMLMs illustrated for each mating-specific association. Dashed red line is $\beta = 0$.

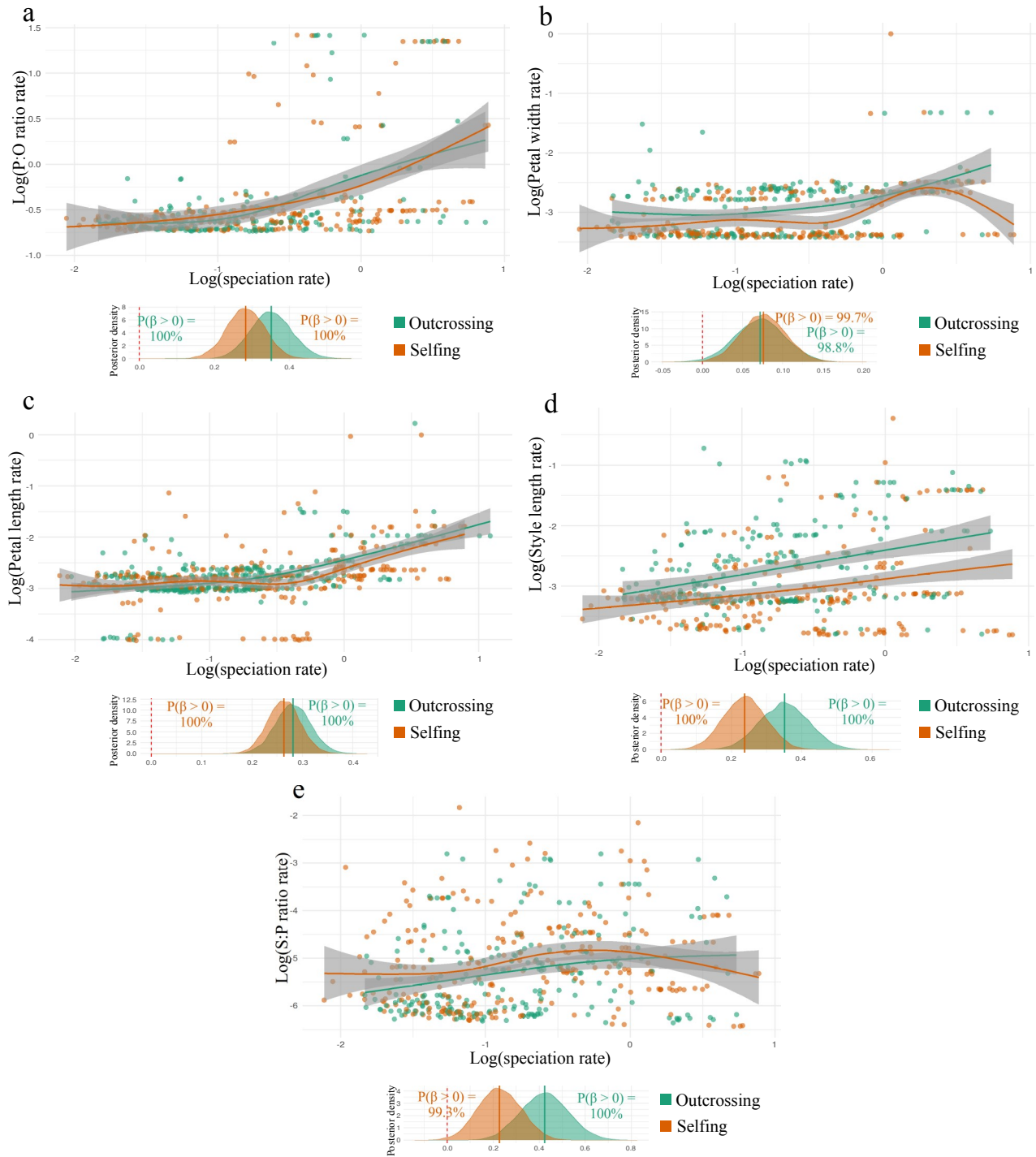


Figure 3 Relationship between log-transformed evolutionary rates (BAMM) of sexual selection proxies and log-transformed speciation rates for a. P:O ratio, b. petal width, c. petal length, d. style length, and e. S:P ratio across mating systems. Scatterplots display species-level relationships, with points coloured by mating system. Solid lines indicate slope with strong posterior support ($P(\beta > 0) > 0.95$). Posterior slope distributions (below each scatterplot) from BMLMs illustrated for each mating-specific association. Dashed red line is $\beta = 0$.

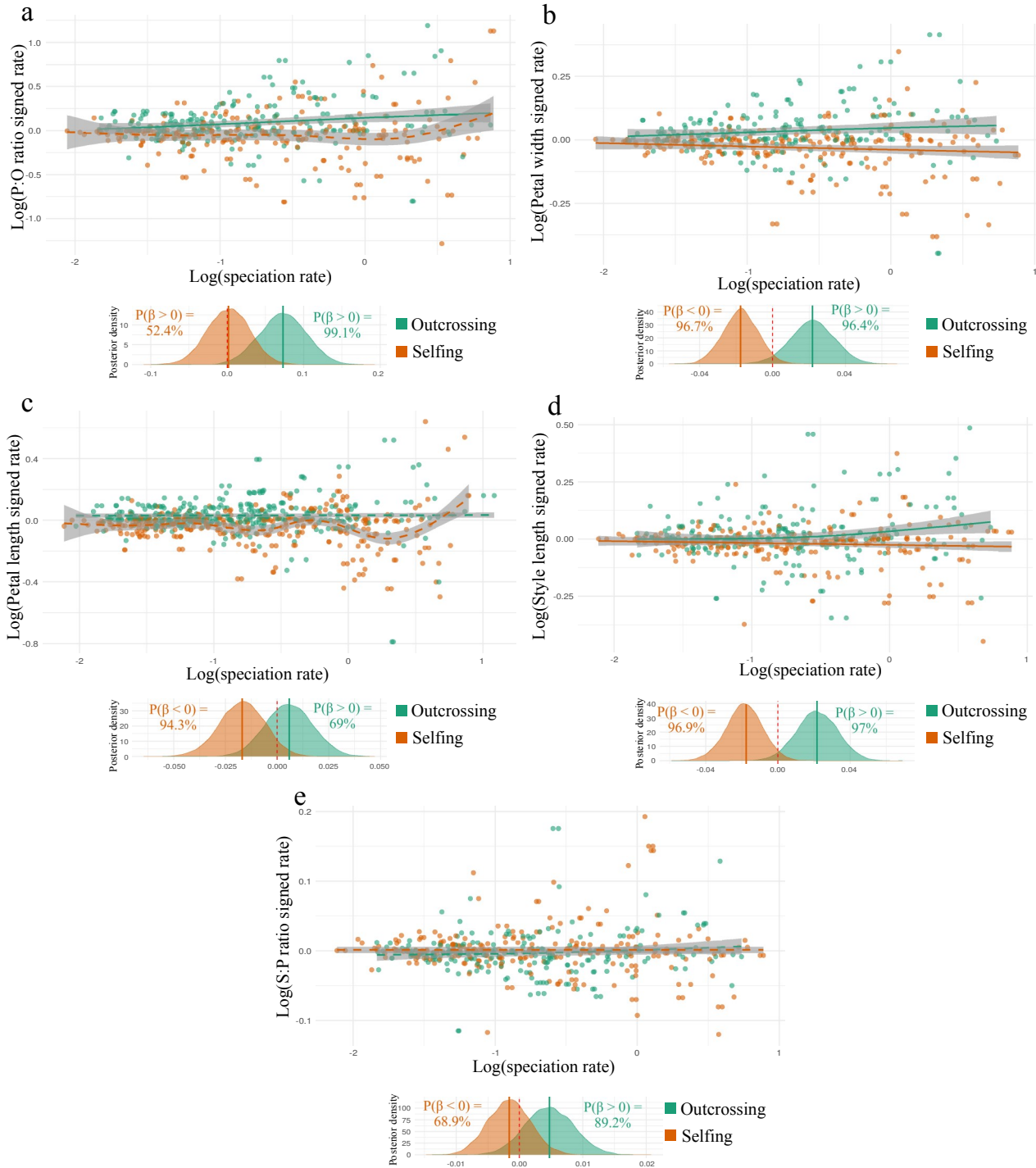
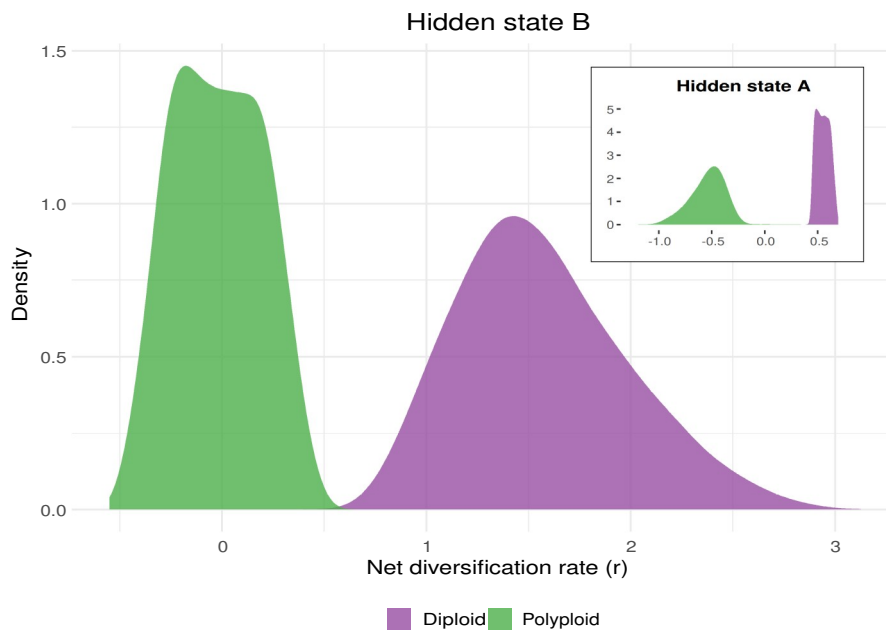


Figure 4. Relationship between signed evolutionary rates (RRPhylo) and log-transformed speciation rates across mating systems. For a. P:O ratio, b. petal width, c. petal length, d. style length, and e. S:P ratio. Scatterplots display species-level relationships, with points coloured by mating system. Solid lines indicate slope with strong posterior support ($P(\beta > 0) > 0.95$); dashed lines indicate no support. Posterior slope distributions (below each scatterplot) from BMLMs illustrated for each mating-specific association. Dashed red line is $\beta = 0$.



Supplementary Figure 1. Concordance of ClaDS speciation rate between phylogenetic trees. Scatterplot shows speciation rates estimated for each species using the ClaDS model, based on two alternative phylogenetic trees: the unconstrained (main) and constrained topologies. Each point represents a species, with a dotted 1:1 reference line indicating perfect agreement between trees. The solid line represents a linear regression. Spearman's rank correlation coefficient (R) and p -values (p) is reported in the plot.



Supplementary Figure 2. Net diversification rate distributions under the best-fitting HiSSE model of ploidy. Posterior-like distributions of net diversification rates (r) for each mating system state (Diploid vs. Polyploid) under hidden state A (main panel) and hidden state B (inset), based on adaptive support region sampling ($n=10,000$) around the maximum-likelihood solution of the best-supported HiSSE model.

Supplementary Table 1. PGLS results of mating system and ploidy effects on reproductive trait variation. The table reports estimates and p-values from phylogenetic generalized least squares (PGLS) models with an interaction term between mating system and ploidy. Columns include the estimated coefficients and associated p-values for the intercept, ploidy, mating system, and their interaction. The final column shows the Shapiro–Wilk p-value for residual normality in each model. Significance levels are indicated as strikes, and no symbol for $p \geq 0.05$.

Trait	Intercept_Est	Intercept_p	Ploidy_Est	Ploidy_p	Mating_Est	Mating_p	Interaction_Est	Interaction_p	Shapiro_p
poratio	7.46	0.00***	0.16	0.33	-1.14	0.00***	-0.52	0.02*	0.14
style_L	0.24	0.76	-0.07	0.5	-0.47	0.00***	-0.05	0.73	0.02
petal_L	0.26	0.63	0.04	0.53	-0.93	0.00***	0.06	0.51	0.02
petal_W	0.42	0.48	0.13	0.18	-1.07	0.00***	-0.01	0.93	0.25
spratio	0.1	0.92	-0.24	0.07	0.28	0.07	-0.17	0.35	0.01

Supplementary Table 2. Model statistics for all fitted HiSSE models for mating system. Columns show the log-likelihood (logL), number of estimated parameters (k), Akaike Information Criterion (AIC), relative Akaike weight, and the difference in AIC (Δ AIC) from the best-fitting model. Models are ranked by increasing AIC.

model	logL	k	AIC	Akaike.weight	deltaAIC
H_irrev_equalTurnover	-1,962.31	8.00	3,940.62	0.94	0.00
H_irrev_allFree	-1,961.99	11.00	3,945.99	0.06	5.37
H_irrev_equalExtinction	-1,973.69	8.00	3,963.37	0.00	22.75
H_irrev_equalTrans	-1,978.94	9.00	3,975.88	0.00	35.26
H_CID2_irrev	-2,000.40	7.00	4,014.80	0.00	74.18
H_irrev_equalTrans_sameTE_byHidden	-2,034.78	5.00	4,079.56	0.00	138.94
H_irrev_equalTE	-2,044.41	5.00	4,098.82	0.00	158.20
BL.qSO0	-2,054.52	5.00	4,119.04	0.00	178.42
BLEqual.t.qSO0	-2,060.95	4.00	4,129.91	0.00	189.29
BLEqual.e.qSO0	-2,083.63	4.00	4,175.27	0.00	234.65
BLEqual.te.qSO0	-2,091.61	3.00	4,189.22	0.00	248.60
H_irrev_equalAll	-2,091.61	3.00	4,189.22	0.00	248.60

Supplementary Table 3. Summary of speciation, extinction, and net diversification of the best-fitted HiSSE model for mating system. Columns show speciation (λ), extinction (μ), and net diversification (net_div) for every state; selfing across hidden states (0A and 0B), and outcrossing (1A and 1B).

model	state	lambda	mu	net_div
H_irrev_equalTurnover	0A	0.91	1.02	-0.11
H_irrev_equalTurnover	1A	1.93	0.00	1.93
H_irrev_equalTurnover	0B	0.48	1.45	-0.96
H_irrev_equalTurnover	1B	1.16	0.77	0.40

Supplementary Table 4. Model statistics for all fitted HiSSE models for ploidy. Columns show the log-likelihood (logL), number of estimated parameters (k), Akaike Information Criterion (AIC), relative Akaike weight, and the difference in AIC (Δ AIC) from the best-fitting model. Models are ranked by increasing AIC.

model	logL	k	AIC	Akaike.weight	deltaAIC
H_trait_hidden_irreversible	-1,607.65	11.00	3,237.30	1.00	0.00
H_irrev_equalTurnover	-1,624.17	8.00	3,264.35	0.00	27.05
H_irrev_equalExtinction	-1,633.20	8.00	3,282.40	0.00	45.09
H_CID2_irrev	-1,649.57	7.00	3,313.13	0.00	75.83
H_irrev_equalTrans	-1,663.03	9.00	3,344.05	0.00	106.75
BL.qSO0	-1,689.03	5.00	3,388.05	0.00	150.75
BLEqual.t.qSO0	-1,708.11	4.00	3,424.22	0.00	186.91
H_irrev_equalTrans_sameTE_byHidden	-1,707.99	5.00	3,425.99	0.00	188.69
H_irrev_equalTE	-1,715.23	5.00	3,440.46	0.00	203.16
BLEqual.e.qSO0	-1,755.44	4.00	3,518.88	0.00	281.58
H_irrev_equalAll	-1,756.46	3.00	3,518.92	0.00	281.62
BLEqual.te.qSO0	-1,756.46	3.00	3,518.92	0.00	281.62

Supplementary Table 5. Summary of speciation, extinction, and net diversification of the best-fitted HiSSE model for ploidy. Columns show speciation (λ), extinction (μ), and net diversification (net_div) for every state; polyploids across hidden states (0A and 0B), and diploids across (1A and 1B).

model	state	lambda	mu	net_div
H_trait_hidden_irreversible	0A	0.26	0.77	-0.51
H_trait_hidden_irreversible	1A	0.54	0.00	0.54
H_trait_hidden_irreversible	0B	2.16	2.16	0.00
H_trait_hidden_irreversible	1B	2.74	1.24	1.50

Supplementary Table 6. Posterior slopes from BMLM models testing associations between log-transformed trait and log-transformed speciation rate. For each trait, columns the posterior mean slope (β), 95% credible interval (CI), and the probability that the slope differs from zero (e.g., $P(\beta < 0)$ or $P(\beta > 0)$), directionally reported according to the sign of β .

Model	Group	Posterior mean (β)	95% CI lower	95% CI upper	$0.05 > P(\beta \neq 0) > 0.95$
poratio	Outcrosser	-0.26	-0.53	0.01	$P(\beta < 0) = 2.8\%$
poratio	Selfer	-0.13	-0.37	0.11	$P(\beta < 0) = 14.1\%$
petal_W	Outcrosser	0.03	-0.11	0.18	$P(\beta > 0) = 67.7\%$
petal_W	Selfer	0.10	-0.03	0.23	$P(\beta > 0) = 93\%$
petal_L	Outcrosser	-0.05	-0.18	0.07	$P(\beta < 0) = 19.2\%$
petal_L	Selfer	-0.09	-0.20	0.02	$P(\beta < 0) = 6.3\%$
style_L	Outcrosser	-0.01	-0.29	0.29	$P(\beta > 0) = 47.7\%$
style_L	Selfer	-0.12	-0.39	0.14	$P(\beta < 0) = 17.9\%$

spratio	Outcrosser	-0.02	-0.29	0.27	$P(\beta > 0) = 45\%$
spratio	Selfer	-0.25	-0.50	0.00	$P(\beta < 0) = 2.6\%$

Supplementary Table 7. BMLM models testing the association between log-transformed trait and log-transformed speciation rate. For each trait model, columns show fixed-effect estimates (intercept and slope), standard deviations (SDs) of random intercepts for genus, tribe, and mating system, and random slopes for mating-specific speciation effects. Residual variance (σ) is also reported. Intra-class correlation coefficients (ICCs) for genus, tribe, and residual components are included to quantify taxonomic signal in trait evolutionary rates. All estimates are presented with standard errors and 95% credible intervals. Components are labelled as Fixed, Random, Residual, or ICC.

Model	Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Component
poratio	Intercept	6.64	1.35	3.87	9.46	Fixed
poratio	log_rate	-0.19	0.75	-1.76	1.42	Fixed
poratio	σ (genus)	0.37	0.12	0.08	0.59	Random
poratio	σ (Intercept mating)	2.22	1.49	0.64	6.07	Random
poratio	σ (slope mating)	0.82	1.09	0.02	3.76	Random
poratio	σ (mating)	-0.09	0.56	-0.96	0.89	Random
poratio	σ (tribe)	0.19	0.11	0.01	0.43	Random
poratio	σ (residual)	NA	NA	NA	NA	Residual
poratio	ICC (Genus)	0.11	NA	NA	NA	ICC
poratio	ICC (Tribe)	0.03	NA	NA	NA	ICC
poratio	ICC (Residual)	0.86	NA	NA	NA	ICC
petal_W	Intercept	0.70	1.23	-1.89	3.22	Fixed
petal_W	log_rate	0.07	0.71	-1.55	1.65	Fixed
petal_W	σ (genus)	0.27	0.05	0.16	0.37	Random
petal_W	σ (Intercept mating)	1.94	1.40	0.51	5.51	Random
petal_W	σ (slope mating)	0.73	1.06	0.01	3.72	Random
petal_W	σ (mating)	-0.07	0.57	-0.96	0.92	Random
petal_W	σ (tribe)	0.15	0.07	0.02	0.29	Random
petal_W	σ (residual)	NA	NA	NA	NA	Residual
petal_W	ICC (Genus)	0.22	NA	NA	NA	ICC
petal_W	ICC (Tribe)	0.07	NA	NA	NA	ICC
petal_W	ICC (Residual)	0.71	NA	NA	NA	ICC
petal_L	Intercept	1.57	1.22	-0.92	4.11	Fixed
petal_L	log_rate	-0.08	0.67	-1.56	1.31	Fixed
petal_L	σ (genus)	0.26	0.05	0.16	0.36	Random
petal_L	σ (Intercept mating)	1.94	1.39	0.50	5.65	Random
petal_L	σ (slope mating)	0.66	1.00	0.01	3.56	Random
petal_L	σ (mating)	0.07	0.58	-0.92	0.96	Random

petal_L	σ (tribe)	0.20	0.07	0.06	0.33	Random
petal_L	σ (residual)	NA	NA	NA	NA	Residual
petal_L	ICC (Genus)	0.18	NA	NA	NA	ICC
petal_L	ICC (Tribe)	0.10	NA	NA	NA	ICC
petal_L	ICC (Residual)	0.72	NA	NA	NA	ICC
style_L	Intercept	-0.26	1.19	-2.74	2.25	Fixed
style_L	log_rate	-0.04	0.77	-1.65	1.73	Fixed
style_L	σ (genus)	0.75	0.09	0.57	0.94	Random
style_L	σ (Intercept mating)	1.82	1.38	0.42	5.44	Random
style_L	σ (slope mating)	0.84	1.09	0.02	3.86	Random
style_L	σ (mating)	0.12	0.57	-0.91	0.97	Random
style_L	σ (tribe)	0.24	0.13	0.02	0.50	Random
style_L	σ (residual)	NA	NA	NA	NA	Residual
style_L	ICC (Genus)	0.35	NA	NA	NA	ICC
style_L	ICC (Tribe)	0.04	NA	NA	NA	ICC
style_L	ICC (Residual)	0.61	NA	NA	NA	ICC
spratio	Intercept	-1.94	0.73	-3.43	-0.34	Fixed
spratio	log_rate	-0.15	0.95	-2.20	1.91	Fixed
spratio	σ (genus)	0.71	0.09	0.53	0.89	Random
spratio	σ (Intercept mating)	0.78	1.04	0.01	3.61	Random
spratio	σ (slope mating)	1.07	1.20	0.04	4.35	Random
spratio	σ (mating)	0.05	0.61	-0.96	0.97	Random
spratio	σ (tribe)	0.26	0.13	0.02	0.51	Random
spratio	σ (residual)	NA	NA	NA	NA	Residual
spratio	ICC (Genus)	0.34	NA	NA	NA	ICC
spratio	ICC (Tribe)	0.05	NA	NA	NA	ICC
spratio	ICC (Residual)	0.62	NA	NA	NA	ICC

Supplementary Table 8. Posterior slopes from BMLM model testing associations between log-transformed trait evolutionary rates (BAMM) and log-transformed speciation rate. Columns show the posterior mean slope (β), 95% credible interval (CI), and the probability that the slope differs from zero (e.g., $P(\beta < 0)$ or $P(\beta > 0)$), directionally reported according to the sign of β .

Model	Group	Posterior mean (β)	95% CI lower	95% CI upper	$0.05 > P(\beta \neq 0) > 0.95$
poratio_rate	Outcrosser	0.35	0.25	0.46	$P(\beta > 0) = 100\%$
poratio_rate	Selfer	0.28	0.18	0.38	$P(\beta > 0) = 100\%$
petalW_rate	Outcrosser	0.07	0.01	0.13	$P(\beta > 0) = 98.8\%$
petalW_rate	Selfer	0.08	0.02	0.13	$P(\beta > 0) = 99.7\%$
petalL_rate	Outcrosser	0.28	0.21	0.35	$P(\beta > 0) = 100\%$
petalL_rate	Selfer	0.26	0.20	0.33	$P(\beta > 0) = 100\%$

styleL_rate	Outcrosser	0.35	0.22	0.49	$P(\beta > 0) = 100\%$
styleL_rate	Selfer	0.24	0.12	0.36	$P(\beta > 0) = 100\%$
spratio_rate	Outcrosser	0.42	0.23	0.63	$P(\beta > 0) = 100\%$
spratio_rate	Selfer	0.23	0.05	0.41	$P(\beta > 0) = 99.3\%$

Supplementary Table 9. BMLM model testing the association between log-transformed trait evolutionary rates (BAMM) and log-transformed speciation rate. Columns report the posterior mean and uncertainty of fixed effects (intercept and slope), the standard deviations (SDs) of random intercepts for genus, tribe, and mating system, and the random slopes for mating-specific speciation effects. Residual variance (σ) is included where available. Intra-class correlation coefficients (ICCs) for genus, tribe, and residual components are also shown to quantify the taxonomic structure in evolutionary rate variation. All estimates are presented with standard errors and 95% credible intervals. Model components are labelled as Fixed, Random, Residual, or ICC.

Model	Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Component
poratio_rate	Intercept	-0.25	0.69	-1.84	1.20	Fixed
poratio_rate	log_rate	0.30	1.05	-1.30	2.10	Fixed
poratio_rate	σ (genus)	0.20	0.03	0.15	0.25	Random
poratio_rate	σ (Intercept mating)	0.76	1.02	0.01	3.60	Random
poratio_rate	σ (slope mating)	0.80	1.27	0.01	3.90	Random
poratio_rate	σ (mating)	0.07	0.63	-0.96	0.98	Random
poratio_rate	σ (tribe)	0.09	0.05	0.01	0.18	Random
poratio_rate	σ (residual)	NA	NA	NA	NA	Residual
poratio_rate	ICC (Genus)	0.20	NA	NA	NA	ICC
poratio_rate	ICC (Tribe)	0.04	NA	NA	NA	ICC
poratio_rate	ICC (Residual)	0.76	NA	NA	NA	ICC
petalW_rate	Intercept	-3.05	0.54	-4.22	-1.86	Fixed
petalW_rate	log_rate	0.09	0.61	-1.15	1.38	Fixed
petalW_rate	σ (genus)	0.15	0.02	0.11	0.19	Random
petalW_rate	σ (Intercept mating)	0.55	0.92	0.00	3.07	Random
petalW_rate	σ (slope mating)	0.57	0.92	0.00	3.09	Random
petalW_rate	σ (mating)	0.03	0.65	-0.98	0.98	Random
petalW_rate	σ (tribe)	0.39	0.05	0.31	0.51	Random
petalW_rate	σ (residual)	NA	NA	NA	NA	Residual
petalW_rate	ICC (Genus)	0.11	NA	NA	NA	ICC
petalW_rate	ICC (Tribe)	0.74	NA	NA	NA	ICC
petalW_rate	ICC (Residual)	0.15	NA	NA	NA	ICC
petalL_rate	Intercept	-2.54	0.57	-3.86	-1.29	Fixed
petalL_rate	log_rate	0.27	0.65	-1.15	1.69	Fixed
petalL_rate	σ (genus)	0.15	0.02	0.11	0.20	Random
petalL_rate	σ (Intercept mating)	0.61	0.93	0.01	3.26	Random

petalL_rate	σ (slope mating)	0.64	1.00	0.01	3.53	Random
petalL_rate	σ (mating)	0.01	0.64	-0.98	0.98	Random
petalL_rate	σ (tribe)	0.21	0.03	0.16	0.28	Random
petalL_rate	σ (residual)	NA	NA	NA	NA	Residual
petalL_rate	ICC (Genus)	0.16	NA	NA	NA	ICC
petalL_rate	ICC (Tribe)	0.31	NA	NA	NA	ICC
petalL_rate	ICC (Residual)	0.53	NA	NA	NA	ICC
styleL_rate	Intercept	-2.81	0.84	-4.69	-1.06	Fixed
styleL_rate	log_rate	0.31	0.87	-1.45	2.23	Fixed
styleL_rate	σ (genus)	0.31	0.04	0.24	0.39	Random
styleL_rate	σ (Intercept mating)	1.02	1.14	0.09	4.15	Random
styleL_rate	σ (slope mating)	0.89	1.14	0.02	4.06	Random
styleL_rate	σ (mating)	0.11	0.60	-0.94	0.98	Random
styleL_rate	σ (tribe)	0.32	0.06	0.21	0.45	Random
styleL_rate	σ (residual)	NA	NA	NA	NA	Residual
styleL_rate	ICC (Genus)	0.25	NA	NA	NA	ICC
styleL_rate	ICC (Tribe)	0.27	NA	NA	NA	ICC
styleL_rate	ICC (Residual)	0.48	NA	NA	NA	ICC
spratio_rate	Intercept	-5.04	0.67	-6.45	-3.62	Fixed
spratio_rate	log_rate	0.31	0.83	-1.47	2.09	Fixed
spratio_rate	σ (genus)	0.56	0.06	0.45	0.69	Random
spratio_rate	σ (Intercept mating)	0.72	1.02	0.01	3.62	Random
spratio_rate	σ (slope mating)	1.01	1.16	0.05	4.21	Random
spratio_rate	σ (mating)	0.05	0.62	-0.96	0.98	Random
spratio_rate	σ (tribe)	0.49	0.10	0.30	0.71	Random
spratio_rate	σ (residual)	NA	NA	NA	NA	Residual
spratio_rate	ICC (Genus)	0.34	NA	NA	NA	ICC
spratio_rate	ICC (Tribe)	0.26	NA	NA	NA	ICC
spratio_rate	ICC (Residual)	0.41	NA	NA	NA	ICC

Supplementary Table 10. Posterior slopes from BMLM model testing associations between signed trait evolutionary rates (RRphylo) and log-transformed speciation rate. Columns show the posterior mean slope (β), 95% credible interval (CI), and the probability that the slope differs from zero (e.g., $P(\beta < 0)$ or $P(\beta > 0)$), directionally reported according to the sign of β .

Model	Group	Posterior mean (β)	95% CI lower	95% CI upper	$0.05 < P(\beta \neq 0) < 0.95$
poratio_rate	Outcrosser	0.07	0.01	0.13	$P(\beta > 0) = 99.1\%$
poratio_rate	Selfer	0.00	-0.05	0.06	$P(\beta > 0) = 52.4\%$
petalW_rate	Outcrosser	0.02	0.00	0.05	$P(\beta > 0) = 96.4\%$
petalW_rate	Selfer	-0.02	-0.04	0.00	$P(\beta < 0) = 3.3\%$

petalL_rate	Outcrosser	0.01	-0.02	0.03	$P(\beta > 0) = 69\%$
petalL_rate	Selfer	-0.02	-0.04	0.00	$P(\beta > 0) = 5.7\%$
styleL_rate	Outcrosser	0.02	0.00	0.04	$P(\beta > 0) = 97\%$
styleL_rate	Selfer	-0.02	-0.04	0.00	$P(\beta < 0) = 3.1\%$
spratio_rate	Outcrosser	0.00	0.00	0.01	$P(\beta > 0) = 89.2\%$
spratio_rate	Selfer	0.00	-0.01	0.01	$P(\beta < 0) = 31.1\%$

Supplementary Table 11. BMLM model testing the association between signed trait evolutionary rates (RRPhylo) and log-transformed speciation rate. Columns report the posterior mean and uncertainty of fixed effects (intercept and slope), the standard deviations (SDs) of random intercepts for genus, tribe, and mating system, and the random slopes for mating-specific speciation effects. Residual variance (σ) is included where available. Intra-class correlation coefficients (ICCs) for genus, tribe, and residual components are also shown to quantify the taxonomic structure in evolutionary rate variation. All estimates are presented with standard errors and 95% credible intervals. Model components are labelled as Fixed, Random, Residual, or ICC.

Model	Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Component
poratio_rate	Intercept	0.04	0.78	-1.67	1.72	Fixed
poratio_rate	log_rate	0.05	0.76	-1.55	1.86	Fixed
poratio_rate	σ (genus)	0.03	0.02	0.00	0.07	Random
poratio_rate	σ (Intercept mating)	1.00	1.12	0.10	3.93	Random
poratio_rate	σ (slope mating)	0.78	1.13	0.02	3.79	Random
poratio_rate	σ (mating)	0.11	0.60	-0.93	0.98	Random
poratio_rate	σ (tribe)	0.02	0.01	0.00	0.06	Random
poratio_rate	σ (residual)	NA	NA	NA	NA	Residual
poratio_rate	ICC (Genus)	0.01	NA	NA	NA	ICC
poratio_rate	ICC (Tribe)	0.01	NA	NA	NA	ICC
poratio_rate	ICC (Residual)	0.98	NA	NA	NA	ICC
petalW_rate	Intercept	-0.01	0.68	-1.49	1.42	Fixed
petalW_rate	log_rate	0.00	0.72	-1.58	1.55	Fixed
petalW_rate	σ (genus)	0.01	0.01	0.00	0.02	Random
petalW_rate	σ (Intercept mating)	0.81	1.04	0.05	3.77	Random
petalW_rate	σ (slope mating)	0.67	1.01	0.02	3.55	Random
petalW_rate	σ (mating)	0.08	0.62	-0.95	0.98	Random
petalW_rate	σ (tribe)	0.01	0.00	0.00	0.02	Random
petalW_rate	σ (residual)	NA	NA	NA	NA	Residual
petalW_rate	ICC (Genus)	0.01	NA	NA	NA	ICC
petalW_rate	ICC (Tribe)	0.00	NA	NA	NA	ICC
petalW_rate	ICC (Residual)	0.99	NA	NA	NA	ICC
petalL_rate	Intercept	-0.02	0.71	-1.51	1.57	Fixed
petalL_rate	log_rate	-0.02	0.53	-1.25	1.13	Fixed

petalL_rate	σ (genus)	0.02	0.01	0.00	0.03	Random
petalL_rate	σ (Intercept mating)	0.83	1.05	0.05	3.66	Random
petalL_rate	σ (slope mating)	0.55	0.85	0.01	3.01	Random
petalL_rate	σ (mating)	0.06	0.63	-0.96	0.98	Random
petalL_rate	σ (tribe)	0.01	0.01	0.00	0.02	Random
petalL_rate	σ (residual)	NA	NA	NA	NA	Residual
petalL_rate	ICC (Genus)	0.02	NA	NA	NA	ICC
petalL_rate	ICC (Tribe)	0.01	NA	NA	NA	ICC
petalL_rate	ICC (Residual)	0.98	NA	NA	NA	ICC
styleL_rate	Intercept	0.00	0.65	-1.42	1.49	Fixed
styleL_rate	log_rate	-0.01	0.68	-1.47	1.41	Fixed
styleL_rate	σ (genus)	0.01	0.01	0.00	0.03	Random
styleL_rate	σ (Intercept mating)	0.74	0.99	0.04	3.57	Random
styleL_rate	σ (slope mating)	0.69	1.01	0.02	3.50	Random
styleL_rate	σ (mating)	0.08	0.62	-0.96	0.98	Random
styleL_rate	σ (tribe)	0.01	0.01	0.00	0.03	Random
styleL_rate	σ (residual)	NA	NA	NA	NA	Residual
styleL_rate	ICC (Genus)	0.01	NA	NA	NA	ICC
styleL_rate	ICC (Tribe)	0.02	NA	NA	NA	ICC
styleL_rate	ICC (Residual)	0.97	NA	NA	NA	ICC
spratio_rate	Intercept	0.01	0.49	-1.03	1.15	Fixed
spratio_rate	log_rate	-0.01	0.55	-1.26	1.06	Fixed
spratio_rate	σ (genus)	0.00	0.00	0.00	0.01	Random
spratio_rate	σ (Intercept mating)	0.45	0.86	0.00	2.95	Random
spratio_rate	σ (slope mating)	0.48	0.89	0.00	3.10	Random
spratio_rate	σ (mating)	0.02	0.65	-0.98	0.99	Random
spratio_rate	σ (tribe)	0.00	0.00	0.00	0.01	Random
spratio_rate	σ (residual)	NA	NA	NA	NA	Residual
spratio_rate	ICC (Genus)	0.01	NA	NA	NA	ICC
spratio_rate	ICC (Tribe)	0.01	NA	NA	NA	ICC
spratio_rate	ICC (Residual)	0.98	NA	NA	NA	ICC